



Supplementary Materials for

Asymmetric ideological segregation in exposure to political news on Facebook

Sandra González-Bailón *et al.*

Corresponding author: Sandra González-Bailón, sandra.gonzalez.bailon@asc.upenn.edu

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The PDF file includes:

Supplementary Text
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Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

S1 Overall Project Description

S1.1 Meta-Academic Collaboration Overview

This paper is part of the *U.S. 2020 Facebook and Instagram Election Study*, a broader set of experimental and observational studies that occurred as a result of a collaboration between academics and Meta. The project was designed to address three intertwined concerns related to scientific understanding of the impact of social media on democratic processes. First, in the aftermath of the 2016 U.S. elections, there was a widely recognized need to understand the impact of social media platforms on U.S. elections. Second, research conducted solely by platform employees could encounter skepticism from the mass public and policy community. And third, outside independent researchers not employed by the platforms faced legal and fiduciary challenges in securing access to the data and research pipelines necessary to conduct rigorous scientific analyses about the impact of social media on elections.

The *U.S. 2020 Facebook and Instagram Election Study* is an attempted solution to this bundle of challenges. The project represents a novel form of collaboration between a team of researchers at Meta and a set of external researchers.^{S1} The costs associated with the research (e.g., participant fees, recruitment, data collection, etc. – see SM section S2.2) were paid by Meta. The academic team members received no form of financial compensation (e.g., support for student assistants, course buyouts, research funds) from Meta for their participation in the project (see S1.2 for other measures applied to ensure the research integrity of the project).

Genealogy of the Project In early 2020, researchers at Facebook (now Meta) approached **Social Science One** about the possibility of jointly organizing a research project around studying the impact of Facebook and Instagram on the November 2020 U.S. elections. Social Science One had been created to facilitate industry-academia collaboration to study social media platforms and their impact on society and, in particular, to make data available to researchers who did not work for Meta. Social Science One at that point consisted of two directors/founders (Gary King and Nate Persily) and a series of advisory committees, each of which had a chair. These included Professor Natalie Jomini Stroud, the Chair of the North America Advisory Committee, and Professor Joshua A. Tucker, the Chair of the Disinformation and Electoral Integrity Committee. As the Chairs of the Social Science One Committees that were most related to the proposed project, Stroud and Tucker **agreed to jointly co-Chair the academic team** to collaborate with Meta on what has come to be called the *U.S. 2020 Facebook and Instagram Election Study*.

In the interest of balancing the competing needs of assembling a research team quickly and ensuring the necessary research expertise for the project, Stroud and Tucker made the decision to recruit the remainder of the academic team from existing Social Science One advisory committee members based on their diverse research expertise. In March of 2020, Stroud and Tucker

^{S1} At the time the project began in the spring of 2020, the company involved was called Facebook. For the sake of simplicity, we refer to the company by its current name, Meta, in the rest of the supplemental materials.

approached 16 of these advisory committee members, 15 of whom agreed to join the academic team for the project. Subsequently, one of the 15 withdrew in the first months of the project because of competing time demands. In addition, one of the advisory committee members requested that a current co-author on very closely related projects and with relevant statistical expertise for the project be allowed to join the team as well, resulting in an academic team of **17 members including Stroud and Tucker**. Meta had no say in who was selected by Stroud and Tucker to be part of the academic team.

Chad Kiewiet de Jonge was the Meta research manager who oversaw day-to-day management of the research project at Meta. Annie Franco and Winter Mason co-led the Meta research team, which grew to include 16 researchers, 2 data engineers, 1 data scientist, and 3 interns working on various parts of the overall project.

Project Execution Once assembled, the team of academics met beginning in March of 2020 to first brainstorm research ideas within the project’s mandate of studying Facebook and Instagram’s impact in the context of the U.S. 2020 elections and then to develop ideas for specific paper proposals. Concurrently, the team of Meta researchers began working with the academic team to provide feedback on research proposals, including the feasibility of possible designs and procedures for collecting the necessary data. As a result of this process, four general areas of inquiry were selected to form the scope of project: (1) dis/mis/information, knowledge, and (mis)perception; (2) political polarization; (3) political participation, both online and offline, and including vote choice and turnout; and (4) attitudes and beliefs about democratic norms and the legitimacy of democratic institutions.

The next step in the project involved identifying specific paper topics within the four general scope conditions. Based on their research interests, a subset of independent academic researchers served as “core authors” of each paper and were given control rights over final versions of the pre-analysis plans (PAPs) and papers.^{S2} Both the academic researchers and the Meta researchers worked together to design the pre-analysis plans.^{S3} The core authors for this paper and associated PAP (<https://osf.io/2y84h>) are Sandra González-Bailón and David Lazer.

Data collection was carried out by Meta and, for off-platform data, by NORC, an independent survey research organization at the University of Chicago.^{S4} Meta recruited most participants and collected on-platform data, while NORC carried out all surveys associated with the project, collected and appended all supplemental data outside of the Facebook/Instagram on-platform data, and recruited additional survey panelists (see SM section **S2** for more de-

^{S2}By *control rights*, we mean that in the event of disagreements between members of the research team, the core authors would have the final say in resolving these disagreements.

^{S3}A total of 16 pre-analysis plans were registered at the Open Science Foundation (<https://osf.io/>).

^{S4}NORC was selected following a competitive bidding process involving other online survey research firms. Employees of NORC who implemented the data collection process were not members of the independent academic research team. More details about NORC can be found at: <https://www.norc.org/Pages/default.aspx>

tails). The academic research team did not contact any human subjects as part of the research efforts. In the rare cases where members of the academic team – whose names **had been publicly announced** – were messaged by study participants, the messages were passed to NORC to respond.

At the data analysis stage, the Meta team produced, and the academics reviewed and approved, pipeline code used to produce the data tables needed for this project from raw platform data (e.g. number of followers) and data created for other internal Meta purposes (e.g. predictions of ideology of U.S. Facebook users) that were employed in the analysis. The Meta researchers and, in some instances, the independent academics, carried out the initial analyses as detailed in the pre-analysis plan and as deemed necessary by the full research team for mutually agreed upon research-relevant analyses. The academics' role in the analysis was to contribute to and monitor the results of data analyses conducted by the Meta research team, including: reviewing and, in some cases writing, code; inspecting de-identified samples or aggregated outputs through screen sharing; and, when possible, replicating the analyses within Meta's secure data-sharing Researcher Platform using data that has been stripped of any individually-identifying information.

Drafts of papers were written by the academic research team members, with feedback from the Meta academic researchers but with final control rights resting with the specified core academic authors (in the case of this paper, Sandra González-Bailón and David Lazer).

Figure **S1** summarizes the genealogy and execution of the project up to the submission of this paper. A full description of the roles and responsibilities of the academic research team, the Meta researchers, and NORC can be found at the Open Science Foundation.^{S5}

S1.2 Research Transparency and Integrity

One of the primary goals in designing the project was to build in transparency concerning the research process given the constraints under which we were operating. With this in mind, five conventions were adopted to guide the research process.

- First, none of the academic researchers nor their institutions received financial compensation (e.g., support for student assistants, course buyouts) from Meta for their participation in the project.
- Second, the analysis for all of the papers resulting from the project, including this one, were pre-registered at the Open Science Foundation (that is, we specified research questions, hypotheses, methods, and planned analysis before data collection). The pre-registrations are embargoed while the research is being carried out, but are publicly released at the time of publication. The pre-analysis plan (PAP) that was registered for this study is included in SM section **S7**.^{S6} A list of deviations from and clarifications of the pre-analysis plan can be found in SM section **S8**.

^{S5}<https://osf.io/7wpgd>

^{S6}Pre-registration for this paper is at <https://osf.io/2y84h>.

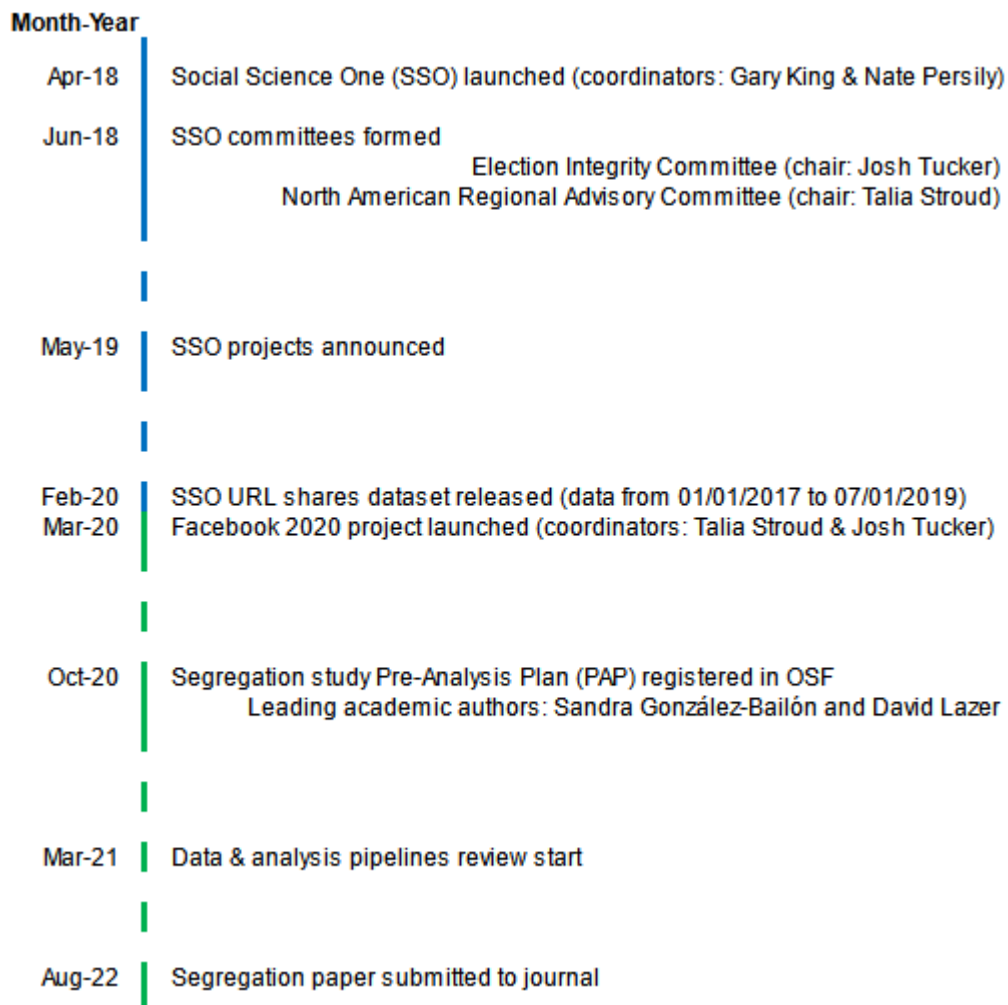


Figure S1: **Project Timeline.** Summary of the genealogy of the *U.S. 2020 Facebook and Instagram Election Study* and key dates in the execution of this paper (pre-registered in October 2020).

- Third, for every paper, a set of core authors with control rights over the final content of the paper were specified in the pre-analysis plan. These core authors consist only of academic researchers (i.e., not employees of Meta). The core authors with control rights for this paper are Sandra González-Bailón and David Lazer.
- Fourth, Meta **publicly** agreed that there would be no pre-publication approval of papers for publication on the basis of their findings. At the time the PAPs were proposed – but before any data analysis was conducted – Meta conducted legal, privacy, and feasibility reviews of the studies. Meta was entitled to review papers prior to publication, but could only request changes to protect confidential or personally identifiable information or to abide by their existing legal obligations.^{S7} For this article, Meta requested that we revise statements about race and ethnicity included in pages SM-14 and SM-15. The revised sentences originally stated that Meta does not have any data on the race or ethnicity of its users. Meta requested the clarification that it had limited race/ethnicity data coverage for its U.S. users and it could not be used to inform large scale probability sampling.
- Fifth, we appointed a rapporteur for the project – Professor Michael Wagner of the University of Wisconsin, Madison – who was neither a paid employee of Meta nor a member of the academic research team. The rapporteur was given access to the project researchers, allowed to join project-related meetings, and had access to project documents. The rapporteur is not a co-author on any of the papers resulting from the study, but the expectation is that he will publish both academic and popular press articles assessing the research process itself.

We also decided that our primary approach to releasing results publicly would be upon completion of the peer review process. The peer review process involves having other scholars, who are not authors on the paper, review the work, provide criticism and feedback, and make a recommendation as to whether the scholarship is worthy of publication. The output from this project was never intended to be a “report” on Facebook and Instagram’s impact on the 2020 U.S. election, but rather a series of peer-reviewed academic publications addressing scientific questions related to the impact of various aspects of the Facebook and Instagram platforms on the 2020 U.S. election.

Finally, Meta plans to make de-identified datasets from each published study conducted under this initiative and designed in collaboration with the academic team available to the broader research community, so that others can reproduce the analyses and conduct further election studies.

Below we also list declarations from the academic author team. For consistency, we use the following key:

^{S7}<https://about.fb.com/news/2020/08/research-impact-of-facebook-and-instagram-on-us-election/>

a	Current employee (Meta)
b	Past employee (Meta)
c	Own individual stocks (Meta)
d	Paid consulting work (Meta)
e	Direct research funding from Meta (grant to you as PI or Co-PI)
f	Received an honorarium/fee (from Meta) for attending or hosting an event/serving as outside expert
g	Attended a Meta event where food, travel, or lodging was paid for by the company
h	Current employee (at a related company: Twitter, TikTok, Google/YouTube)
i	Past employee (at a related company)
j	Own individual stocks (at a related company)
k	Paid consulting work (at a related company)
l	Direct research funding from a related company (grant to you as PI or Co-PI)
m	Received an honorarium/fee (from a related company) for attending or hosting an event/serving as outside expert
n	Attended an event (at a related company) where food, travel, or lodging was paid for by the company

Author declarations: Hunt Allcott: former Microsoft employee; none of the above. Deen Freelon: g. Matthew Gentzkow: f, g, m, n. Sandra González-Bailón: g, l. Andrew Guess: e, g. Shanto Iyengar: e, g. Young Mie Kim: g. David Lazer: g, n. Neil Malhotra: g, n. Brendan Nyhan: e, g, n. Jennifer Pan: e, f, g. Jaime Settle: c, e, g, j. Natalie Jomini Stroud: d, e, g, l, n. Emily Thorson: g. Rebekah Tromble: e, g, l. Joshua A. Tucker: e, f, g, n. Magdalena Wojcieszak: e, g, n.

We also provide the following additional disclosures:

- Ancillary support (e.g., research assistants, course buyouts, etc.) was sourced, where applicable, by academics from the Democracy Fund, the Guggenheim Foundation, the John S. and James L. Knight Foundation, the Charles Koch Foundation, the Hewlett Foundation, Hopewell Fund, the Alfred P. Sloan Foundation, the University of Texas at Austin, New York University, Stanford University, the Stanford Institute for Economic Policy Research, and the University of Wisconsin-Madison.
- Hunt Allcott received compensation from another source for consulting on a matter related to Meta’s business. H.A. was an employee of Microsoft Research until December 2022.
- Matthew Gentzkow received compensation from Compass Lexecon for advice on a project for which Meta was the client. He has been a paid consultant for Amazon and done economic consulting for Analysis Group for clients including Google. He has received compensation as a member of the Toulouse Network for Information Technology, a research group funded in part by Microsoft.

- Brendan Nyhan has served as a member of the Misinformation Interventions Working Group at Meta since 2022. Meta has donated the honorarium he would receive for participating in this group to Doctors Without Borders on his behalf.
- Jennifer Pan served as a member of Meta’s News Integrity Circle from fall 2020 to fall 2022. She received an honorarium for participating in the fall 2020 and spring 2021 session; the honorarium she would have received for participating was donated to Reporters without Borders for the fall 2021, spring 2022, and fall 2022 sessions.
- Natalie Jomini Stroud provided consultant services to Facebook in 2018–2019 on a prior research project; she was paid before the current project began.
- Joshua A. Tucker received a fee from Facebook to compensate him for administrative time spent in organizing a 1-day conference for approximately 30 academic researchers and a dozen Facebook product managers and data scientists that was held at NYU in the summer of 2017 to discuss research related to civic engagement; his fee was paid before the current project began. He is currently a Kroll Institute Fellow and Senior Advisor at Kroll.

S1.3 Ethical Considerations

Researchers involved in the project considered a number of ethical concerns related to the research and designed the studies to minimize potential harms to the respondents involved in them, as well as any broader social harms.

Unlike other studies in the project, which required informed consent, the study reported in this paper involved analyzing platform data for *all* adult U.S. users, in aggregate form. As explained in SM section [S2](#), we aggregated the data following privacy protection rules (e.g., only URLs shared at least 100 times are analyzed). The part of the analyses that relies on the Web panel data includes only consenting participants. Individual-level participation in the panel was compensated and required informed consent (see SM section [S2](#) for more details on overall study design).

Meta sought review from and was granted approval to conduct the experimental studies by the NORC Institutional Review Board (Protocol number 20.08.10, Project number 8870). Academic collaborators worked with their respective university IRB’s to ensure compliance with Human Subjects Research regulations in their authorship of papers, including analysis of aggregated, de-identified data collected by Meta and NORC.

S1.4 Professional Ethics Advice

Meta retained the services of *Ethical Resolve*, a data ethics firm that was consulted by both Meta and academic researchers at various stages of the project prior to implementation of the research

to evaluate whether it met long-running traditions of research ethics as well as emerging norms and best practices for conducting digital research.^{S8}

S2 Study Design for this Paper

In SM section S1, we introduced the scope and characteristics of the *U.S. 2020 Facebook and Instagram Election Study*. This paper is one of the multiple studies that, as discussed, were pre-registered as part of the larger project. In this section, we offer details of the specific data that was used to produce the results in this paper. The four figures reported in the main text, and most of the results discussed (i.e., research questions 1-3 and 5-6), are based on aggregated platform data tracking *all* political news URLs posted privately or publicly by *all* active, adult U.S.-based users. In other words, the results we report are *not* based on recruited participants but on *all* adult U.S. users that were active on the platform. This is the reason why we aggregate the data, as explained in subsection S2.1: we wanted to preserve the privacy of users who did not opt into the *U.S. 2020 Facebook and Instagram Election Study* (which are the majority of Facebook users).

Some of our analyses, briefly mentioned in the main text and only reported in full in this SM, make use of the panel data collected by NORC to track off-platform behavior and attitudes (as explained in SM section S1 and discussed in more detail in subsection S2.2). In particular, we use this panel data for two purposes: (1) to test the accuracy of the on-platform ideology classifier (as discussed in SM section S3.4); and (2) to calculate segregation measures for news exposure on the web, which allows us to address research question 4 (we discuss in depth the analyses to answer this question in SM section S5). In SM section S2.2 we offer more details about this panel. In particular, we discuss information on sampling; participant recruitment and consent; response rates; and passive web tracking, the part of most relevance for this paper and the analyses comparing on-platform and off-platform (i.e., web) segregation.

S2.1 Aggregated Platform Data

Our platform-wide data includes aggregated activity generated by *all* adult, U.S.-based, active Facebook users whose ideology could be predicted using Facebook’s internal classifier (see SM section S3.4), which corresponded to $N \sim 208$ million users during our observation period. This activity included the creation of, impressions on, and engagement with political news URLs between September 2020 and February 2021. Most results reported in the main text, and the four main figures, rely on platform-wide data.

We intentionally built privacy protections into the research design of this paper that involved preventing the academic research team from accessing individual-level data of exposure to political news URLs. All data used in this paper were shared with the academic team in aggregate form. Code to collect and process the data from Meta servers was written by Meta researchers

^{S8}<https://ethicalresolve.com/>

and then reviewed by at least one member of the academic research team. The academic researchers only had access to code that was implemented within Meta specifically for this project (e.g., the pre-existing code to aggregate the ratings from third-party fact checkers into misinformation ‘strikes’, as defined under the company’s Misinformation Repeat Offender policy, was not provided for review).

While the use of platform-wide data is permissible under Meta’s Data Policy, care was nevertheless taken to ensure that user privacy was protected. These procedures included the following:

- First, all requests were reviewed by Meta Legal and Privacy to ensure that no individual-level or identifying data were shared with the academic research team.
- Second, techniques were used to reduce the risk of any individually identifying data being shared. In the case of this paper, privacy-protection measures restricted data in two ways: (1) none of the analyses involve individual-level behavioral data; all data analyzed are aggregated at the domain and URL levels; and (2) the URL- or domain-level analyses only consider URLs or domains that were shared more than 100 times (for platform data; the threshold for the web panel data described in [S2.2](#) excludes domains with less than 20 unique panelist visits). In SM section [S3.2](#) we offer some analyses of how these thresholds affect the patterns we analyze.

S2.2 Panel Data

The analyses comparing on- and of- platform segregation rely on data from a set of consenting participants for which both Facebook log data and passive web tracking data were available. This sample was initially recruited for the purposes of studying how changes in platform features affect users attitudes and behavior, but it is used here for descriptive purposes to directly compare web browsing patterns and on-platform exposure to content for the same exact set of users. To avoid the possibility that any impact of these interventions on respondents’ exposure to news may affect our conclusions, in this study we only rely on the control group of this sample, i.e., users not affected by the experimental interventions applied as part of other studies. Figure [S2](#) summarizes the selection of panel users and participant flow through each stage of the selection process. Participants generating the data we analyse in this study were paid \$5 for completing the surveys (waves 1 and 2, see below). The total sum of incentives paid to participants whose data is used in the ideology analysis (N = 19,543; see SM section [S3.4](#)) is \$97,715. The total sum of incentives paid to participants whose data is used in the analysis of web activity (N = 2,724, see SM section [S5](#)) is \$13,620. The rest of this section describes in more detail the recruitment process and completion rates.

Sampling The sampling approach was designed to achieve specific sample targets across different stages of a six-wave survey panel (this panel data is more extensively used in other papers

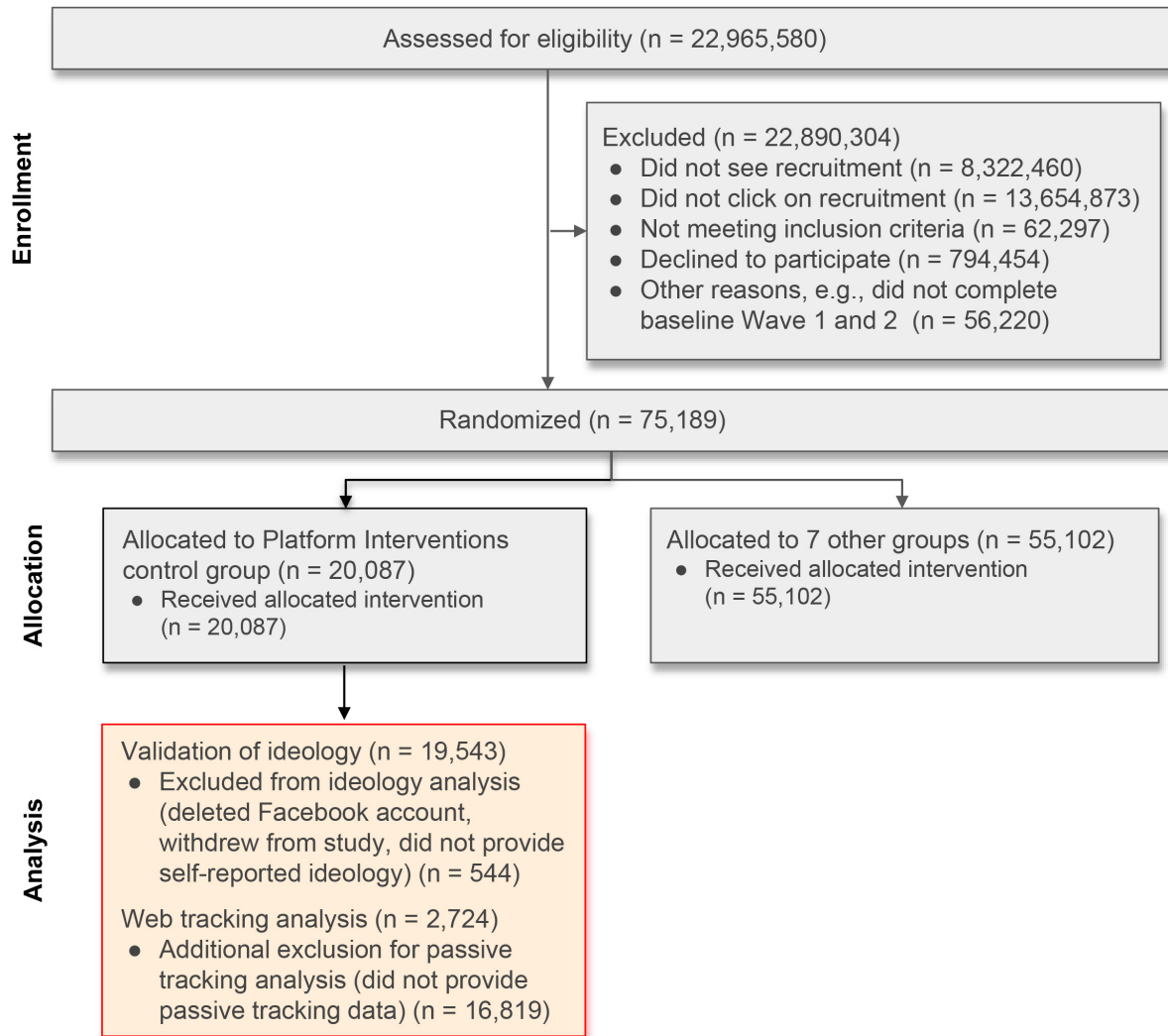


Figure S2: Recruitment Flow Diagram. A total of 75,189 users agreed to participate in the platform interventions conducted as part of the *U.S. 2020 Facebook and Instagram Election Study* and were eligible (e.g., completed baseline surveys in Waves 1 and 2). Of these, 20,087 were randomly allocated to the control group, i.e., this group did not see any experimental intervention and therefore their user experience did not change. Our analyses are based on this group of recruited participants and, in particular, on those who provided self-reported ideology (N = 19,543) and those who provided web tracking data (N = 2,724). We use self-reported ideology to test the performance of the on-platform ideology classifier (S3.4) and to compare on-platform segregation with off-platform, web segregation (S5).

within the larger project). The sample targets for this study are provided in Table S1. These targets were chosen to achieve desired minimum detectable effect sizes (MDEs) for the project’s experimental interventions (reported in separate papers) across different subgroups and among the set of respondents participating in wave 1 (recruitment) and wave 2 (baseline) surveys in addition to at least one of waves 4 or 5 of the study (we refer to these participants as *three-wave completes*). We worked backward from our target MDEs in waves 4 and 5 to determine the number of respondents we would need at the recruitment stage (initial 3-wave). The three-wave completes target sample for on-platform interventions was adjusted after observing the level of attrition between the wave 1 and wave 2. In response, and prior to treatment randomization, the academic researchers in the study decided to reduce the set of planned platform interventions, which also reduced the target sample size. The revised targets are provided in the last column of Table S1 (final 3-wave).

Table S1: Target size for the sample used in this study

Apps	Study sample	<i>Sample targets</i>		
		Wave 1	Initial 3-wave	Final 3-wave
FB	Platform interventions	119,000	71,118	54,740

The three-wave sample targets were revised after observing the wave 1 and wave 2 survey completions and adjusting the proposed number of treatment arms prior to randomizing participants into condition.

Sampling Frames and Stratification The sampling frames included all Facebook monthly active U.S.-based users 18 years of age or older eligible to receive general surveys as of August 17, 2020. Participants were asked to confirm they were over 18 years of age and lived in the United States as part of the recruitment process. The Facebook sampling frame was trimmed by removing predicted fake accounts, employees, and advertisers.

The sampling frames were stratified along several covariates of interest, i.e., number of days a user was active on the platform (the number of days a user logged in to Facebook in the 30 days on or before August 17, 2020, classified into three categories: 1–14 days, 15–29 days, and 30 days); a user’s predicted census region (East, Midwest, South, West)^{S9}; whether the user is predicted to live in a battleground state^{S10}; a user’s predicted ideology (liberal, moderate, or

^{S9}The classification of states across each of the census regions is available [here](#)

^{S10}Following the two most recent [Electoral College Ratings](#) by the Cook Political Report prior to August, we defined as battleground states those whose complete electoral geography was considered in the “Toss Up”, “Lean Democrat”, or “Lean Republican” in at least one of the reports. “Toss Up” states included: Arizona, Georgia, Maine, North Carolina; “Lean Democrat” or “Lean Republican” states included: Florida, Michigan, Minnesota, New Hampshire, Pennsylvania, Wisconsin, Iowa, Ohio, and Texas. Nebraska was excluded because only one of three congressional districts was identified as a battleground district.

Table S2: Target sample demographic distribution of survey sample

Demographic	Target Distribution
Number of days user logged in to app	Less than 15 days (4%), between 15 and 29 days (24%), and 30 days (72%)
Minority users (Black or Hispanic)	56% in battleground states and 58% in non-battlegrounds states.
Users in battleground states	40%
Predicted ideology	Conservative, liberal, moderate. No initial target.

conservative)^{S11}; and the census ethnic/racial composition in the zip code in which a user is predicted to live (percent of Hispanic residents and Black residents)^{S12}. The stratification of the sampling frame for this sample generated 621 population cells.

Sampling Probabilities and Target Distributions Sampling probabilities were computed to achieve specific sample distributions for the set of demographics encoded in the stratification step across each of the samples of interest. The sampling probabilities took into account (a) differential non-response across different demographics^{S13} and (b) the desired sample size across the different studies.

The Wave 1 target sample size took into consideration a total attrition rate of 40% between Wave 1 and the combination of Wave 2 and at least one of Waves 4 or 5 (three-wave completes) of the study while ensuring our ability to detect an MDE of 1.5 percentage points on turnout and vote choice for the experiments (reported in separate papers). The specific distributions we aimed for the sample used in this paper are included in Table S2.

As shown in Table S2, the target distribution included 4% of those logging on for fewer than 15 days, 24% between 15 and 29 days, and 72% for 30 of the past 30 days. This includes an oversample of those using the platform more frequently. In battleground states, the target sample of minority (Black or Hispanic) users was 56%. In non-battleground states, the target sample was 58% minority users. The remaining rows show the targets based on geography and predicted ideology.

^{S11}See SM section S3.4 for additional details on how ideology is predicted. Liberal users have predicted ideology score below 0.35, conservative users have a predicted ideology score above 0.65; moderates have a predicted ideology score between 0.35 and 0.65.

^{S12}Some fields had missing values (e.g., predicted ideology, state, and zip code). Individual values were imputed probabilistically using the distribution of demographics in the population. In general, the percent of missing values for a given demographic was quite small, never exceeding more than a few percentage points of the population.

^{S13}Responses to Facebook surveys with a similar design were used to model differential response rates.

In meeting these targets, it is important to take into account three considerations:

1. All else equal, the probability of a user being invited to participate in the study in a given strata is proportional to its size in the sampling frame.
2. There was no initial target distribution for ideology. We incorporated this dimension to the stratification in the second week of recruitment after seeing that self-reported white liberal users were more likely to consent to participate in the study. We did so by oversampling, based on their predicted ideology, moderate and conservative users. No specific targets were identified, but the proportion of users who self-identified as Democrats was reduced.
3. Meta informed the academics that it had limited race/ethnicity data coverage for its U.S. users and it could not be used to inform large scale probability sampling. Instead the probability that a given survey respondent identifies with a given ethnic/racial category was derived based on the ethnic/racial distribution of a user's predicted zip code. The implementation of this approach is presented in more detail in the 'Sampling for Race and Ethnicity' subsection below. As the approach to sample minority users had mixed results in the early stages of the recruitment period, and in light of observed imbalances in ideology, ideology targets were prioritized.

Implementation of Sampling Scheme After defining the sampling probabilities, the sampling scheme was executed as planned. In the implementation of the sampling scheme two additional steps were taken:

1. The sampling probabilities were adjusted when the size of a given stratum was exhausted. The adjustment to the sampling probabilities for non-exhausted cells across the sampling frame was done in proportion to their size.
2. Sampling was executed sequentially to avoid users being invited to more than one experimental intervention within a given app. This left a small probability that users of Facebook and Instagram could have been invited to participate in a similar or different experience across the two apps. This issue does not affect this study since we only rely on the sample recruited on Facebook.

Sampling for Race and Ethnicity The project aimed for specific sample proportions of Black and Hispanic users in the sample. Meta informed the academic team that Facebook and Instagram do not have data on the race/ethnicity of U.S. users that could have informed our sampling decisions. Therefore, the probability a given user identified with one of the categories of interest within a given strata of the sampling frame was modeled. The focus was on achieving targets for the White and Other ethnic categories, which allowed us to obtain the desired distribution for minorities as the residual categories, based on people's predicted zip codes. The implementation of this approach involved the following steps:

1. Let i represent one of the stratum of the S strata in the sampling frame of a given app. Let X_i represent the vector of covariates defining the characteristics of strata i . The vector X_i has the following components: number of days users were active on the platform in the last 30 days, census region, battleground state, predicted ideology, zip code percent Black, zip code percent Hispanic.
2. Within a given X_i we compute the probability that a user belongs to one of three different categories: Black, Hispanic, White or Other. The following steps were followed:
 - (a) We use the quartiles of percent Black and Hispanic in a given zip code. For these two categories we have four potential values 0–25%, 25–50%, 50–75%, 75%+.
 - (b) The lower and upper bounds of the residual category are backed out of the combination of these quartiles. For example, in a zip code with 0–25% Hispanic and 0–25% Black, we know that the White or Other population represents at least 50% or at most 100%.
 - (c) Based on this information, the probability of a user’s race/ethnicity was derived using the midpoints of each of the quartiles across all ethnic categories. In the example introduced in the previous bullet, the probabilities were as follows: $\Pr(\text{Hispanic}) = \Pr(\text{Black}) = 0.125$ and $\Pr(\text{White or Other}) = 0.75$.
 - (d) The only exception is when the bounds for White or Other are 0–25%. In this situation, we set $\Pr(\text{White or Other}) = 0.125$ and define $\Pr(\text{Hispanic}) = \text{Hispanic midpoint} * (1 - \Pr(\text{White or Other}))$ and $\Pr(\text{Black}) = \text{Black midpoint} * (1 - \Pr(\text{White or Other}))$. (This solution applies to cases when the percent Hispanic is 0–25% and the percent Black is 75–100%.) Otherwise, if we simply used midpoints to assign probabilities would yield $\Pr(\text{Hispanic}) = 0.125$, $\Pr(\text{Black}) = 0.875$, and $\Pr(\text{White or Other}) = 0.125$, which would sum to more than one.
3. For a given attrition and click-through rate we back out the number of wave 1 respondents in the White or Other category needed to achieve the target for this group in the three-wave complete sample. In particular, the number of wave 1 respondents in the White or Other category is:

$$n_o|stratum_i = pr(\text{White or Other}|stratum = i) * pr(s = 1|stratum = i) * N_i$$

where $pr(s = 1|stratum = i)$ represents the probability a user in stratum i is invited to participate in the study and N_i denotes stratum i ’s population size.

Solving for the sampling probability in stratum i yields:

$$pr(s = 1|stratum = i) = n_o|stratum_i / (pr(\text{White or Other}|stratum = i) * N_i)$$



Figure S3: Image shown to recruit participants on Facebook

4. Note that the $pr(s = 1|stratum = i)$ derived in the previous step implies that we have a total number of invitations of Black and Hispanic users. These are given by:

$$n_{Black}|stratum_i = pr(Black|stratum = i) * pr(s = 1|stratum = i) * N_i$$

$$n_{Hispanic}|stratum_i = pr(Hispanic|stratum = i) * pr(s = 1|stratum = i) * N_i$$

From these quantities we can back out the implied numbers of respondents in the wave 1 and three-wave sample in a given stratum.

5. We repeated steps 1–4 to obtain across all strata S of the sampling frame of a given app to try to obtain the desired targets. However, as noted above, we decided to prioritize over-sampling users on predicted ideology because (a) our initial approach exhibited mixed results in terms of meeting the specified minority targets, and because (b) the early recruitment numbers showed a skew towards Democrats. To correct the partisanship skew, we simply repeated steps 1–4 across each of the three predicted ideology strata (L, M, and C) with a higher proportion of survey invitations allocated to the M and C strata.

Participant Recruitment and Consent At the top of their Facebook Feed, randomly selected participants saw a recruitment message asking them if they would like to share their opinion as shown in Figure S3. Those clicking “Start Survey” were directed to a consent form. Participants gave their consent to participate in the on-platform experiments using an IRB-approved consent form, as follows:

***Do You Want to Participate in a Research Study About the U.S. Election in November?**
Your participation in this research will help researchers at New York University, The University of Texas at Austin, and other academic institutions, as well as Facebook, understand more about how people’s experience with Facebook and Instagram affects their opinions and behaviors on elections.*

How it Works Over the next four months, you'll be asked to fill out a short survey each month. This monthly survey will take about 15 minutes, for a total of 60 minutes over four months. Our partner, NORC at the University of Chicago, will administer this research. During this time, your [Facebook/Instagram] experience may be different than what you're used to. For example, you might:

- See more or fewer ads in specific categories such as retail, entertainment, or politics
- See more or fewer posts in [News Feed / your feed] related to specific topics
- See more content from some [friends / connections] and less content from other [friends / connections]
- See more or less content about voting and elections

You'll be paid at least \$30 for participating in this study and completing all four surveys, including \$5 for each of the first two surveys and \$10 for each of the final two surveys.

- You will receive your reward as an electronic gift card, delivered within 1 day of completing each survey
- You can only take each survey once
- If you do not complete the first survey, you will be removed from this study

If you choose to participate in this study, your survey responses will be linked with your Facebook and Instagram activity data from the 2020 calendar year.

Benefits, Alternatives, and Risks There are no benefits to participating in this research, nor are there risks greater than those encountered in everyday life, including risks related to the loss of confidentiality. You can learn more about how we're keeping your information safe in the Data Collection and Your Privacy section below. You can choose not to participate in this study.

Data Collection and Privacy If you choose to participate in the study, the following will happen:

- NORC will join your survey responses to publicly available third-party data like if you've voted or made a political contribution, if this data is available
- Facebook will combine this data with your activity on Facebook and Instagram from the 2020 calendar year, collectively called Combined Data
- This Combined Data will only be used for research purposes and will not be used to show you ads

- *This Combined Data will be shared with our academic partners and, if legally required, with the Institutional Review Board (IRB) that reviewed this study*
- *All access to this Combined Data will be monitored and logged*
- *Once this study is over, de-identified data (i.e. data where identifiers such as your name and other information that could reasonably be linked to you are removed) will be stored and shared for future research on elections, to validate the findings of this study, or if required by law for an IRB inquiry*

You can decide to stop participating in this study at any time, for any reason, and without consequences. You may withdraw by visiting the study website hosted by our survey administrator, NORC at the University of Chicago, at 2020erp.norc.org. If you have any questions related to this research, you can email NORC at erpStudy@norc.org, or call toll-free at (866) 270-2602 between 9:00 AM - 10:00 PM ET.

If you are a research participant and have questions about your rights, or have concerns or complaints about this research, you can email the NORC Institutional Review Board (IRB) at surveyhelp@norc.org or call (866) 856 - 6672 between 9:00 AM and 10:00 PM ET. Please note that by contacting or providing information to NORC IRB, NORC IRB may obtain information about you, including any personal information that you share. Even though NORC IRB is affiliated with Facebook as this research study's IRB, Facebook's Data Policy does not apply to any information about you shared with NORC IRB when you initiate contact.

If you join this study, you affirm that you are at least 18 years of age and live in the United States. Once you join this study, you'll be sent off [Facebook/Instagram] to a site hosted by our study administrator, NORC, to complete a 5-minute enrollment form.

Data Collection Timeline Data collection began with a soft launch on August 31, 2020 and continued through March 2, 2021. In this study we only use data from the first two survey waves, which were fielded as follows:

- **Wave 1:** A subsample of Facebook-recruited respondents were invited to the survey on August 31 in a soft-launch. The remainder of sampled Facebook/-recruited respondents were invited to the survey on September 1. The recruitment of the sample continued until Saturday, September 12. The wave included the recruitment and consent processes and a short survey.
- **Wave 2:** The field period for Wave 2 started on September 8 and continued through September 23. The wave included a baseline survey and was conducted prior to randomization into a control group and a series of different platform intervention treatment groups. As noted above, in this study we only use data from the control group.

Response Rates Table S3 contains details on the survey response rates. Eligibility was determined based on respondents' age (18 or above) and removing duplicate cases. While the study continued for up to six survey waves and contained other samples and platforms, here we only report the rates that are relevant to the data we use in our study: the Facebook sample recruited for on-platform interventions that completed waves 1 and 2 of the survey. The control group that we analyze in this study corresponds to a randomly selected subset of 35% of participants within this sample.

Table S3: Survey completion and response rates

Row	Measure	FB Intervention sample	Definition / Formula
Pre-NORC Recruitment			
A	QP eligible users sampled	22,965,580	
B	QP viewers	14,643,120	
C	QP clickers	988,247	
D	DE willing	n/a	
E	DE amount selected	n/a	
F	Consented to full study	193,880	
G	Did not consent to full study	794,367	C-F
H	Subsample of study non-consenters asked to consent to surveys only	n/a	
I	Consented to surveys only	n/a	
J	QP viewers among sampled	63.8%	B/A
K	QP clickers among QP viewers	6.7%	C/B
L	DE willing among QP clickers	n/a	
M	DE amount selected among DE willing	n/a	
N	Consented among asked	19.6%	F/C
O	Response rate before pass to NORC	0.8%	J*K*N
Wave 1			
P	Passed to NORC	189,792	
Q	Screened for eligibility	163,207	
R	Confirmed eligible	162,698	
S	Completed wave 1	139,193	
T	Screening completion rate among passed to NORC	86.0%	Q/P
U	Eligibility rate among screened	99.7%	R/Q
V	Interview completion rate among eligible	85.6%	S/R
W	Response rate among those passed to NORC	73.6%	T*V
X	Cumulative response rate, pre-NORC recruitment wave 1	0.6%	O*W
Wave 2			
X	Invited	139,193	
Y	Screened for eligibility	77,438	
Z	Confirmed eligible	77,405	
AA	Completed wave 2	75,318	
AB	Screening completion rate among invited	55.6%	Y/X
AC	Eligibility rate among screened	100.0%	Z/Y
AD	Interview completion rate among eligible	97.3%	AA/Z
AE	Response rate among invited, wave 2	54.1%	AB*AD
AF	Cumulative response rate, pre-NORC recruitment waves 1 2	0.3%	X*AE

Weighting for Panel Data For the on-platform interventions sample we use in this study, survey weights were created to generalize treatment effects to the best estimate of adult monthly active users: all U.S. Facebook monthly active users 18 years of age or older eligible to receive general surveys as of August 17, 2020. The general approach to creating the weights was to reduce bias while maintaining a low design effect.

Inverse Propensity Scores Weights (IPSW) using LASSO regression with Facebook log data were built. Covariates used for block randomization in the experiments and variables presumed to predict treatment heterogeneity were prioritized. The weights calibrate to the full population of Facebook users and they were built using:

- Predicted ideology (divided into liberal, moderate, and conservative using the ideology classifier described in SM [S3.4](#) with cut points of 0.35 and 0.65).
- Friend count, divided based on terciles.
- Political Pages followed, divided by tercile.
- The number of days a user logged on to their account in the 30 days prior to August 17, 2020, divided into 29 or less vs. 30.

We used raking to create the set of final weights that calibrate to population estimates of race (white vs. non-white), party ID (Democrat, Independent, or Republican, including leaners as partisans), and education (less than a college degree vs. a college degree or more). The specific targets are based on the distribution of these characteristics among self-reported Facebook users in a separate dataset (the AmeriSpeak panel) that is representative of the U.S. adult population.

Our final step was to trim the weights. Following the [Cooperative Election Study](#), which trims weights above a particular threshold, and the [Pew Research Center](#), which has trimmed weights at the 1st and 99th percentiles, we trimmed the top 1% of the survey weights.

We did not include design weights in the computation of the survey weights as the weights increase the design effect significantly without appreciably decreasing the bias.

Passive Web Tracking Participants in the panel data (described in the previous section) were asked for their consent to track their mobile and desktop internet browsing behavior. To collect the passive measurement data, NORC partnered with two vendors: MDI Global and RealityMine. Users who consented to passive data tracking were asked to install an app and use a virtual private network (VPN) on their mobile or desktop devices to collect data about the number of visits and time spent on different web domains, as well as usage and time spent on apps on their mobile device.^{S14} The app was developed by MDI Global and the VPN was developed and maintained by RealityMine. Both firms collected the passive tracking data and

^{S14}Note, however, that in this study we only analyze their web visits, which we compare to their exposure logs on Facebook for the same set of domains.

sanitized, truncated, and/or categorized the URLs to minimize the risk of sharing any additional personally identifiable information (PII).

The passive measurement software collected data on which applications participants were using and for how long on mobile devices, but no data were collected on what participants were doing within those applications. On all devices, information was collected on the websites that participants were visiting, but no additional information was collected beyond the domain name. For instance, a participant searching for information on “election night meals” on Google would be logged as www.google.com rather than <https://www.google.com/search?q=election+night+meals>. Further, k-anonymization was applied to the domain-level data by excluding domains that had visits from fewer than 20 unique panelists. The applications and VPNs only collected data while installed on a participant’s device. The software could be uninstalled at any time and data collection could be paused using functionality in the apps.

Participants were invited to enroll using the email address provided in wave 1 of the Facebook/Instagram sample survey. They were invited to download the passive monitoring software between September 11 and September 21, 2020. The recruitment language was as follows:

Subject: 2020 Election Research Project: Additional Study Opportunity

As a member of the 2020 Election Research Project, you have been selected to participate in an additional study to learn more about the apps you use and sites you visit.

You can earn up to \$90 for choosing to participate in this additional study. To participate, you only need to install the software and keep it active for the 3 month study.

NORC at the University of Chicago and the study sponsor, Facebook, would like to understand more about how you’re using your device during this study. To participate, you’ll need to download software to your device. When installed, this software will automatically collect data about your device and the websites you visit and apps you use. The data will only be used for research purposes. Please note that passwords, and other information you might enter on websites, like your banking details, will not be collected.

Those who clicked to learn more were provided with the following additional information:

NORC at the University of Chicago and the study sponsor, Facebook, would like to understand more about how you’re using your device during this study. To participate, you’ll need to download an app, install a Virtual Private Network (VPN), or a browser plugin to your device. This software is developed by NORC’s partners, MDI and RealityMine. When installed, this software will automatically collect data about your device and how you use it as further specified below, and no further action will be required from you. Please note that passwords, and other information you might enter on websites, like your banking details, will not be collected. You may install the software on one or more devices.

Installing this software is completely optional. Should you choose to install it, researchers at New York University, The University of Texas at Austin, and other academic institutions, as well as Facebook will use the data to better understand how online behavior changes in response to events during the course of the study. More information on this software can be found here: (FAQ information appended below).

Earn \$5 per device just for installing and setting up the software. For your first 2 weeks of data sharing, earn an additional \$5 per device (maximum 2 devices). You'll then earn another \$5 per device at the end of your first month, if your devices are still sharing data. That means you could earn up to \$30 in your first month! Keep participating and you'll earn \$10 per month, per device. If you complete all 3 months, you'll receive a bonus \$20 for 2 devices, or \$15 for one device. This means you can earn up to \$90 for 3 months of participation in this study! You will be paid for a maximum of 2 devices, though you may install the software on as many devices as you'd like. You may forfeit the monthly payment if you fail to send data from your mobile device for 3 days in a row or from your computer for 8 days in a row.

- *All of your mobile device's data will flow through a VPN connection on iOS*
- *Web data for specific browsers will flow through a VPN on Android*
- *All of your desktop or laptop data will flow through a browser plugin*
- *Of the data that flows through the VPN or browser plugin, NORC will collect data on:*
 - *Your operating system, device model and manufacturer, and device type (e.g., mobile, tablet, desktop)*
 - *Which apps you use, including app name and category, the date and time you use the app, and for how long*
 - *Which browser you use and technical details about your session such as your IP address*
 - *What websites you visit, the date and time you visit a website, when and for how long*
- *NORC will use this data in order to facilitate the research and for data quality assurance purposes*
- *Of the data collected, the following device data will be shared with Facebook and Facebook's academic research partners:*
 - *Your operating system, model and manufacturer, and device type (e.g., mobile, tablet, and desktop)*
 - *Which apps you use, including app name and category, and for how long*

- Which browser you use
- What websites you visit, the date and time you visit a website, when and for how long
- Your device data will be linked to your survey responses as well as publicly available third-party data, like if you've voted or made a political contribution, if this third-party data is available
- Your device data will not be used for ads
- Facebook will also combine your device data, your survey responses and the third-party data with your activity on Facebook and Instagram from the 2020 calendar year, collectively called Combined Data
- This Combined Data will be shared with Facebook's academic partners and, if legally required, with the Institutional Review Board (IRB) that reviewed this study.
- Once this study is over, de-identified data (i.e. data where identifiers such as your name and other information that could reasonably be linked to you are removed) will be stored and shared for future research on elections, to validate the findings of this study, or if required by law for an IRB inquiry.

The FAQ included the following:

Along with your traditional surveys, this study gives you the opportunity to add software to your online devices to understand mobile and desktop behavior.

How does it work? *On mobile (Android/iOS) we use VPN services to understand web data usage on those devices, from which we can understand what sort of websites you visit. We also use this web data on iOS, or OS information on Android, to see what your favorite applications are. You can also download the application onto Windows and Mac which installs Browser Extensions onto Chrome, Firefox and Safari depending on what browsers you have installed.*

What data do you collect? *We can collect data on what apps you are using and for how long on mobile, but we do not see what you do within those applications. Across all platforms we also collect information on what websites you have been on, for instance we could see if you have been using Google or YouTube on your browser; however we would not be able to see what you were searching or viewing on those channels, e.g. www.google.com was used for 5 minutes*

Can I stop data being collected? *The applications and VPNs will only collect data whilst installed on your device. You may uninstall at any time or even pause data collection using functionality in the apps.*

***What else do the apps do?** The apps will sit in the background and passively collect data, meaning all you need to do is keep them installed to earn your rewards. The Android and iOS applications will send you notifications periodically to keep you up to date with the study, for instance letting you know when there is a survey to complete.*

Participants also were given a link to more privacy details and terms, as well as provided with a website allowing them to withdraw from the study at any time.

The passive measurement recruitment phase closed on September 21, 2020. A total of 20,077 Wave 1 respondents consented to participate in passive tracking, of which 7,730 (41%) downloaded, installed, and activated the software on at least one device on a total of 10,421 devices. The participation rates are included in Table S4. The final sample we analyze in this study is further restricted to respondents in the control group, which contains a total of 2,721 users (approximately 35% of the total sample).

Table S4: Online passive tracking participation rates

Invited sample	125,694
Completed enrollment survey	20,822
Enrollment survey yield (complete/invited)	17%
Consent to participate	20,077
Consent rate (consent/complete)	96%
Total compliant users – Sept. 21	7,730
Total compliant users – Oct. 19	7,104
Total compliant users – Nov. 23	6,980
Total compliant users – Dec. 23	6,485
Active rate (active users/consent) – Sept. 21	39%
Active rate (active users/consent) – Dec. 23	32%
Total active devices – Sept. 21	10,421
Total active devices – Oct. 19	9,016
Total active devices – Nov. 23	9,078
Total active devices – Dec. 23	8,219
Deactivated (users) – Dec. 23	123
Non-compliant (users) – Dec. 23	5,358
Total user universe (Active+Deactivated+Non-Compliant)	11,966
Rate of dropout (users)	46%

Weighting for Passive Web Tracking To examine the extent to which any estimates derived from the sample of participants who provided passive tracking data can be generalized to the entire Facebook population, we report both unweighted and weighted results for any quantity computed using the tracking data. The weights we use are the same as described in SM S2.2.

We chose to employ these weights, as opposed to building a new set of weights for this specific subsample, because we found that selecting into the web tracking sample could not easily be predicted by the variables we used to create the weights. Although user characteristics (e.g. ideology, income, and race) do predict whether respondents consented to sharing their passive tracking data, overall these factors only explain a small proportion of the variance regarding who consents. In addition, we found that the subsample with web tracking data deviates from the overall Facebook population in similar ways as the overall sample of respondents. For example, the correlation of feature bias across the two samples is 0.98. We also discovered that it deviates from the targets in the overall U.S. population (self-reported party ID, education, and income) in similar ways as the overall sample of respondents. As a result, we considered that using the same set of weights was adequate.

We were not able to use this same set of weights for metrics based on Facebook exposure logs for the subsample that also provided web tracking data (see SM section S5). The reason is that the aggregation of the platform data had to be conducted on Meta servers (see SM section S2.1), but the respondent-level weights could not be uploaded to Meta servers as established by the project’s data collection and privacy framework.^{S15} Instead, we adopted a simpler approach that consisted on weighting each subgroup, as defined by their political ideology and political interest (a set of 3×3 combinations), and by their relative prevalence in the population. Additional details on this approach are provided in SM section S5.

S3 Materials and Methods

S3.1 Observation Period

The analyses in this paper are based on all URLs shared in posts classified as political news that were posted, seen or engaged with by all U.S. adult users on Facebook during the period from September 1 2020 to February 1 2021 (see SM section S3.4 for details on the classifiers we use to select URLs) .

S3.2 Audience Measures

The key measure in this research is “views”, which we use to count impressions and the *exposed audience* (i.e., unique users) exposed to domains/URLs on a daily basis and for the full period (September 1 to February 1). A view on a post is counted whenever the post renders in the visible portion of the user’s Web browser or mobile app for more than 250 milliseconds (i.e. a “validated viewport view” or VPV). In order to be consistent with other papers in this project, we only analyze exposure to content on Facebook’s Feed, which represents approximately 90%

^{S15}The weights are partially derived from survey data; and when respondents gave consent to have their survey data collected, we guaranteed that it would not be used to show them ads. Meta committed to keeping the survey data off its servers entirely to conclusively demonstrate that it was not used in any way for advertising purposes.

of all views of political news URLs on Facebook. Other surfaces (e.g. user or Page profiles and Groups tab) are excluded from our dataset.^{S16}

We also calculate metrics of *potential audience* (i.e., users who had the domains/URLs in their inventory but may not have viewed the content). The inventory of available news changes from user to user, and it is composed of all the political news domains/URLs posted by their friends, Pages they follow, and Groups they are members of. The composition of the inventory is determined, in other words, by the content posted by the networks users build on Facebook. As figure S4 illustrates, this means that the inventory results from another curation process determining the structure and composition of the Facebook graph, which itself results from social and algorithmic dynamics. The data we have access to does not allow us to analyze this part of the process, and thus we cannot determine how much of the content in the inventory is driven by choices made by users versus platform affordances (including algorithmic recommendations on who to follow). Our data, however, allows us to measure the potential audience for each political news URL based on the underlying network. This is operationalized as the set of users that are either friends with another user that shared the URL, follow a Page that shared it, or are members of a Group in which it was shared.

Finally, we also compute metrics of *engaged audience*, which adds a layer of “meaningful exposure” to news (“views” offer a less strict definition of news exposure than “engagement”, which presumes a deeper examination of the content by the user). Engagement includes clicks, reactions, likes, reshares and comments of any post containing a given political news URL.

The key type of content we analyze are URLs predicted to be political news (see SM section S3.4), which we examine on a daily basis but also aggregated for the full period. Our analyses also aggregate the data to the domain level. Table S5 displays summary statistics for each of our audience measures and aggregations, both in total and for each of the ideology and political interest subgroups. Only URLs shared more than 100 times during this period, which account for 3% of all unique URLs, 83% of shares, and 58% of views of political news URLs, are included in the data – see figure S6 for details on how this threshold affects the aggregated segregation measure (i.e., not drastically).

S3.3 Ranking Algorithm

Figure S5 offers a schematic representation of the logic behind Facebook’s Feed ranking algorithm. (See ref. (27) for additional information.). The process involves collecting all relevant information about who publishes posts and analyzing all their features (e.g., how many people have liked a post before). This information is used to predict the post’s value to a given user. The prediction requires processing data on three levels: (1) Inventory: this is the collection of all posts that could be ranked for a given user. The eligible inventory includes any non-deleted post shared with the user by a friend, Group, or Page that the user is connected to and that was made since the user’s last login. To make sure unseen posts are also reconsidered, the algorithm

^{S16}Our analysis is also limited to posts shared by users or Pages, as well as posts shared in Groups; in total, these represent approximately 99.8% of the content that users saw on their Feed during the study period.

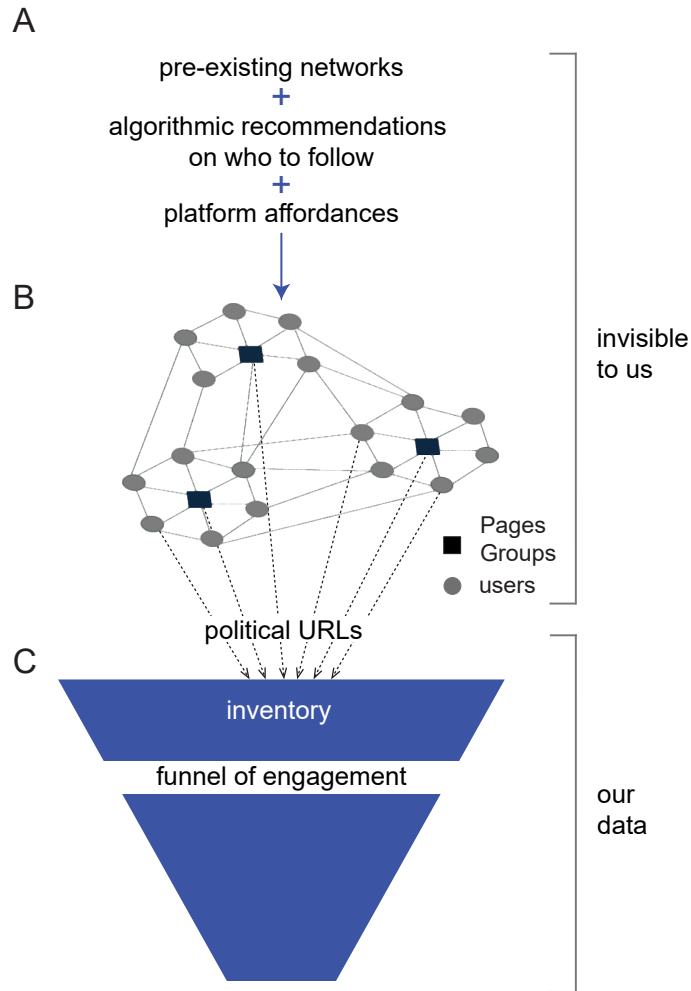


Figure S4: Source of Inventory. This figure is an extension to Figure 1B in the main text. For privacy reasons, we were only granted access to aggregated measures of audience composition at the URL and domain levels. However, the inventory (and subsequent funnel of engagement we analyze in the main text) results from another type of curation operating at the network level. This curation results from a combination of (1) people choosing their ties based on pre-existing (off-platform) networks/interests; (2) algorithmic recommendations on who to follow; and (3) platform affordances (i.e., Pages and Groups). The content in the inventory results from these social and algorithmic processes (panel A) determining the structure and composition of the underlying network topology (panel B).

	N	Median count	Ideology			Pol.Int.	
			Lib.	Mod.	Cons.	Low	High
Potential (URLs)	639090	407116	91714	37298	139078	197078	125883
Exposed (URLs)	639090	47538	9264	3114	16514	20046	12952
Engaged (URLs)	639089	2834	674	140	1055	818	1076
Potential (Domains)	35058	624159	191297	76830	230210	343762	186718
Exposed (Domains)	35058	67083	17732	7983	23984	36400	15448
Engaged (Domains)	35051	5184	1664	556	2114	2323	1837

Table S5: Descriptive statistics: URL- and domain-level datasets at each audience level.

This table displays the count of URLs and domains included in each audience measure, the median audience size, and the median audience size disaggregated by ideology and political interest subgroups. Only URLs and domains with $k > 100$ shares are included in the analysis. We use 0.35 and 0.65 as thresholds to categorize users into liberals, moderates, and conservatives. The audience with high political interest is defined as users in the top 10% of engagement with political content; the bottom 90% is considered low political interest.

follows an unread bumping logic: posts that a user has not yet seen but that were ranked for the user in previous sessions are eligible again for the user to see. The algorithm also follows an action-bumping logic: if any posts a user has already seen have triggered an interesting conversation among friends, the user may be eligible to see the post again as a comment-bumped post. (2) Posts: once the inventory is identified, the algorithm scores each post in the inventory using multitask neural nets, which process features such as type of post, embeddings (i.e., feature representations generated by deep learning models), and what the viewer tends to interact with. (3) Aggregated score: this is the single score assigned to posts based on multiple predictions at the user level. First, integrity processes are applied to every post. These are designed to determine which integrity detection measures (such as removal or demotion) need to be applied to the stories selected for ranking, if any. Then, a lightweight model is run to select approximately 500 of the most relevant posts for a given user that are eligible for ranking. Then each story is scored independently and all 500 eligible posts are ordered by score. Finally, the contextual pass is applied. Here, contextual features, such as content-type diversity rules, are added to help diversify users' Feed. More information on the ranking algorithm can be found at <https://engineering.fb.com/2021/01/26/ml-applications/news-feed-ranking/>.

S3.4 Meta Classifiers and Categorization Methods

The following classifiers, concepts, and categorization methods were either developed at Meta or are defined under Facebook's platform policies.

Civic and News Classifiers We use Meta's internal classifiers to restrict our analysis to URLs shared in posts identified as civic by the Civic classifier *and* as news by the News classifier. The

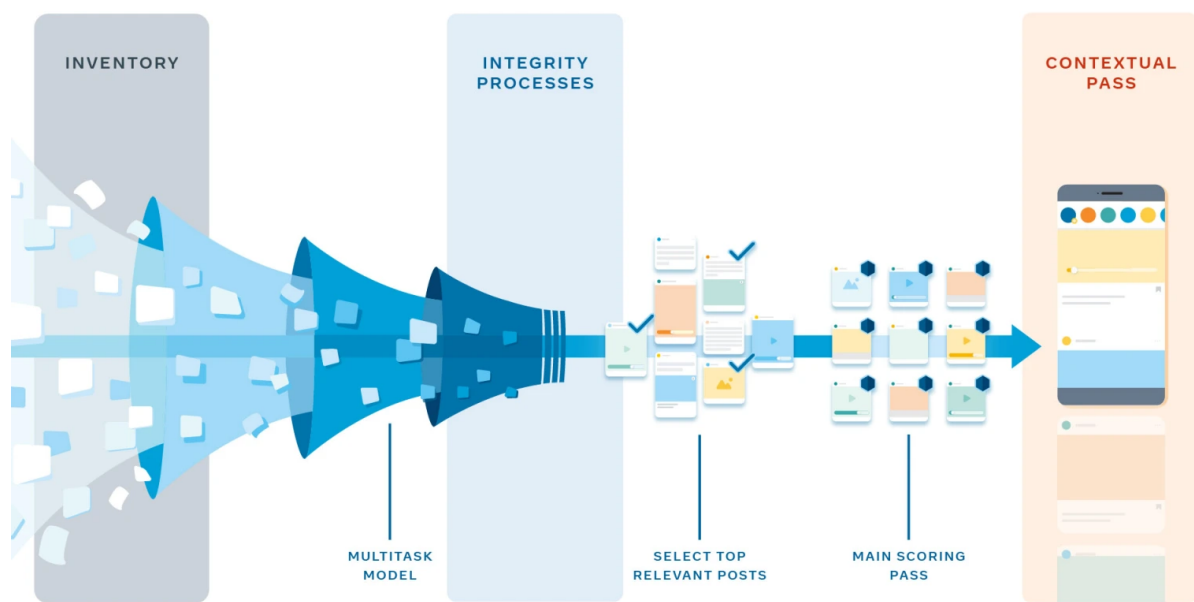


Figure S5: Schematic Representation of Facebook’s Feed Ranking. The ranking algorithm processes data on three levels: (1) inventory (the collection of all posts that could be ranked for a given user); (2) posts (using features such as embeddings and engagement and integrity processes); and (3) user-level aggregated score (using contextual features, such as amount of diversity in user’s Feed). More information on the ranking algorithm can be found at <https://engineering.fb.com/2021/01/26/ml-applications/news-feed-ranking/>.

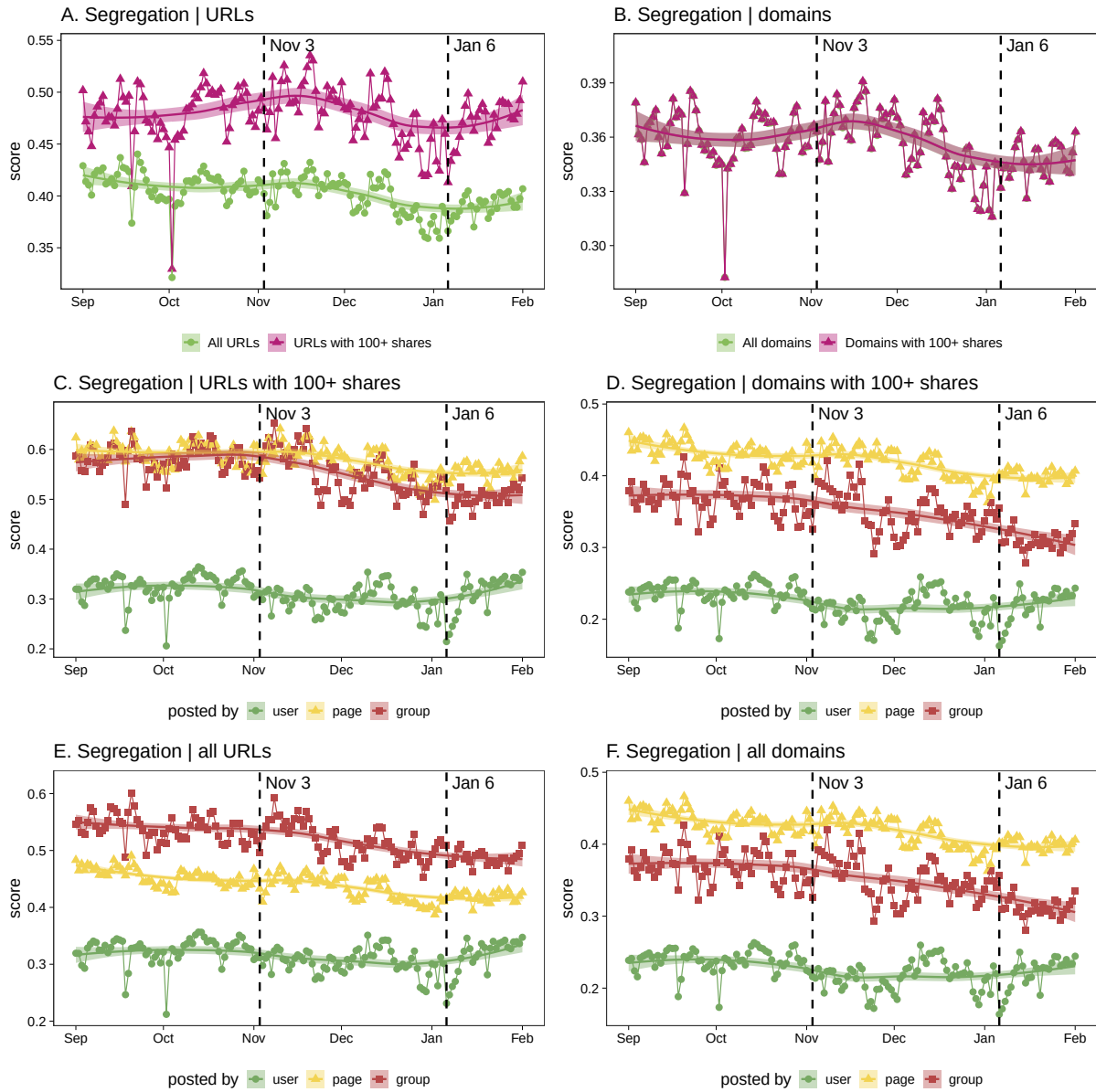


Figure S6: Impact of Shares Threshold ($k > 100$) on the Segregation Index. For privacy reasons, our URL- and domain-level analyses only include URLs and domains that were shared more than 100 times during the observation period. Here we show the impact that this threshold has on the aggregate segregation index measure. The left column plots the index for URL-level data, the right column plots the index for domain-level data. The curves in panel B have a nearly perfect overlap, which gives the impression that only one curve is plotted. For the URL-level aggregation, focusing on the most shared URLs increases the segregation score by roughly 10 points; however, both sets of URLs are consistent in identifying Pages and Groups as driving segregation upwards. The threshold does not impact domain-level measures noticeably.

main text refers to these URLs and associated domains as “political news”.

- Civic classifier
 - **Definition:** This classifier predicts whether a given post is related to *politics* (government, elections, politicians, activism, etc.) or *social issues* (major issues that affect a large group of people, such as the economy, inequality, racism, education, immigration, human rights, the environment, etc.). In this paper, we refer to any content that is classified as being in either of these two categories as “political”; otherwise they are not political.
 - **Usage:** We use the classifications for Facebook link posts that were created, seen or engaged with by U.S. users during the U.S. 2020 Facebook and Instagram Election Study (FIES). We use the classifications for both English- and Spanish-language content.
 - **Performance:**
 - * Based on a sample of approximately 10k labeled posts, the classifier has 83% precision and 82% recall on English-language Facebook content.
 - * Based on a sample of approximately 17k labeled posts, the classifier has 81% precision and 85% recall on Spanish-language Facebook content.
- News classifier
 - **Definition:** This binary classifier predicts whether content is about current events, timely information, and follows journalistic standards such as citing sources and having a byline. See <https://www.facebook.com/business/help/224099772719228> for additional information.
 - **Usage:** We use the classifier predictions for Facebook posts with a link that were created, seen or engaged with by U.S. active users during the U.S. 2020 FIES. We use the predictions for both English- and Spanish-language content.
 - **Performance:** Classification thresholds were chosen to yield 80% recall. Based on a sample of approximately 52k labeled links, precision at this threshold is 90% for U.S. English-language News links. Based on a sample of approximately 36k labeled links, precision at this threshold is 42% for U.S. Spanish-language News links.

To illustrate the type of URLs that are captured by each of these classifiers, as well as their combination, Table S6 offers a random sample of 20 URLs in each of the four possible combinations of these two classifiers. These URLs were selected randomly from the set of URLs that were shared 100 or more times by US adult users during our period of analysis. Note that, because classifications are at the post level, as explained above, a URL can appear in multiple categories of this table if it was shared in posts with different predicted categories. The URLs were manually reviewed and, where needed, sanitized to ensure that no personally-identifiable

information is contained in the URL address. We note that these lists represent a random selection of URLs that were shared during this period and are not necessarily a representative sample of the content that users saw on the platform.

We find that most URLs in posts classified as not civic and not news are links to music videos, prayers, recipes, GIFs, memes, etc. Most URLs in posts classified as civic but not news are links about COVID and other health issues, civic organizations, and political opinion. Most URLs in posts classified as news but not civic are links to sports news, entertainment news, crime news, local events, etc. Finally, most URLs in posts classified as civic and news correspond indeed to news about content that relates to either politics (government, elections, etc.) or social issues (the economy, inequality, racism, etc.). As detailed above, the empirical analyses reported in this paper only consider URLs in this last category (“URLs in posts classified as news and civic”), a subset to which we refer with the label “political news.”

Table S6: **Examples of URLs classified as civic or not; news or not**

URLs in posts classified as <i>not</i> civic and <i>not</i> news
https://timelesslife.info/be-honest
https://www.dogsaddict.com/14-reasons-you-cant-keep-german-shepherds-indoor/
https://grandbaby-cakes.com/lazy-strawberry-cinnamon-rolls/
https://www.youtube.com/watch?v=Vqbk9cDX0I0
https://catholicsay.com/the-prayer-john-paul-iis-dad-taught-him-that-he-prayed...
https://kittencatpost.com/rescues/born-without-a-tail/
https://www.youtube.com/watch?v=ZGhJKiMR4XY
https://media1.giphy.com/media/l1BgT1rs5YTOzvMHu/giphy.gif?cid=120...
https://wakeupyourmind.net/4-people-can-see-odd-one-images-can.html
https://kitchencookbook.net/chiffon-cake-recipe/
https://www.gofundme.com/f/mom-and-twins-lost-everything-in-fire
https://www.bigdoghunting.com/
https://www.youtube.com/watch?v=ZhIsAZO5gl0
https://media1.tenor.co/images/dcd288ea5b0df65020fd95f2f913e906/...
https://www.taliadinapoli.com/
https://media3.giphy.com/media/3ohhwmQ0xIg8W3pHd6/giphy.gif?cid=...
https://media1.tenor.co/images/b7497c5ffd054e58977609fef18b029e/...
https://media0.giphy.com/media/v1.Y2lkPTEyMGMwMTQ3c3Z0dXB1ZzR...
https://myfox8.com/news/oorah-happy-birthday-to-the-u-s-marine-crops/
https://www.youtube.com/watch?v=WAzg09vpT0g
URLs in posts classified as civic but <i>not</i> news
https://www.youtube.com/watch?v=a1qLR7ADYUs
https://www.merithealthwesley.com/hospital-opportunities
https://xemphimtop1.com/mexico/actores-de-televisan-amenazan-fuertemente...
https://apply.mrelief.com/screener
https://ffrf.org/

<https://secure.winred.com/save-america-joint-fundraising-committee/2024-...>
<https://homevaccineeducationnetwork.com/flu-vaccine-and-covid-19>
<https://www.greatlakesmichiganjobs.com/line5>
<https://www.unicefusa.org/stories/unicef-launches-online-tool-tracking-covid-...>
<https://www.lc.org/Churches>
<https://notthebee.com/article/the-washington-post-wants-you-to-know-that-in-...>
<https://www.youtube.com/watch?v=eOHxpd80srY>
<https://www.youtube.com/watch?v=eQPUTMU4Lho>
<https://www.youtube.com/watch?v=7uAUoz7jimg>
<https://www.usatoday.com/story/sponsor-story/astrazeneca/2020/12/04/woman-...>
<https://www.youtube.com/watch?v=Fl-x4Vlx4T8>
<https://actionnetwork.org/forms/we-need-to-add-seats-to-the-supreme-court-2/>
<http://health.mo.gov/communitytest>
<https://www.youtube.com/watch?v=P0-eNyOjtf0>
https://www.youtube.com/watch?v=_O9ltm_Gml0

URLs in posts classified as news but *not* civic

<https://www.fox26houston.com/news/hobby-lobby-raises-minimum-wage-...>
<https://petrescuereport.com/2021/stray-dog-led-motorcycle-driver-to-...>
<https://comicbook.com/tv-shows/news/marvel-chris-evans-captain-america-...>
<https://www.hotnewhiphop.com/ti-gracefully-responds-to-busta-rhymes-verzuz-...>
<https://www.mexicodesconocido.com.mx/la-luna-azul-iluminara-la-noche-...>
<https://losangeles.cbslocal.com/2020/12/05/actor-david-lander-known-as-...>
<https://www.israel365news.com/160829/first-lab-grown-meat-restaurant-opens-...>
<https://robbreport.com/motors/cars/bugatti-recall-77-cars-1234585887/>
<https://www.salisburypost.com/2020/11/15/tallest-cone-in-town-new-...>
<https://fox8.com/news/missing/south-euclid-police-issue-alert-for-missing-...>
<https://houseofheat.co/jordan/air-jordan-1-zoom-comfort-pink-glaze-ct0978-...>
<https://www.usmagazine.com/entertainment/news/ellen-pompeo-teases-...>
<https://serveandconnect.networkforgood.com/projects/111120-officer-...>
<https://www.bestproducts.com/fashion/g28609306/90s-stores-no-longer-exist/>
<https://www.khou.com/article/news/local/2-killed-in-wrong-way-crash-on-...>
<https://www.imdb.com/title/tt7907922/>
<https://ultimateclassicrock.com/alex-van-halen-david-lee-roth-sammy-hagar/>
<https://myfox8.com/news/oorah-happy-birthday-to-the-u-s-marine-crops/>
<https://abcnews.go.com/US/women-girls-found-dead-arkansas-home-...>
<https://petrescuereport.com/2021/man-arrested-and-charged-for-killing-a-...>

URLs in posts classified as news *and* civic

<https://www.usatoday.com/story/entertainment/celebrities/2021/01/20/trump-...>
<https://www.newschannel5.com/news/national/president-jimmy-carter-...>
<https://crooksandliars.com/2021/01/money-trail-insurrection-rally-leads>
<https://www.fox2detroit.com/news/michigan-house-and-senate-approve-covid-...>

<https://www.politico.com/states/florida/story/2020/10/19/final-tally-group...>
<https://www.politico.com/states/new-jersey/story/2020/10/05/murphy-...>
<https://www.spokesman.com/stories/2021/jan/14/state-senate-passes...>
https://apple.news/AbrGVExSOQ9C_x_GrlHGhLg
<https://www.rawstory.com/2020/10/republicans-upset-with-steve-mnuchin...>
<https://www.youtube.com/watch?v=kos034rs2zs>
<https://jacobinmag.com/2020/10/us-art-museums-workers-organizing-racism...>
<https://uniteamericafirst.com/steven/democrat-leaders-have-encouraged...>
<https://www.forbes.com/sites/siladityaray/2020/12/08/armed-police-raid-home...>
<https://www.msnbc.com/ayman-mohyeldin/watch/khizr-khan-on-trump-his...>
<https://nativenewsonline.net/arts-entertainment/indigenous-futurism-ushers...>
<https://www.khou.com/article/news/local/2-killed-in-wrong-way-crash-on...>
<https://truthout.org/articles/in-the-face-of-far-right-violence-we-must-organize-...>
<https://theconservativenews.org/hillary-clinton-tries-to-paint-trump-as-a-traitor...>
<https://www.cleveland.com/open/2020/11/more-legal-battles-likely-if-ohio-gov...>
<https://www.vladtv.com/article/265115/trump-says-kenosha-shooter-was-in-big...>

COVID-19 Classifier We use Meta’s internal COVID-19 classifier to determine whether the URLs posted are predicted to be related to the COVID-19 pandemic.

- **Definition:** This classifier predicts whether a given post is related to COVID-19; i.e. health concepts related to the virus, policy or public health guidance or the economic impact of the virus, as well as people’s reactions to the pandemic.
- **Usage:** We use the classifier predictions for Facebook link posts that were created, seen or engaged with by U.S. users during the U.S.2020 FIES. We use the predictions for both English- and Spanish-language posts.
- **Performance:**
 - Based on a sample of approximately 30k labeled posts, 94% precision and recall for English-language Facebook posts.
 - Based on a sample of approximately 31k labeled posts, 94% precision and 95% recall for Spanish-language Facebook posts.

Misinformation Misinformation refers to content (in our case, URLs) that is directly rated “false” by one of Meta’s independent fact-checking partners. See <https://www.facebook.com/business/help/2593586717571940?id=673052479947730> for a detailed description of how these rating categories are defined.

Meta’s third-party fact-checkers are certified by the nonpartisan International Fact-Checking Network. A full list of these partners (all independent to Meta) is available at <https://ifc>

cncodeofprinciples.poynter.org/signatories. In the U.S., this list includes organizations such as Snopes, Reuters, The Washington Post, Fact Checker, FactCheck.org, and PolitiFact. All fact-checks are publicly available on the websites of these organizations and can be reviewed by any external source for accuracy.

These ratings are assigned for specific URLs, which we use in the part of our analyses that rely on URL data. For domain-level analyses, see [S3.5](#). Since fact-checker ratings can change over time as new information emerges, we decided to use the list of ratings as of 2021-02-15, i.e., the list of what was considered misinformation at the end of our study period.

U.S. Ideology Classifier The core of our analyses rely on measuring audience composition for domains/URLs in terms of the ideology of their audiences on the platform. The labels “liberal” and “conservative” are assigned to users based on Meta’s internal U.S. ideology classifier.

- **Definition:** This classifier predicts adult U.S. active Facebook users’ political ideology.
- **Methodology**
 - This classifier is trained to predict the self-reported ideology of adult U.S. monthly active Facebook users based on their demographics, preferred language, location, and engagement with content, Pages, and Groups. It outputs a numeric score ranging from 0 (indicating a user is predicted to be liberal) to 1 (indicating a user is predicted to be conservative).
 - The classifier has high coverage – it is able to place up to 95% of adult U.S. monthly active Facebook users on this numeric scale. The classifier also has high week-over-week stability – on average, the weekly scores for individual users have a correlation of 0.96.
 - To further categorize adult U.S. monthly active Facebook users into distinct ideological groups, we discretize the continuous ideology scores as follows: users with a score less than or equal to 0.35 are categorized as liberal. Those with a score greater than or equal to 0.65 are categorized as conservative. The remaining users with a score between 0.35 and 0.65 are categorized as moderate. However, to demonstrate the robustness of our findings, we also consider two alternative sets of thresholds: (a) 0.10 and 0.90; and (b) a single threshold of 0.50, which assigns all users on each side of the threshold either a conservative or liberal label.
- **Usage:**
 - We use the ideological classifications of U.S. monthly active Facebook users 18 years or older who were active during the U.S. 2020 FIES.
- **Performance:** We evaluated how well the user-level predictions performed at inferring self-reported ideology by comparing our classifications to the survey responses of U.S.

2020 FIES panelists (see SM section S2). Precision and recall for each ideological group are as follows:

- 3-class classification (using 0.35 and 0.65 as thresholds)
 - * Self-reported Liberals: 62% precision and 77% recall
 - * Self-reported Moderates: 53% precision and 30% recall
 - * Self-reported Conservatives: 59% precision and 74% recall
- 3-class classification (using 0.10 and 0.90 as thresholds)
 - * Self-reported Liberals: 80% precision and 55% recall
 - * Self-reported Moderates: 50% precision and 71% recall
 - * Self-reported Conservatives: 71% precision and 59% recall
- 2-class classification (using 0.50 as threshold)
 - * Self-reported Liberals: 85% precision and 86% recall
 - * Self-reported Conservatives: 83% precision and 83% recall
- We note that the reason precision is lower across all groups and recall is lowest for self-reported moderates in the three-class classification task is that the model tends to predict self-reported moderates to actually be left- or right-leaning. This is consistent with external research finding self-reported moderates often have non-centrist ideological positions on various issues (28).

As part of the analysis in this paper we also separately evaluated the performance of this classifier on subsets of respondents who are regular consumers of specific news outlets. In particular, we compared the ideology classifier with survey self-reports for survey participants that frequently watched Fox News, MSNBC, and CNN. The results of this comparison are displayed in table S7. These analyses suggest that errors in the classification are not correlated with ideology in a way that would distort the segregation index. In particular, we find that the classifier does not underpredict the prevalence of liberals among Fox News watchers or the prevalence of conservatives among MSNBC and CNN watchers, as indicated by the corresponding calibration ratios being above 1. This helps address the concern that the use of this classifier induces an upward bias in the segregation index by predicting that anyone that consumes partisan outlets has an ideology that is aligned with that outlet.

S3.5 Other Categorization Methods

The following categorization methods were either proposed by the academic team or adapted from published academic research.

		Conservatives					Liberals		
	N	Acc.	Prec.	Rec.	Cal.	Prec.	Rec.	Cal.	
Fox News; every day	2172	0.65	0.77	0.84	1.09	0.35	0.61	1.76	
Fox News; several times per week	4052.00	0.58	0.65	0.82	1.26	0.46	0.70	1.54	
MSNBC; every day	1556	0.62	0.44	0.53	1.21	0.68	0.88	1.29	
MSNBC; several times per week	3776	0.60	0.41	0.64	1.56	0.70	0.83	1.18	
CNN; every day	2579	0.58	0.41	0.52	1.24	0.65	0.82	1.26	
CNN; several times per week	5495	0.61	0.43	0.66	1.55	0.72	0.82	1.14	

Table S7: **Performance of the ideology classifier for study participants that frequently watch cable news.** This table displays performance metrics (overall accuracy; precision, recall, and calibration ratio for liberals and conservatives) comparing the ideology classifier with survey self-reports for the sample of participants in the control group. Self-reported ideology is recoded into three categories (liberals, moderates, and conservatives). Predicted ideology is recoded into categories using 0.35 and 0.65 as thresholds. The calibration ratio is computed as the proportion of respondents predicted to be in a given category (e.g. liberal) over the total of respondents who self-report being in that category. Generally, although the classifier is less accurate for ideologues who consume counter-attitudinal cable news, these errors do not appear to be correlated with ideology in a way that would inflate the segregation index, as indicated by the fact that the calibration ratio is higher than 1 for such groups.

Untrustworthy Domains For our domain-level analysis, we implemented a metric that identifies domains that repeatedly share misinformation (see S3.4). In particular, we label a domain as “untrustworthy” if 2+ of their political news stories (URLs) were rated as “false” by Meta’s independent fact-checkers from the beginning of the Misinformation Repeat Offender program in August 2018 until the end of our study period (2021-02-15). We exclude from this list of domains other social media platforms, hosting sites, and URL shorteners, even if they may host URLs that have been fact-checked as false, in order to avoid identifying as untrustworthy domains for which misinformation may represent a small proportion of their URLs.

Political Interest Measure To add more granularity to our analyses (without compromising privacy) we segment exposure to domains/URLs in two groups: users with high and low political interest. High political interest users are those in the top 10% (and, in an alternative operationalization, top 1%) of engagement (i.e., comments, likes, reactions, reshares) with content classified as political on Facebook during the 90 days prior to our study period (2020-06-01 to 2020-08-31). The rest of users (bottom 90 and 99%) are considered to have low political interest. In addition to engagement, we also operationalize political interest using views of political content. The first operationalization (based on engagement) gives us a more stringent measure of ‘political interest’. In SM section S4.7. we present results for the two operationalizations and percentile cut-offs.

S3.6 Segregation Index and Favorability Scores

As we explain in the main text, one of our main metrics is the segregation score, which we use to determine common exposure to news. In figure S7 we plot separately the two components of the score plotted in figures 2A-C of the main text. Table S8 shows the same two components, but aggregated for our entire period of analysis. What this decomposition of the overall score shows is that segregation (and its increase as we move from potential, to exposed, to engaged audiences) is driven mostly by conservative users. See also figure S6 for data on how the segregation index is affected by the $k > 100$ shares threshold we impose at the URL level.

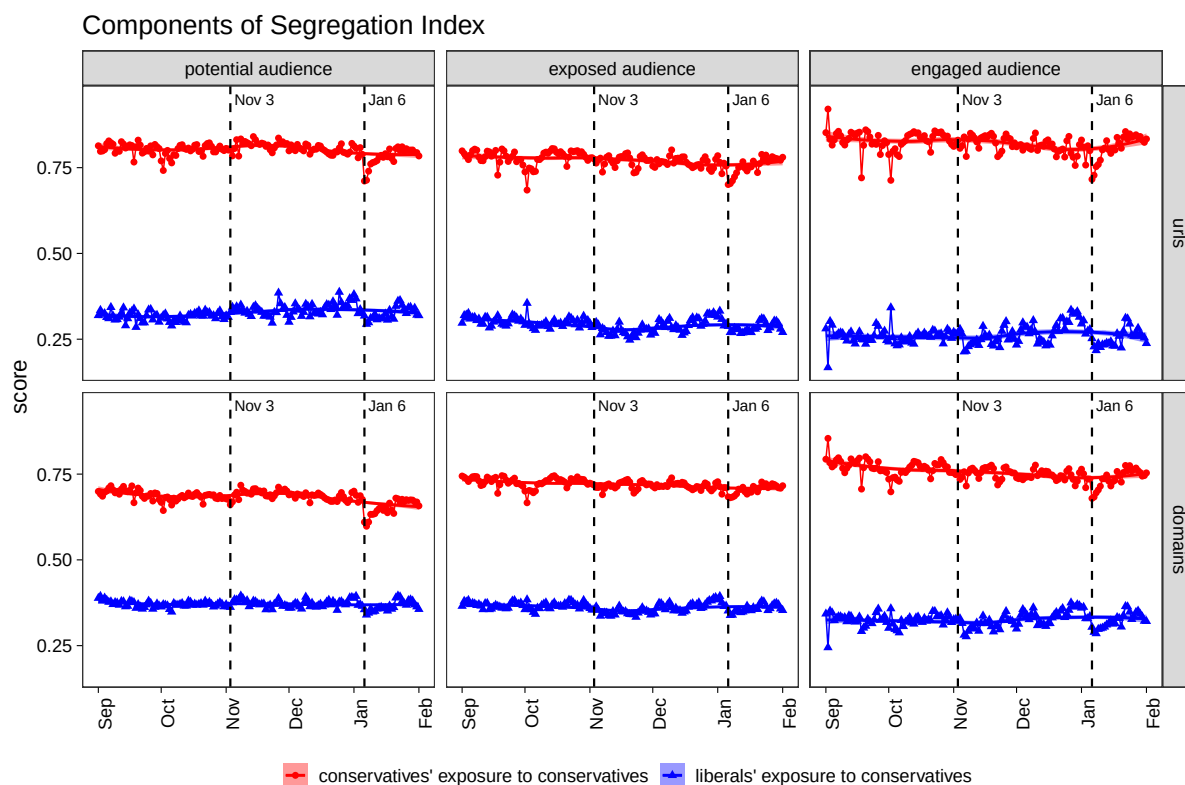


Figure S7: **Components of the Segregation Index.** This figure expands on figure 2A-C in the main text. The decomposition of the overall score into its two components shows that segregation (and its increase as we move from potential, to exposed, to engaged audiences) is driven mostly by conservative users.

The favorability scores are (unsurprisingly) more sensitive than the segregation index to different group sizes (in this case, the number of conservatives and liberals). In figure S8 we show the results of a simulation exercise where we manipulate (a) the fraction of conservative users and how much more active they are compared to liberals (panels A-B); (b) the total number of users and total number of domains (panels C-D); and (e) the strength of selective exposure

	URLs			Domains		
	Potential Exposed	Engaged	Potential Exposed	Engaged	Potential Exposed	Engaged
Term 1: Conservatives’ exposure to conservatives	0.80	0.77	0.82	0.67	0.71	0.75
Term 2: Liberals’ exposure to conservatives	0.33	0.30	0.26	0.38	0.37	0.33
Segregation index (term 1 - term 2)	0.47	0.47	0.55	0.29	0.35	0.42

Table S8: **Components of the Segregation Index.** This table displays the two components of the overall score for the entire period, for both URLs and domains, and for each of the audience measure.

to specific subsets of domains (panels E-F).

S4 Additional Results

We run an exhaustive set of additional analyses to add insights on the data we analyze and test the robustness of our results. The following sections summarize the results of these analyses. All the analyses reported in this section rely on platform-wide data (see SM section S2).

S4.1 Top Domains and URLs

In Figure S9 we show the list of top 100 domains (panel A) and top 100 URLs (panel B) ordered by views (note that for URLs, the parent domain is used as the label). Panels C and D show the scatterplots for ideology $\sim$ views (log transformed), suggesting a low correlation. One thing to note about this figure is that the most viewed news stories (URLs) are not necessarily published by the most viewed domains. For instance, *pjmedia.com* published the most viewed news story during our observation period (it was viewed 113,272,405 times) but it only ranks 51 when views are aggregated at the domain level (as the lower panels of the figure show the top 5 most viewed domains are, in decreasing order, *cnn.com*, *dailywire.com*, *foxnews.com*, *nytimes.com*, and *nbcnews.com*). This top-1 news story published by *pjmedia.com* runs under the headline “Military Ballots Found in the Trash in Pennsylvania—Most Were Trump Votes”. This story was not flagged as misinformation, i.e., the URL was evaluated but not rated “false” by the Third Party Fact-Checking Program (3PFC). A closer read of this particular news article reveals that the claims made are, in fact, accurate (the fact check with additional context can be found here: <https://factcheck.thedispatch.com/p/clearing-up-what-happened-with-the>). However, this example also illustrates that actors can find ways around the 3PFC program to spread factual claims that can easily be misunderstood or taken out of context.

In general, we are most likely undercounting the total volume of false news circulating on the platform (i.e., figure 1D in the main paper offers a low estimate) because the most visible stories are more likely to be assessed by fact-checkers. It is worth noting, however, that while

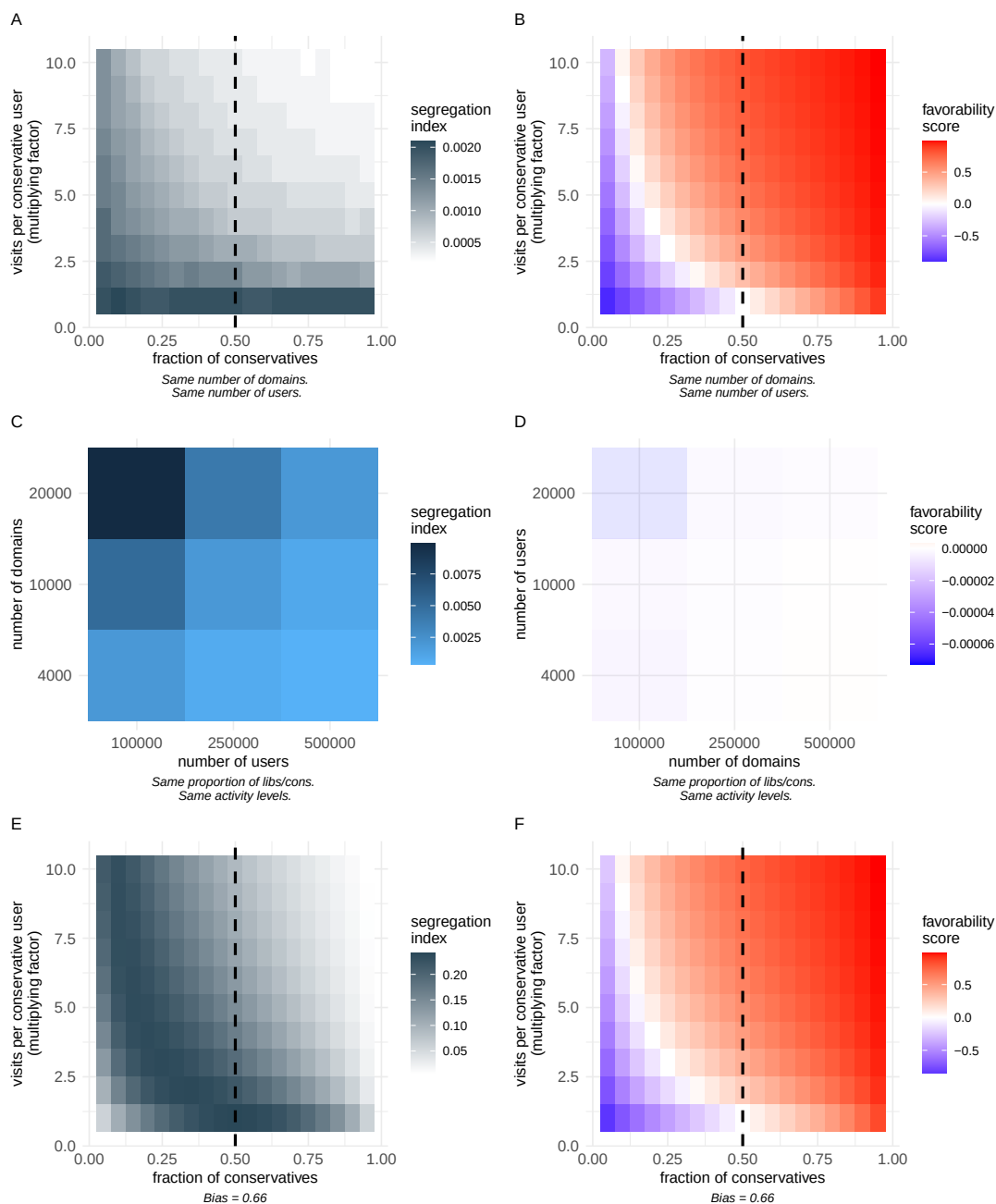
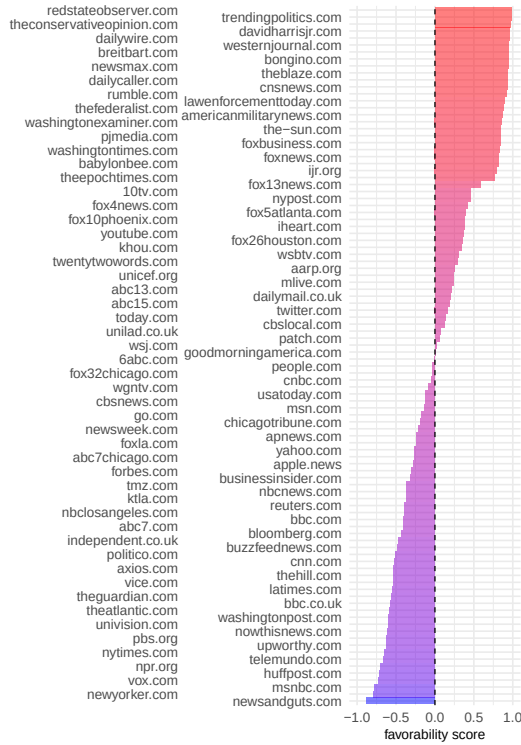


Figure S8: Changes in Segregation and Favorability Scores as Key Parameters Vary. In panels A-B we manipulate the fraction of conservative users and how much more active they are (in terms of visits to domains) compared to liberals. In panels C-D we manipulate the total number of users and the total number of domains. And in panels E-F we enable selective exposure to specific subsets of domains (bias ranges from 0 to 1 and it is set to 0 in panels A-D).

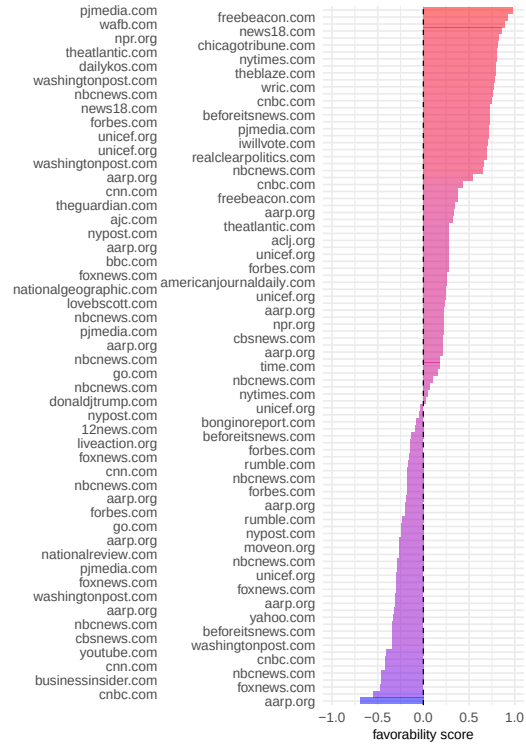
specific “false” news stories may go undetected, trustworthy labels at the domain label are more reliable. To follow up with this example in the previous paragraph, *pjmedia.com* is flagged as an untrustworthy source, which means that 2+ of their news stories (URLs) were rated as “false” as of 2021-02-15. This makes us think that the coverage of false-content labels is likely to be higher for domains than for URLs.

In figure **S10** we examine how much variation, in terms of ideology, we find within the top 100 domains by views. The favorability scores at the URL level only take into account the users who see that specific news story (here we refer to exposed audience only). It is empirically possible that different news stories circulate among different populations even if they are published by the same source (i.e., even if they are nested within the same domain). What the figure shows is that, for the most part, URL-level ideology estimates are consistent in their liberal-conservative slant. Some sources, like *msn.com*, have a mean favorability score leaning liberal, but this leaning differs widely depending on which subset of news stories we consider. Consistent with our main message of asymmetrical segregation, the figure also reveals a cluster of news sources with extreme conservative scores.

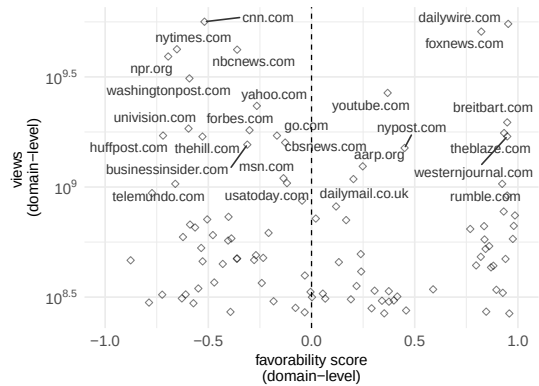
A. Top 100 Domains by Views



B. Top 100 News Stories (URLs) by Views



C.



D.

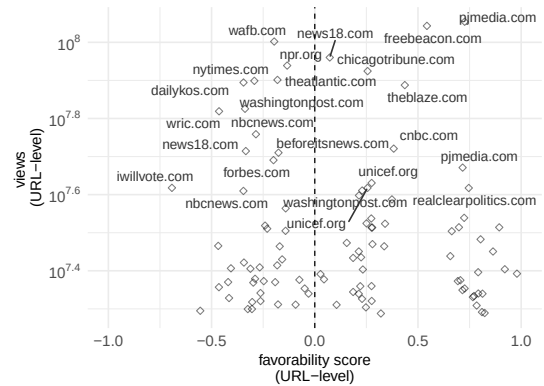


Figure S9: Top 100 Domains and URLs (Parent-Domain) Ranked by Views. Panel A shows the favorability scores of the top 100 domains according to total unique views during the period we consider. Panel B shows the parent domain of the 100 most viewed URLs (note that there is some overlap with panel A but these are two different sets of sources). In both panels A and B the first column corresponds to odd rows, and the second column to even rows (we split the labels in two columns so that they are all legible). Panels C and D show the scatterplots views $\sim$ favorability scores, with labels marking the domains/URL-parent-domains at the top of the views distribution.

URL Ideology Variation within Domains (Top 100)

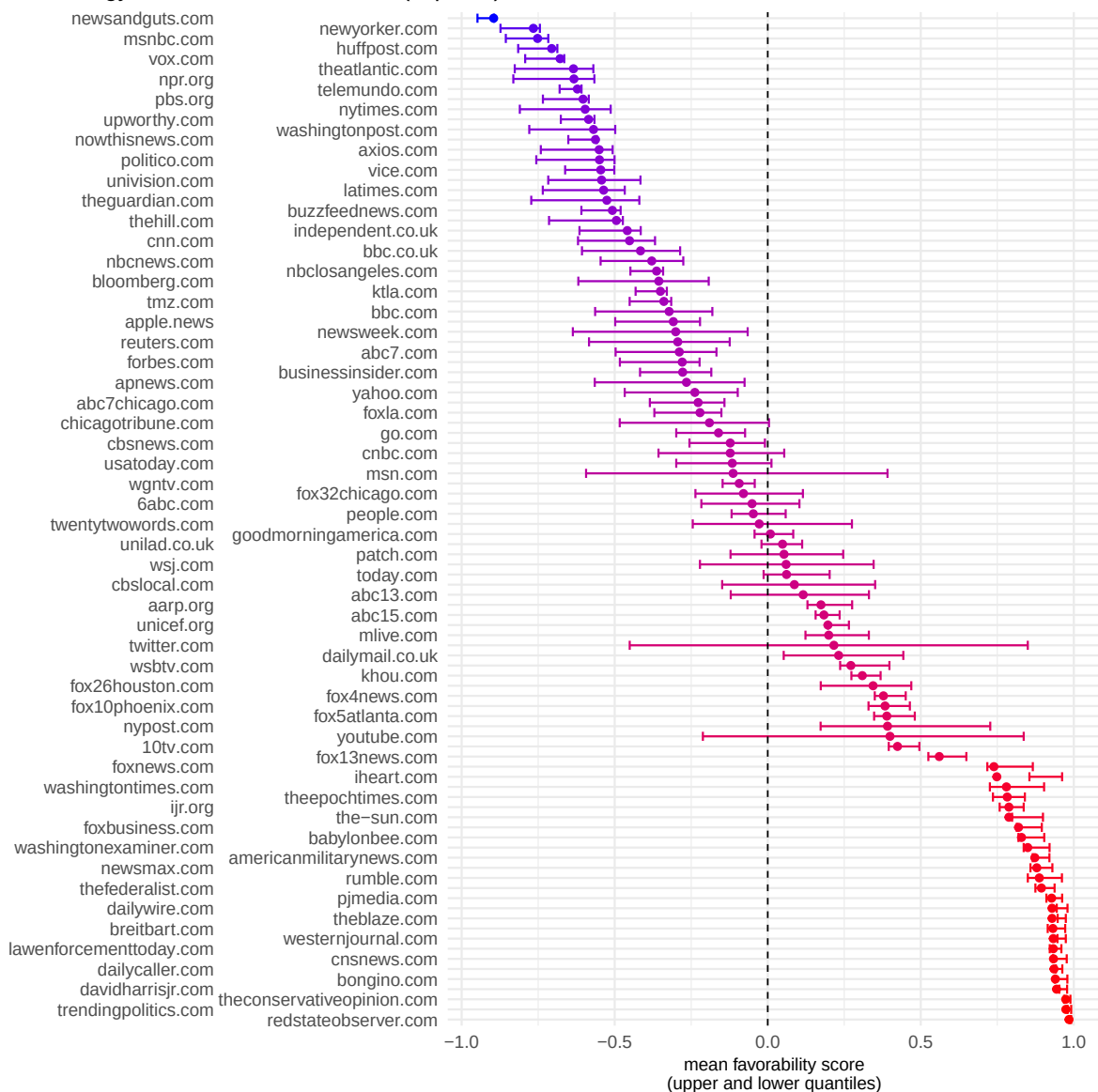


Figure S10: **Variation of URL Ideology Scores within Domains.** The figure suggests that there is some variation in the favorability scores at the news story level, but overall most scores stay below/above the 0 line. A few domains (e.g., *msn.com*, *mlive.com*) publish news stories with a wider appeal across left-right audiences. Note there is a sizable group of domains with consistently extreme conservative scores (lower right corner).

S4.2 Comparison with Prior Research

There is only one study published to date analyzing platform-wide exposure to news on Facebook for U.S.-based active users (3). In addition to the fact that this study is now quite dated (the data are from 2014), our study differs from this prior work in three significant ways: (1) we look at all political news URLs (i.e., content posted by friends, Pages, and Groups), not just URLs posted by friends; (2) we do not use users' self-reported affiliation on their profile as a way to categorize users, and instead rely on a classifier with much higher coverage: in (3), only 9% of U.S. users 18+ self-reported their ideology whereas the classifier we use here is able to place up to 95% of adult U.S. monthly active users (i.e., we have 10x higher coverage); and (3) we use three different measures of ideological segregation instead of the cross-cutting exposure statistic. However, in order to compare the results of our analyses as directly as possible with this prior work, in figure S11 we show the plot reported in the published article (panel A) with an approximation to the same cross-cutting measure calculated with our data, restricted only to URLs posted by friends (B) and including also URLs posted by Pages and Groups (C).

Like (3), we find that there are social and algorithmic effects directing ideologically compatible content to users: the volume of cross-cutting content is reduced as we move from random exposure, to potential exposure, to actual exposure, to engagement. (The random benchmark measures the amount of cross-cutting exposure we should expect in the absence of the underlying network, which is the source of the inventory determining potential exposure). Similarly, here we also find that the largest drop in cross-cutting content takes place between the random benchmark and potential exposure, which suggests that the composition of user networks (in itself driven by a combination of user choices and algorithmic recommendations) is the most important factor limiting exposure to political diversity. However, the comparison also reveals important differences. In particular, we find reversed trends in terms of ideology: in the case of our data, liberals have more cross-cutting exposure to ideologically dissimilar content than conservatives (which is the opposite of what Bakshy et al. showed). The results also underline one of our core findings: cross-cutting exposure is much lower for both liberals and conservatives once we include URLs shared by Pages and Groups.

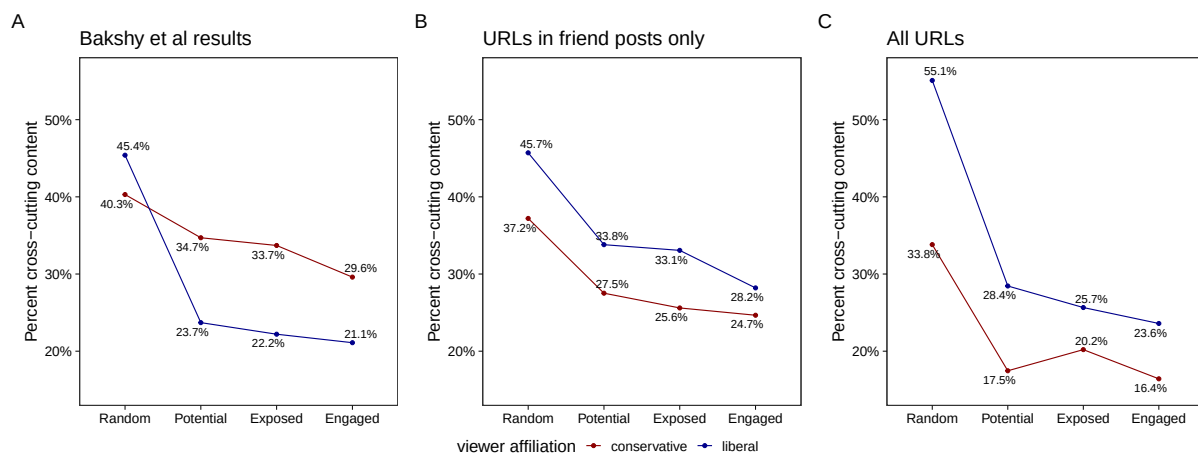


Figure S11: Comparison with Cross-Cutting Exposure Estimates. Here we use our data to calculate estimates comparable to those in (3).

S4.3 Segregation Scores Aggregated for the Entire Study Period

Tables S9 and S10 complement the results in Figures 2 and 3 of the main paper by computing the same metrics (segregation index and average favorability scores) aggregated over our entire period of analysis - from 2020-09-01 to 2021-02-01. Table S9 displays these metrics for domains; whereas Table S10 does so for URLs. These tables also display the results of computing these metrics for two other subsets of URLs or users: COVID-related news and high vs low political interest users. We present further evidence of how our results vary for these subsets in SM sections S4.5 and S4.7, respectively.

	Segregation			Favorability		
	Potential	Exposed	Engaged	Potential	Exposed	Engaged
All domains and users	0.29	0.35	0.42	0.05	0.12	0.07
Subset: High pol. int. users	0.40	0.48	0.52	0.01	0.01	-0.00
Subset: Low pol. int. users	0.21	0.27	0.28	0.08	0.18	0.16
Subset: Untrustworthy domains	0.16	0.27	0.32	0.32	0.40	0.41

Table S9: **Segregation and favorability scores for three base audiences (Domains).** This table displays the same segregation and favorability metrics reported in Figures 2 and 3 of the main text, but computed when aggregating data over the entire study period.

	Segregation			Favorability		
	Potential	Exposed	Engaged	Potential	Exposed	Engaged
All URLs and users	0.47	0.47	0.55	0.12	0.16	0.13
Subset: High pol. int. users	0.60	0.63	0.67	0.08	0.08	0.07
Subset: Low pol. int. users	0.35	0.35	0.35	0.16	0.23	0.23
Subset: COVID-related URLs	0.36	0.33	0.38	0.04	0.08	0.06
Subset: Misinformation URLs	0.10	0.19	0.17	0.73	0.79	0.81

Table S10: **Segregation and favorability scores for three base audiences (URLs).** This table displays the same segregation and favorability metrics reported in Figures 2 and 3 of the main text, but computed when aggregating data over the entire study period.

S4.4 Segregation Scores for Pages, Groups, and Users

In Figure 2 of the main paper we show the segregation scores for potential, exposed, and engaged audiences (panels B and C). Here, in figure S12 we show the exposed audience scores for content posted by Pages, Groups, and users; and in figures S13 and S14 we further disaggregate these patterns to analyze the funnel of engagement for the subset of domains/URLs posted by users, Pages, and Groups. Tables S11 and S12 further display segregation and favorability metrics computed using aggregated data for the entire period of analysis.

Pages and Groups are clearly increasing levels of segregation and polarization (i.e., in the case of URLs, the segregation score for Pages and Groups is twice as large as for content posted by users, S12B). However, when we increase the granularity of the analysis, we see that the process of algorithmic curation (i.e., the change from potential audience to exposed audience) operates differently for Groups: segregation for exposed and engaged audiences is actually lower than for potential audiences (S13C and S14C). There are several possible explanations that could account for this. One is that the inventory of content for Groups that post political news URLs is more extreme, and the process of algorithmic curation can only increase cross-cutting exposure (thus reducing the segregation present at the inventory level). A second possible explanation is that there is a stronger self-selection process for Groups compared to Pages or to user networks (again, explaining the higher segregation at the inventory level). A third possible explanation is that there is more content from untrustworthy sources being posted on Groups, which means Groups post a higher proportion of low-quality posts that are likely to be demoted by the Feed ranking algorithm (S3.3). These three factors are likely interacting in producing the patterns in panel C of figures S13 and S14. Given the aggregated nature of our observational data, we cannot disentangle these possible explanations.

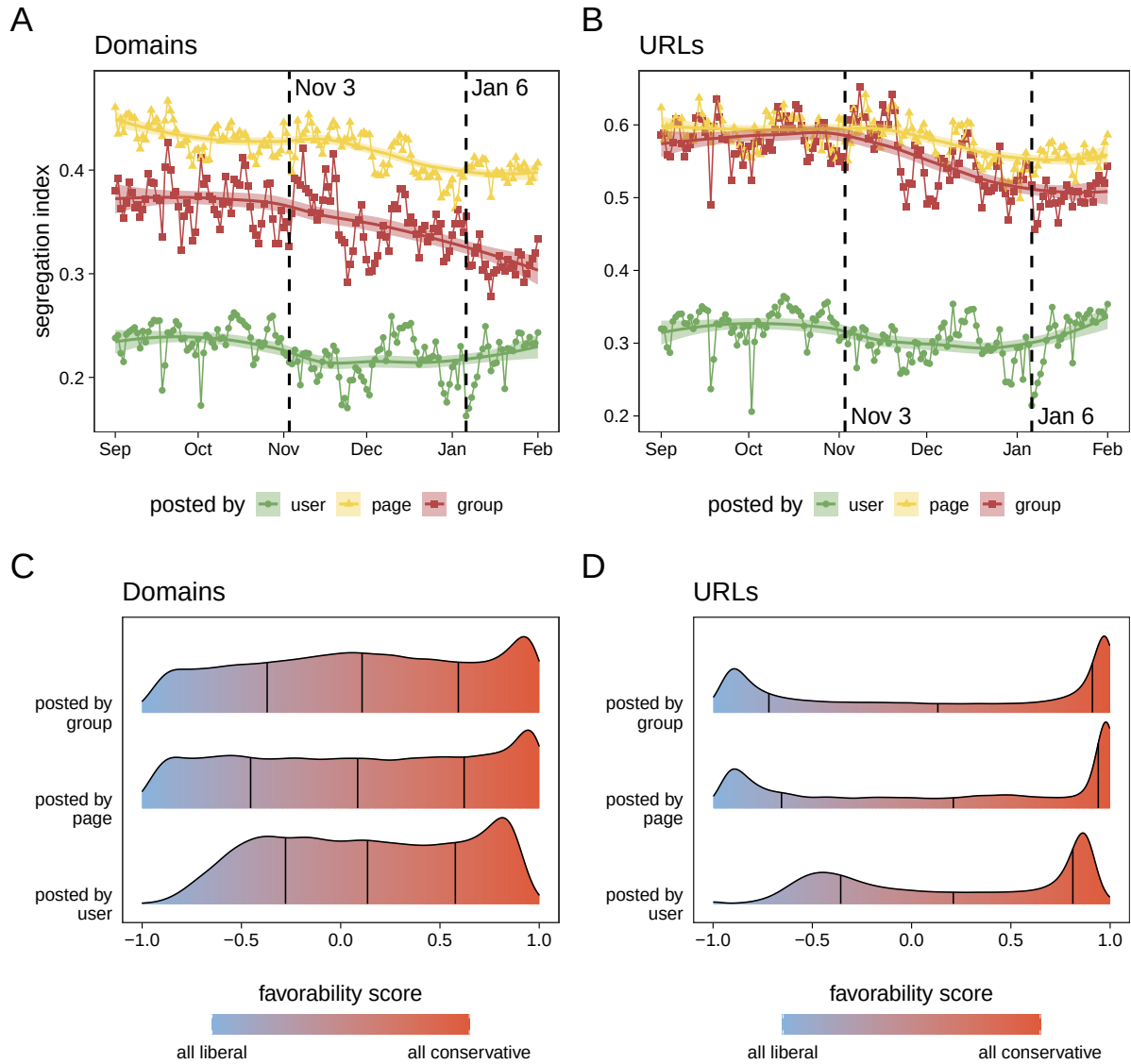


Figure S12: Segregation and Polarization at the Domain and URL level (Content posted by Pages, Groups, and Users). This figure is an extension to figure 2 in the main paper. Here content is segmented by post owner (Pages, Groups, or users). Pages and Groups are driving segregation levels up, a pattern that is especially visible at the URL-level. They are also increasing polarization, as assessed via the distribution of favorability scores.

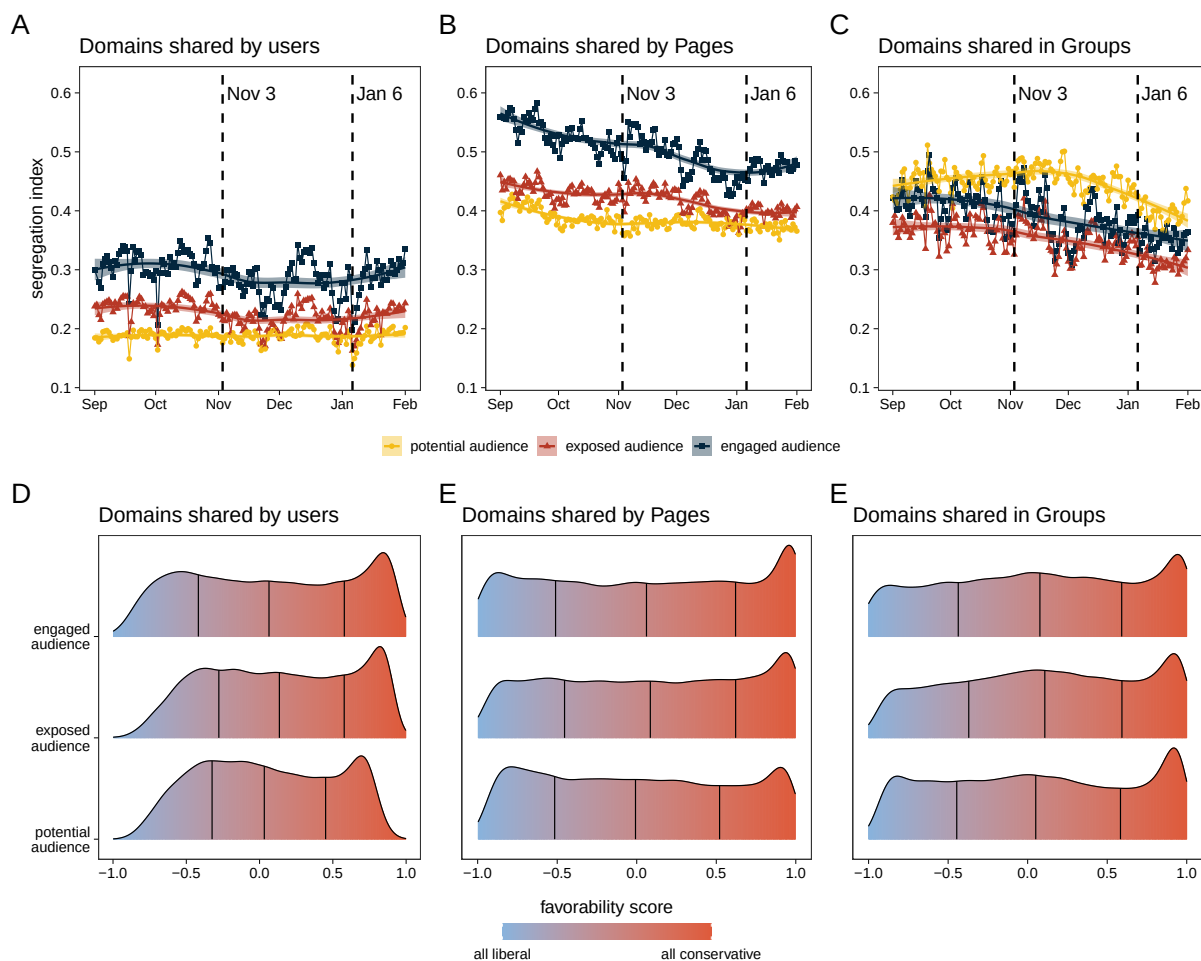


Figure S13: Funnel of Engagement for Domains posted by Users, Pages, and Groups. This figure is an extension to figure 2 in the main paper and to figure S12 above. Here domains posted by users, Pages, and Groups are segmented by potential, exposed, and engaged audiences. Pages and Groups substantially increase segregation and polarization, but in Groups the process of algorithmic curation operates differently, diversifying the content that users would otherwise see.

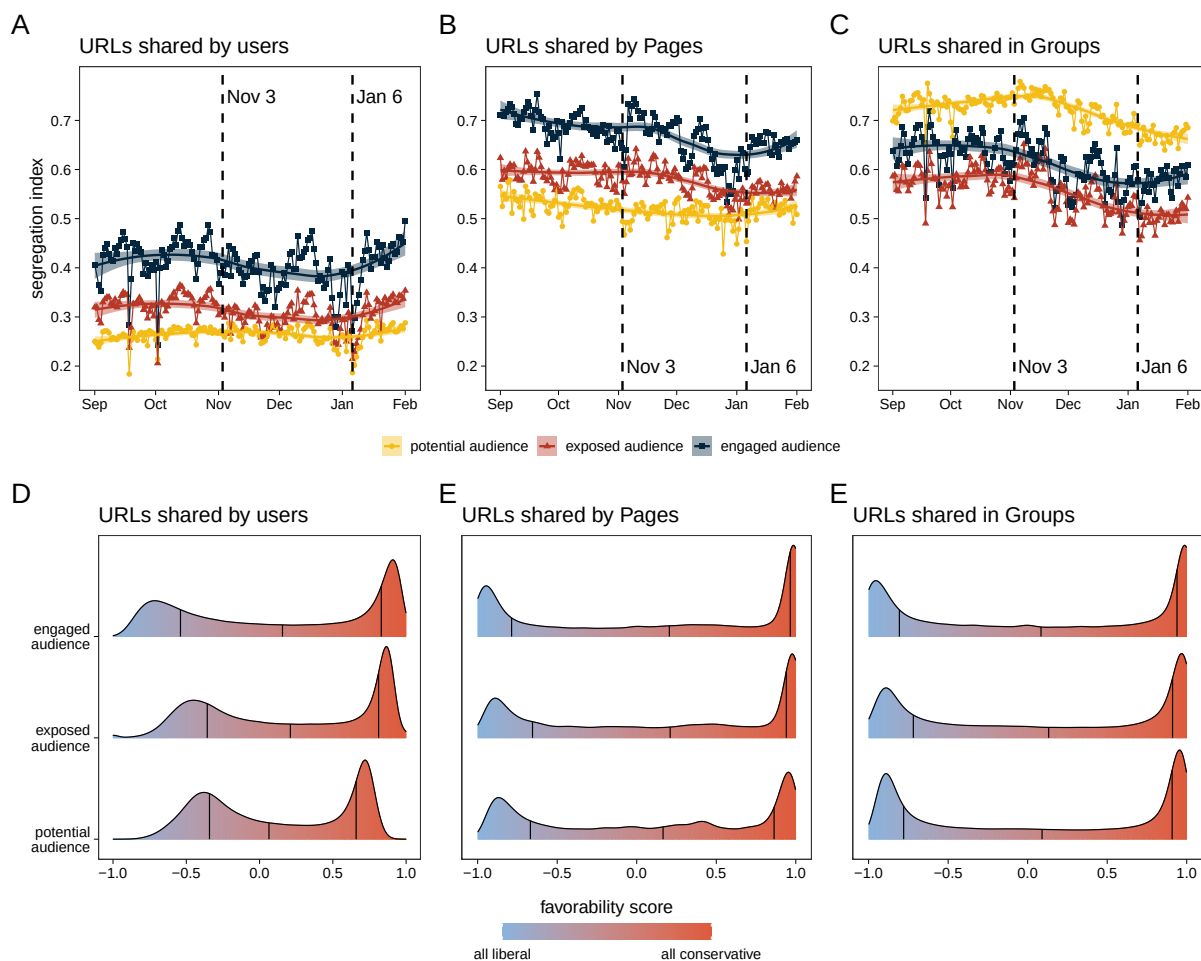


Figure S14: Funnel of Engagement for URLs posted by Users, Pages, and Groups. This figure is an extension to figure 2 in the main paper and to figure S12 above. Here URLs posted by users, Pages, and Groups are segmented by their potential, exposed, and engaged audiences. As in figure S13, these patterns reveal that algorithmic curation operates differently for content posted by Groups.

	Segregation			Favorability		
	User	Page	Group	User	Page	Group
All domains and users	0.21	0.41	0.32	0.14	0.08	0.10
Subset: High pol. int. users	0.34	0.54	0.43	-0.00	-0.01	0.01
Subset: Low pol. int. users	0.14	0.34	0.19	0.22	0.12	0.16
Subset: Untrustworthy domains	0.13	0.34	0.36	0.39	0.39	0.43

Table S11: **Segregation and favorability scores in exposure to political news domains in user, Page, and Group posts.** This table displays segregation and favorability metrics in exposure, segmented by poster (users, Pages, and Groups) and computed using data aggregated over the entire study period.

	Segregation			Favorability		
	User	Page	Group	User	Page	Group
All URLs and users	0.31	0.57	0.54	0.20	0.12	0.09
Subset: High pol. int. users	0.48	0.70	0.67	0.09	0.07	0.04
Subset: Low pol. int. users	0.21	0.46	0.35	0.29	0.17	0.16
Subset: COVID-related URLs	0.21	0.41	0.37	0.12	0.04	0.02
Subset: Misinformation URLs	0.17	0.31	0.34	0.77	0.83	0.85

Table S12: **Segregation and favorability scores in exposure to political news URLs in user, Page, and Group posts.** This table displays segregation and favorability metrics in exposure, segmented by poster (users, Pages, and Groups) and computed using data aggregated over the entire study period.

S4.5 COVID-Related Content

In figure **S15** we show the segregation and favorability scores for COVID-related content. The figure shows no evidence of algorithmic and social amplification (the segregation lines for potential, exposed and engaged audiences overlap for most of the period) but Pages and Groups are still driving segregation upwards. Favorability scores also reveal asymmetric polarization, but it is slightly less prominent compared to Figure 2 (panels E and F) in the main paper.

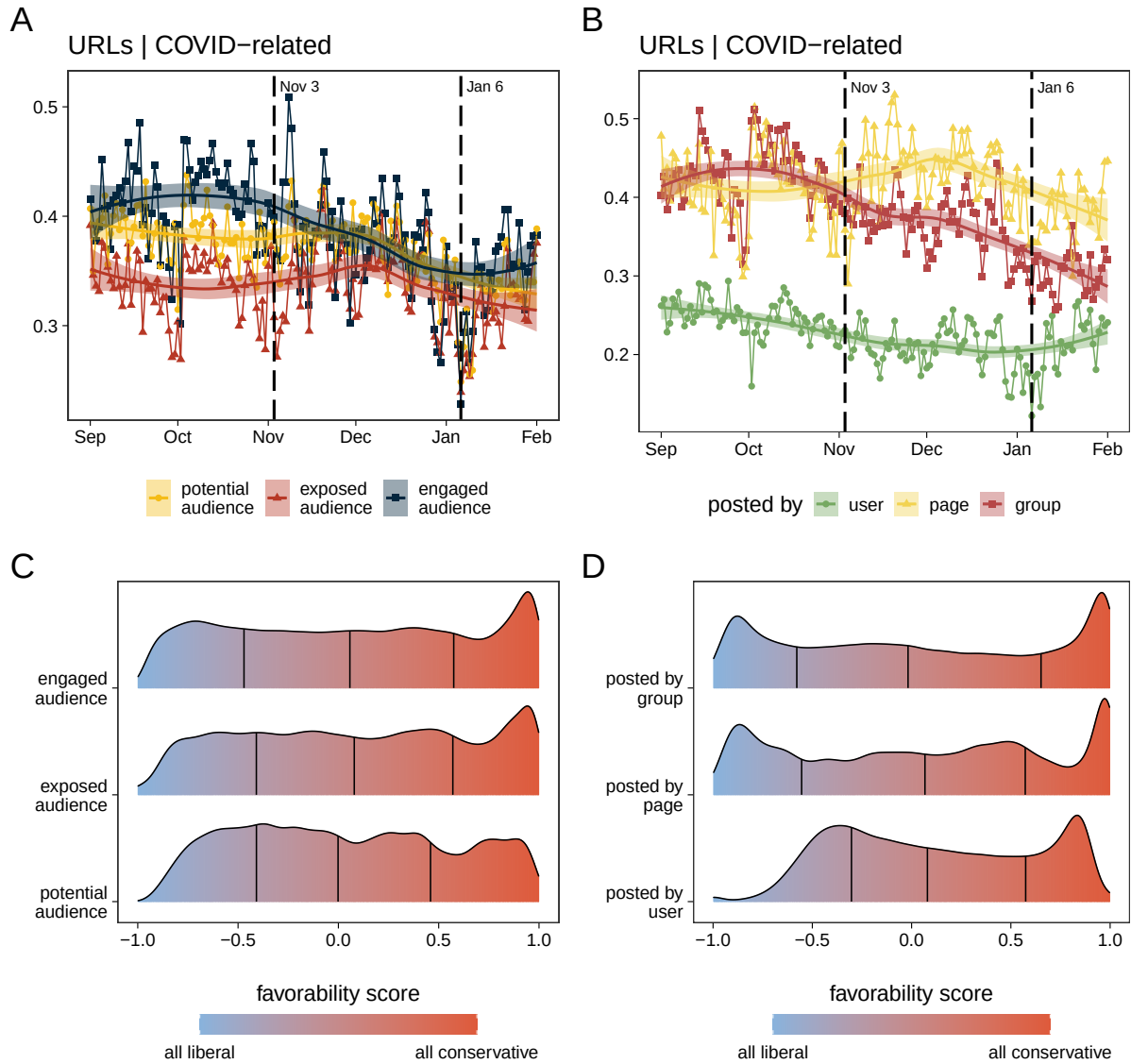


Figure S15: Segregation and Polarization around COVID-related Content. This figure is an extension to figure 2 in the main paper. It shows the segregation and favorability scores for news classified as being related to the pandemic. Here we see no evidence of algorithmic and social amplification (the segregation lines for potential, exposed and engaged audiences overlap for most of the period) but Pages and Groups are still driving segregation upwards. Favorability scores also reveal asymmetric polarization, but it is less prominent.

S4.6 Weekly Aggregations

Figure **S16** shows the weekly aggregations for the domain favorability distributions (an extension to Figure 2E in the main paper) and figure **S17** shows the same for URLs (an extension to Figure 2F in the main paper). In figure **S18** we apply a different visualization strategy to ease visual assessment of temporal changes.

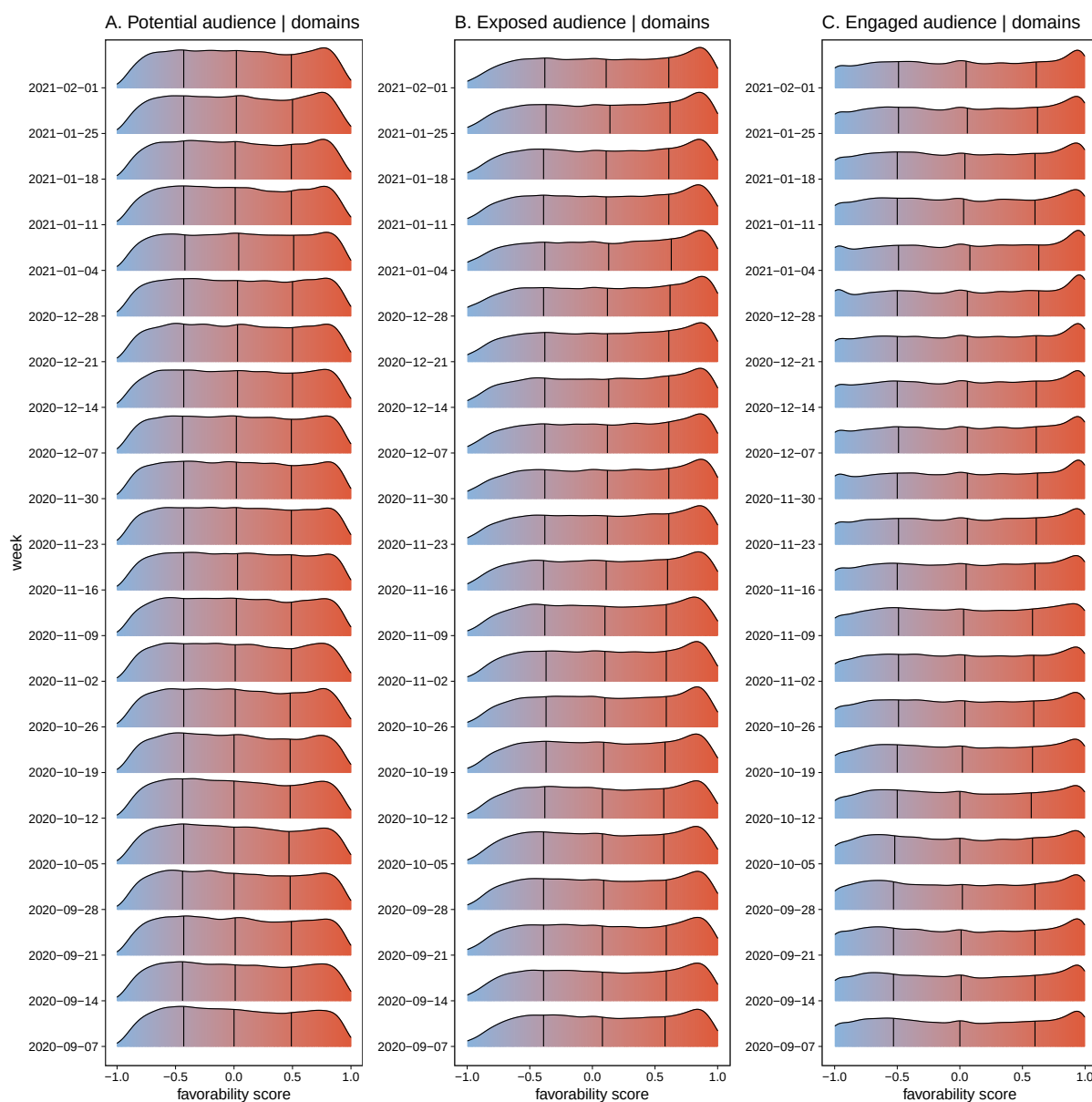


Figure S16: Weekly Aggregations for Domain-Level Favorability Distributions. This figure is an extension of Figure 2E in the main paper. The plots show no substantive variation in overall distribution shape.

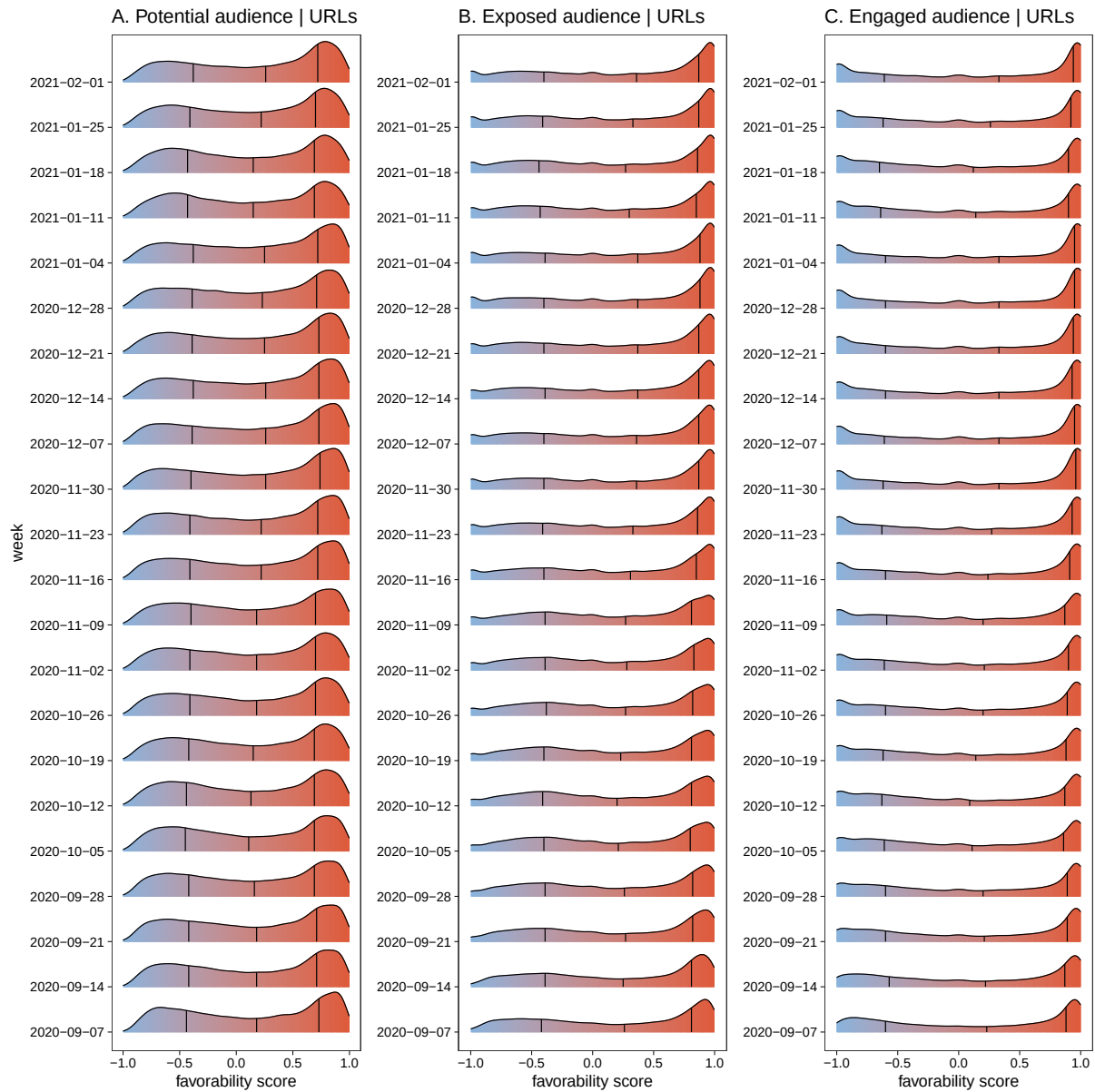


Figure S17: Weekly Aggregations for URL-Level Favorability Distributions. This figure is an extension of Figure 2F in the main paper. The plots show no substantive variation in overall distribution shape.

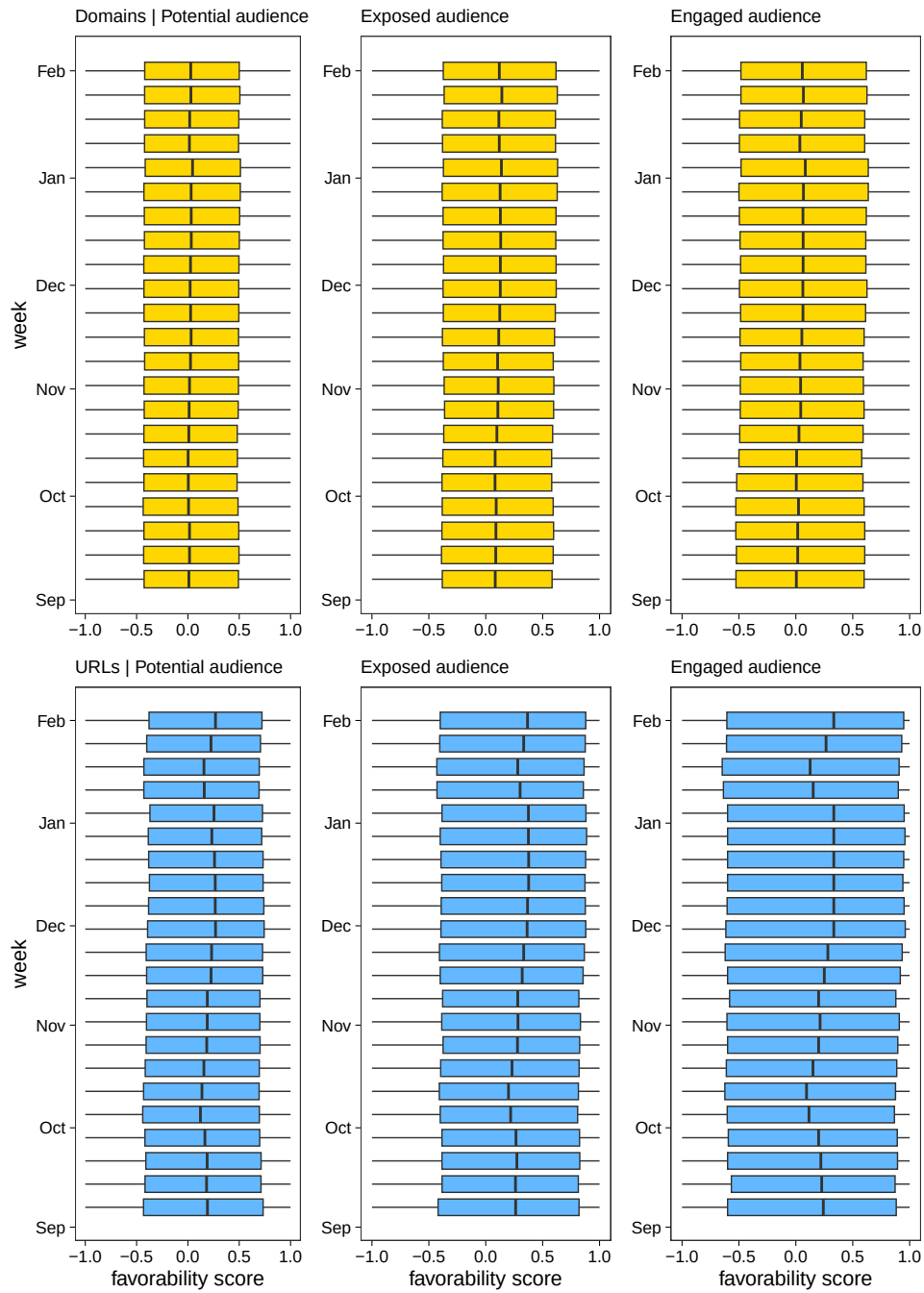


Figure S18: Weekly Aggregations for Domain and URL-Level Favorability Distributions. This figure is an extension of Figures 2E-F in the main paper. This visualization makes it easier to assess the lack of substantive temporal variation in overall distribution.

S4.7 Subsets of Data by Levels of Political Interest

As explained in SM section S3.5, we define as ‘high-political interest’ the audiences that are formed by users in the top 10% / top 1% of on-platform political engagement (i.e., engagement with political content in the form of comments, likes, and shares); the bottom 90 / 99% are classified as ‘low-political interest’. As an additional robustness test, we also operationalized ‘political interest’ using viewing activity (or exposure) instead of engagement.

Tables S13 and S14 replicate the topline results we showed in Tables S9 and S10 using our alternative operationalizations of political interest. The rest of this section discusses these results with greater length and offers additional figures to visualize these differences.

	Segregation			Favorability		
	Potential	Exposed	Engaged	Potential	Exposed	Engaged
All users	0.29	0.35	0.42	0.05	0.12	0.07
Top 1% (engagement)	0.50	0.59	0.63	0.01	-0.01	-0.03
Top 10% (engagement)	0.40	0.48	0.52	0.01	0.01	-0.00
Bottom 90% (engagement)	0.21	0.27	0.28	0.08	0.18	0.16
Top 1% (views)	0.47	0.52	0.60	-0.06	-0.03	-0.07
Top 10% (views)	0.39	0.41	0.49	0.02	0.06	0.02
Bottom 90% (views)	0.25	0.24	0.30	0.07	0.20	0.14

Table S13: **Segregation and favorability scores for three base audiences (Domains), by levels of political interest.** This table displays the same segregation and favorability metrics reported in Tables S10, but computed using alternative operationalizations of political interest.

	Segregation			Favorability		
	Potential	Exposed	Engaged	Potential	Exposed	Engaged
All users	0.47	0.47	0.55	0.12	0.16	0.13
Top 1% (engagement)	0.71	0.73	0.79	0.07	0.06	0.05
Top 10% (engagement)	0.60	0.63	0.67	0.08	0.08	0.07
Bottom 90% (engagement)	0.35	0.35	0.35	0.16	0.23	0.23
Top 1% (views)	0.69	0.67	0.75	0.03	0.06	0.03
Top 10% (views)	0.60	0.55	0.64	0.09	0.13	0.10
Bottom 90% (views)	0.40	0.33	0.41	0.14	0.24	0.19

Table S14: **Segregation and favorability scores for three base audiences (URLs), by levels of political interest.** This table displays the same segregation and favorability metrics reported in Tables S10, but computed using alternative operationalizations of political interest.

Political Interest Measured as Engagement In figure S19 we show an extension of Figures 2B-C in the main text based on the segmentation of audiences as having high/low political

interest (according to engagement levels). What the figure suggests is that high-political interest users are more strongly segregated: the magnitude of the score is nearly double compared to low-interest users. The plots also suggest that exposed and engaged audiences are more segregated than potential audiences, regardless of political interest (although this trend is only clear for domain-level data).

In figure S20, we show similar information but this time segmenting content in terms of who posted it (users, Pages, or Groups). Consistent with other figures, we find that Pages and Groups are driving up segregation levels, regardless of how the data are aggregated (domain or URL level).

In figure S21 we extend Figure 2E-F in the main paper to show the favorability scores for high-low political interest audiences. The figure shows that it is high-political interest users who are driving extreme polarization. This is particularly visible at the URL-level. The figure also shows that, overall, right-leaning content prevails, regardless of level of political interest.

In figure S22 we show the favorability distributions for high-low political interest and content posted by users, Pages, and Groups. Again, the figure suggests that high-political interest users are driving extreme polarization, a pattern that is particularly visible at the URL-level. Overall, extreme right-leaning content is mostly posted by Pages and Groups.

In figures S23 and S24 we show the same plots but classifying as high-political interest only those users in the top 1% of the distribution of on-platform engagement with political content. Segregation levels are higher for this group of highly-engaged users, and polarization becomes even more visible, and less asymmetrical, which suggests that segregation dynamics are driven by a very small fraction of highly committed users, but also that the asymmetry we identify on the right is driven by the mass of users with lower levels of political interest.

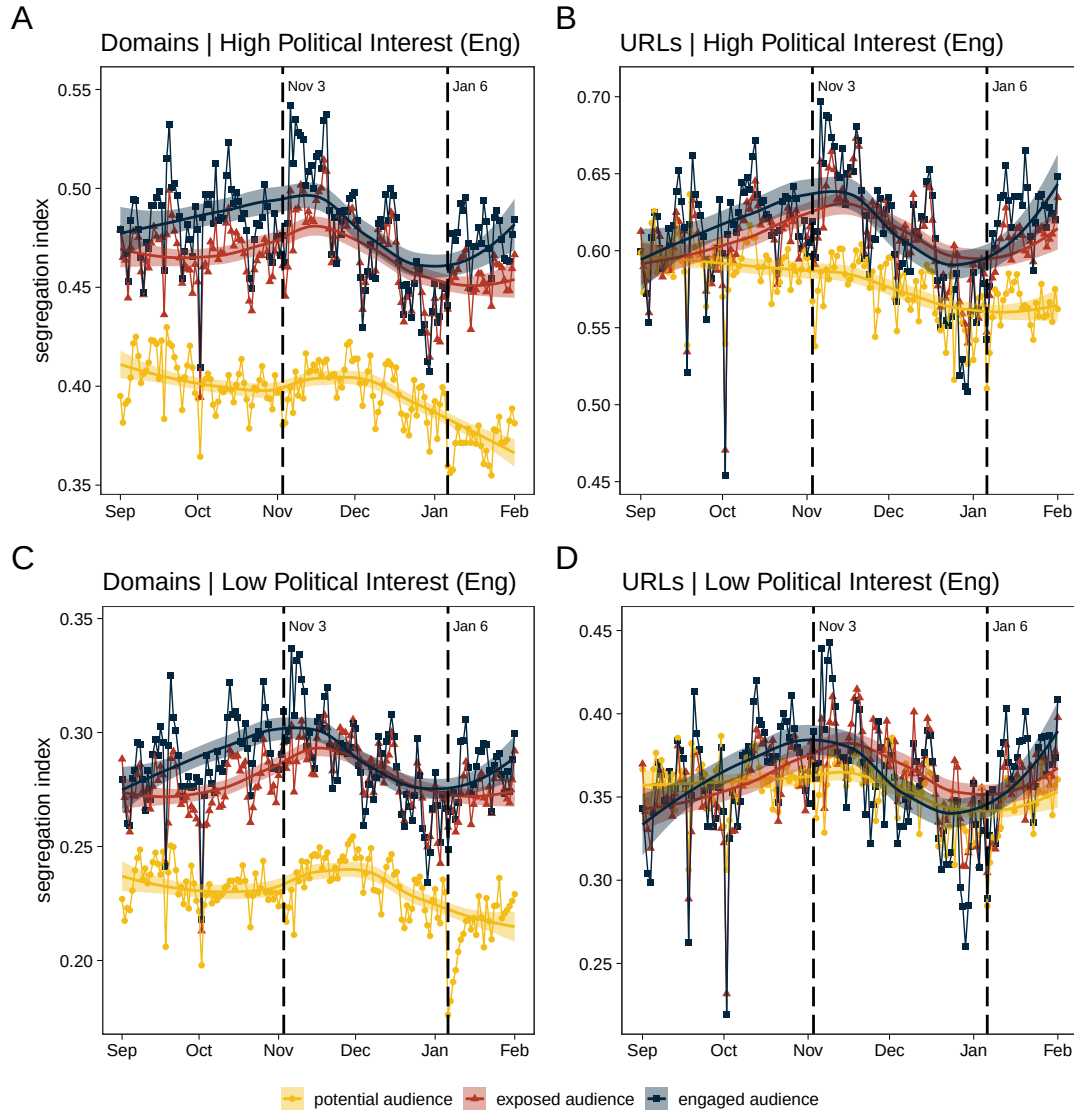


Figure S19: Segregation Scores for High (top 10%) and Low Political Interest Audiences (based on Engagement Levels for Potential, Exposed, and Engaged Audiences). This figure is an extension to Figures 2B-C in the main paper. The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. A vertical comparison of the plots suggests that high-political interest users are more strongly segregated: the magnitude of the score is nearly double. It also suggests that exposed and engaged audiences are more segregated than potential audiences, regardless of political interest (although this trend is only clear for domain-level data).

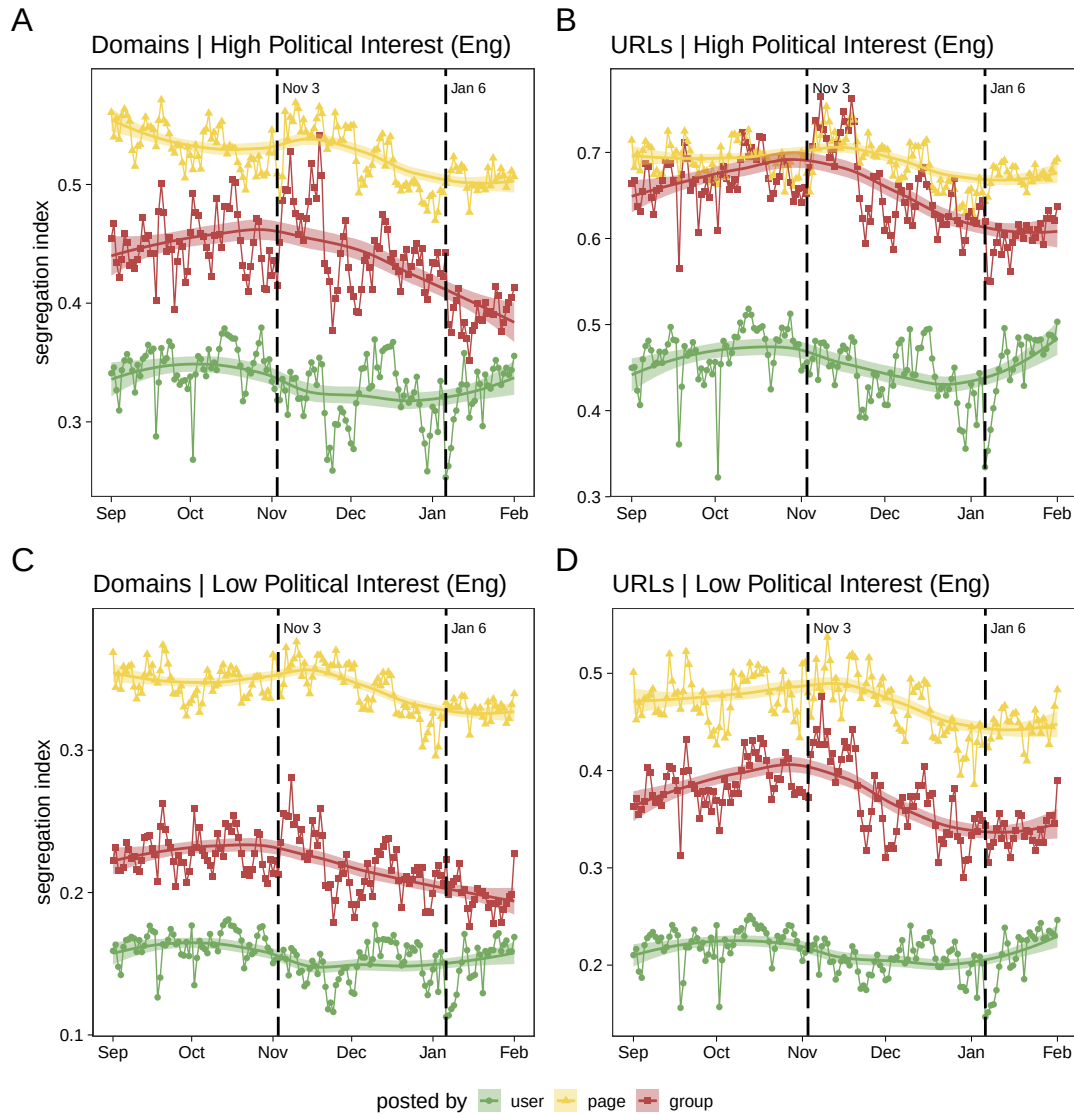


Figure S20: Segregation Scores for High (top 10%) and Low Political Interest Audiences (based on Engagement Levels for Users, Pages, and Groups). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. A vertical comparison of the plots suggests that high-political interest users are more strongly segregated: the magnitude of the score is nearly double. It also suggests that Pages and Groups are driving up segregation levels regardless of how the data are aggregated (domains/URLs).

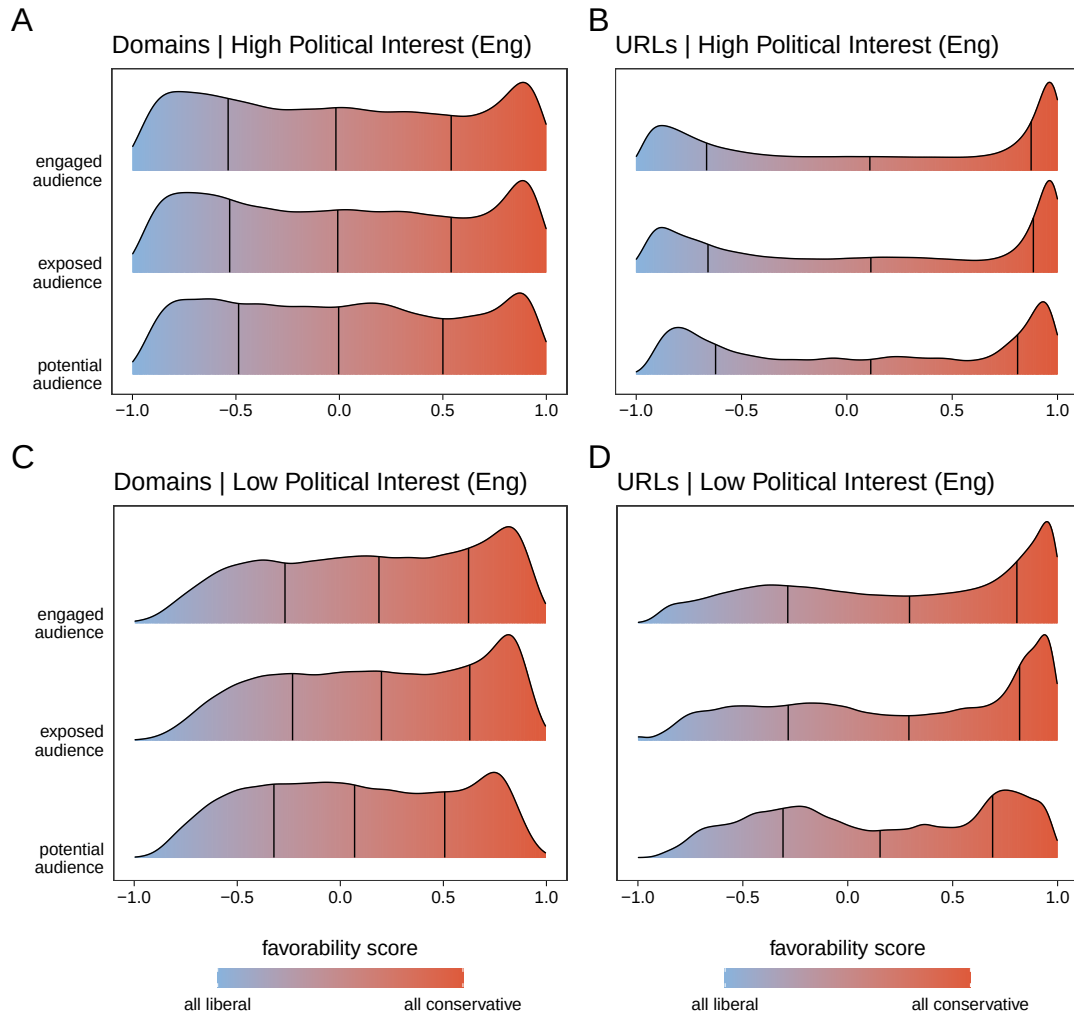


Figure S21: Favorability Scores for High (top 10%) and Low Political Interest Audiences (based on Engagement Levels for Potential, Exposed, and Engaged Audiences). This figure is an extension to Figures 2E-F in the main paper. The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. A vertical comparison of the plots suggests that high-political interest users are driving extreme polarization. This is particularly visible at the URL-level. Overall, right-leaning content prevails in all audiences, regardless of level of political interest.

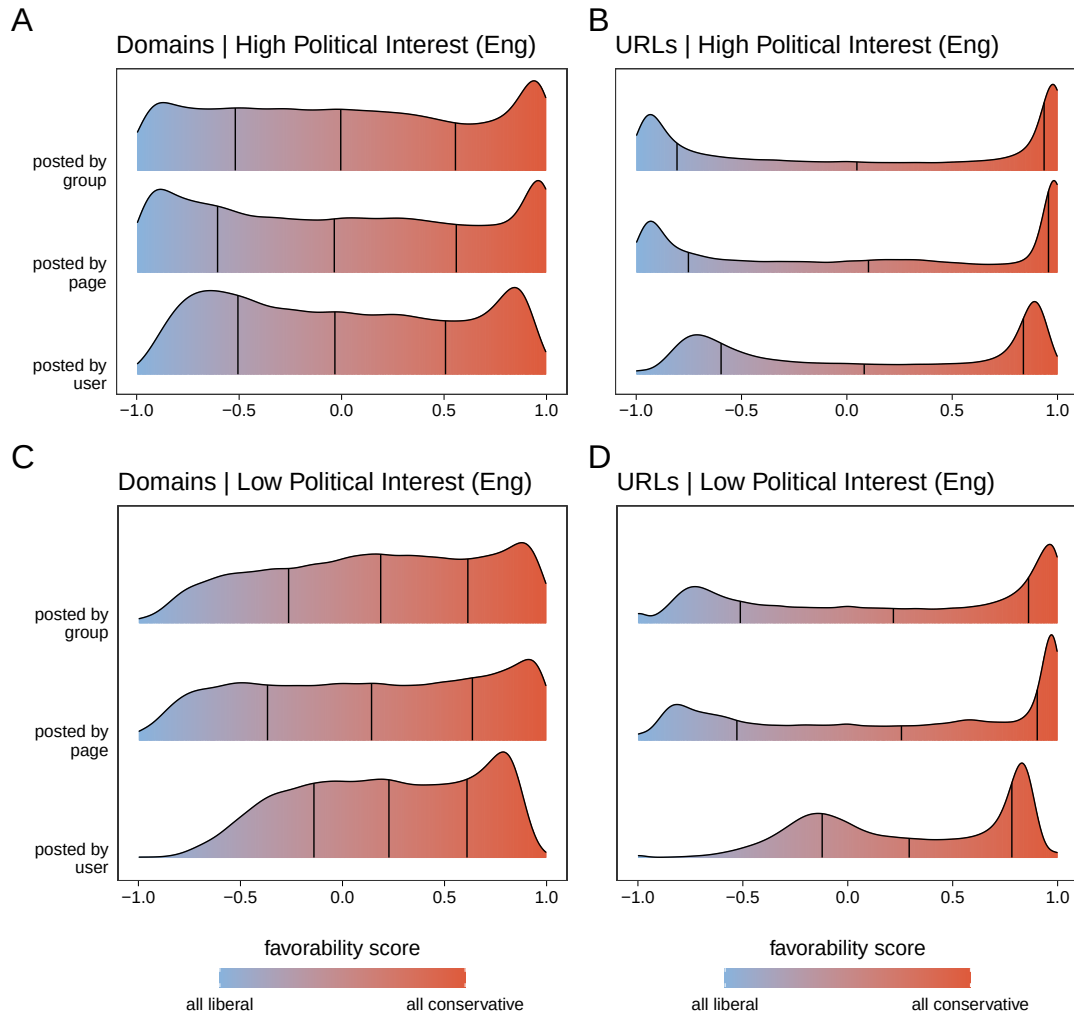


Figure S22: Favorability Scores for High (top 10%) and Low Political Interest Audiences (based on Engagement Levels for Users, Pages, and Groups). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. A vertical comparison of the plots suggests that high-political interest users are driving extreme polarization. This is particularly visible at the URL-level. Overall, extreme right-leaning content is mostly posted by Pages and Groups.

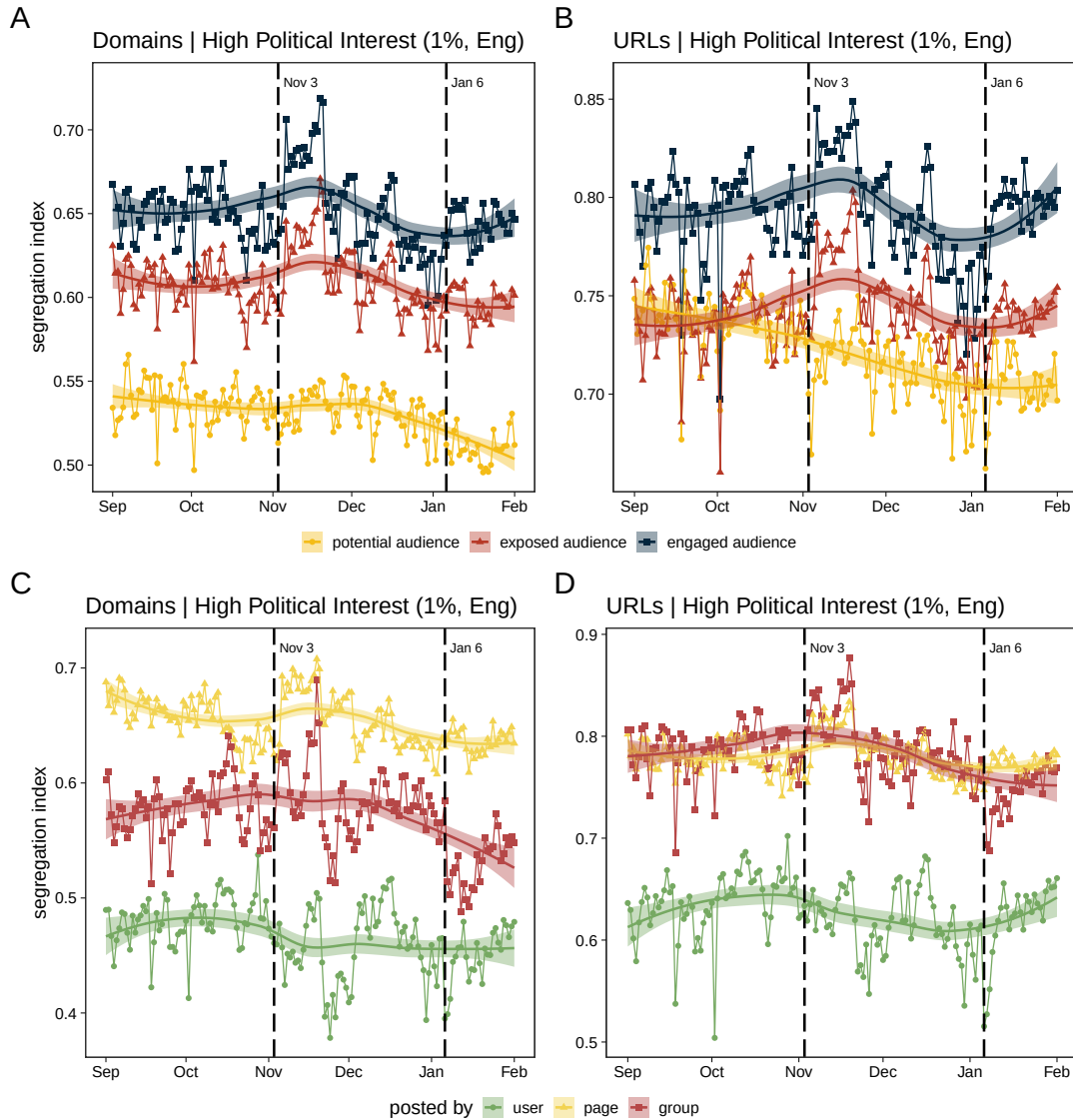


Figure S23: Segregation Scores for High (top 1%) Political Interest Audiences (based on Engagement Levels). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. The plots suggest that, under this definition of high-political interest, users are even more strongly segregated; they also suggest that Pages and Groups are, again, driving up segregation levels regardless of how we aggregate the data (domains/URLs).

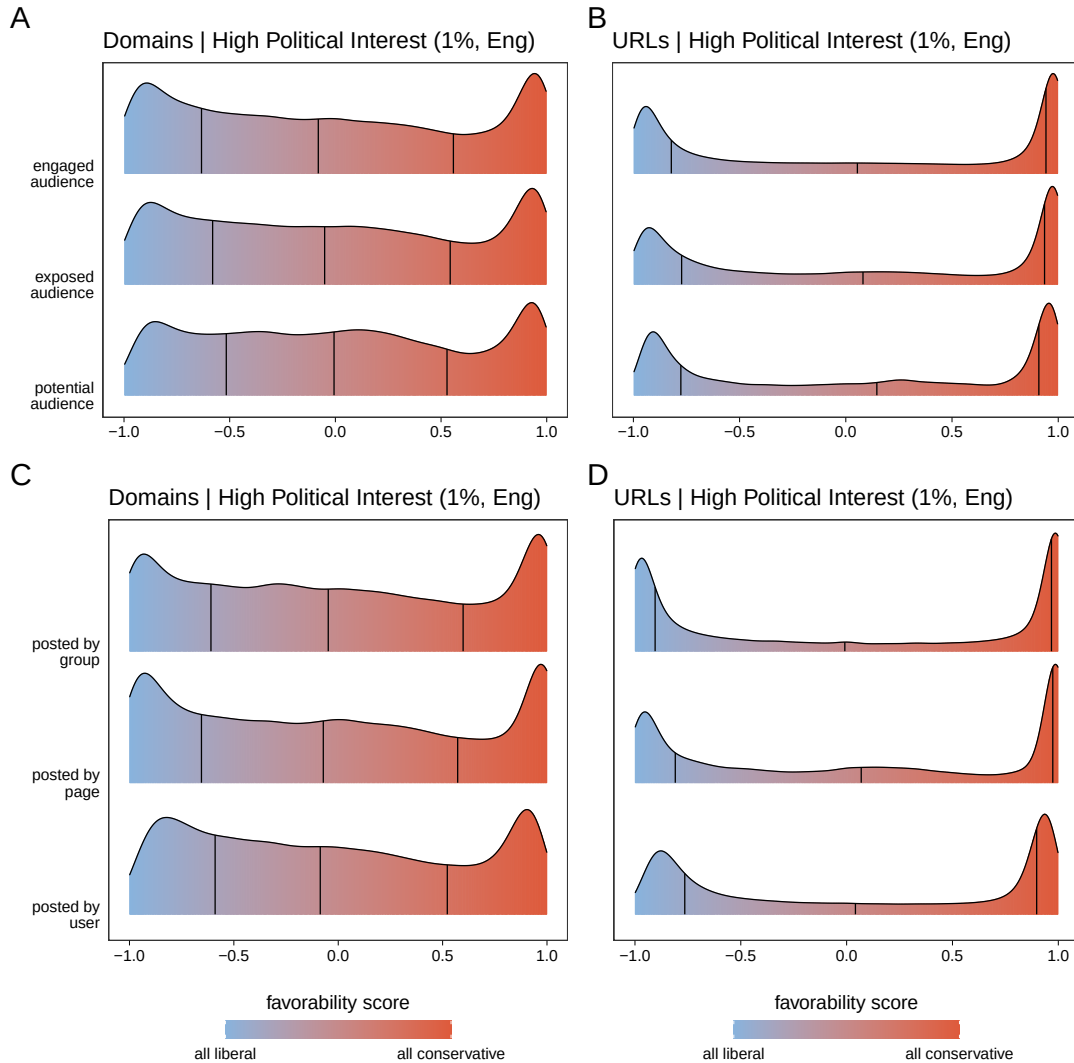


Figure S24: Favorability Scores for High (top 1%) Political Interest Audiences (based on Engagement Levels). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. Under this definition of high-political interest users, polarization is even more extreme, and more clearly visible on the left side of the ideological scale.

Political Interest Measured as Views In figure S25 we show segregation levels for high/low political interest operationalized as ‘views’ (exposure). A vertical comparison of the plots confirms that high-political interest users are more strongly segregated. However, under this operationalization of political interest, the difference between potential and exposed audiences disappears. For data aggregated at the URL-level we see a reversal of trends in the potential

and exposed audience scores. One potential explanation is that for this subset of high political users (based on views) the inventory of potential content is not that different from the content they actually see (since, being high interest, these users access Facebook so often that essentially all the material in their inventory is consumed). Another possible explanation points at the non-engagement signals used by the Feed Ranking and content-type diversity rules: algorithmic downranking of clickbait may reduce segregation for users whose entire inventory is low-quality news (likely for many in the high-political interest category); and segregation is likely higher for low-quality news. Similarly, if users only follow News Pages, content diversity rules may limit how much News they see, and the lowest quality News would be the first to go down. Ultimately, however, it is difficult to disambiguate these potential explanations given that all our data are aggregated.

Figure S26 confirms what we already showed in the previous section, i.e., using “engagement levels” to operationalize political interest: high-political interest users are more strongly segregated; and Pages and Groups are driving up segregation levels regardless of how the data are aggregated (domains/URLs).

Figures S27 and S28 confirm that high-political interest users are driving extreme polarization, and that extreme right-leaning content is mostly posted by Pages and Groups.

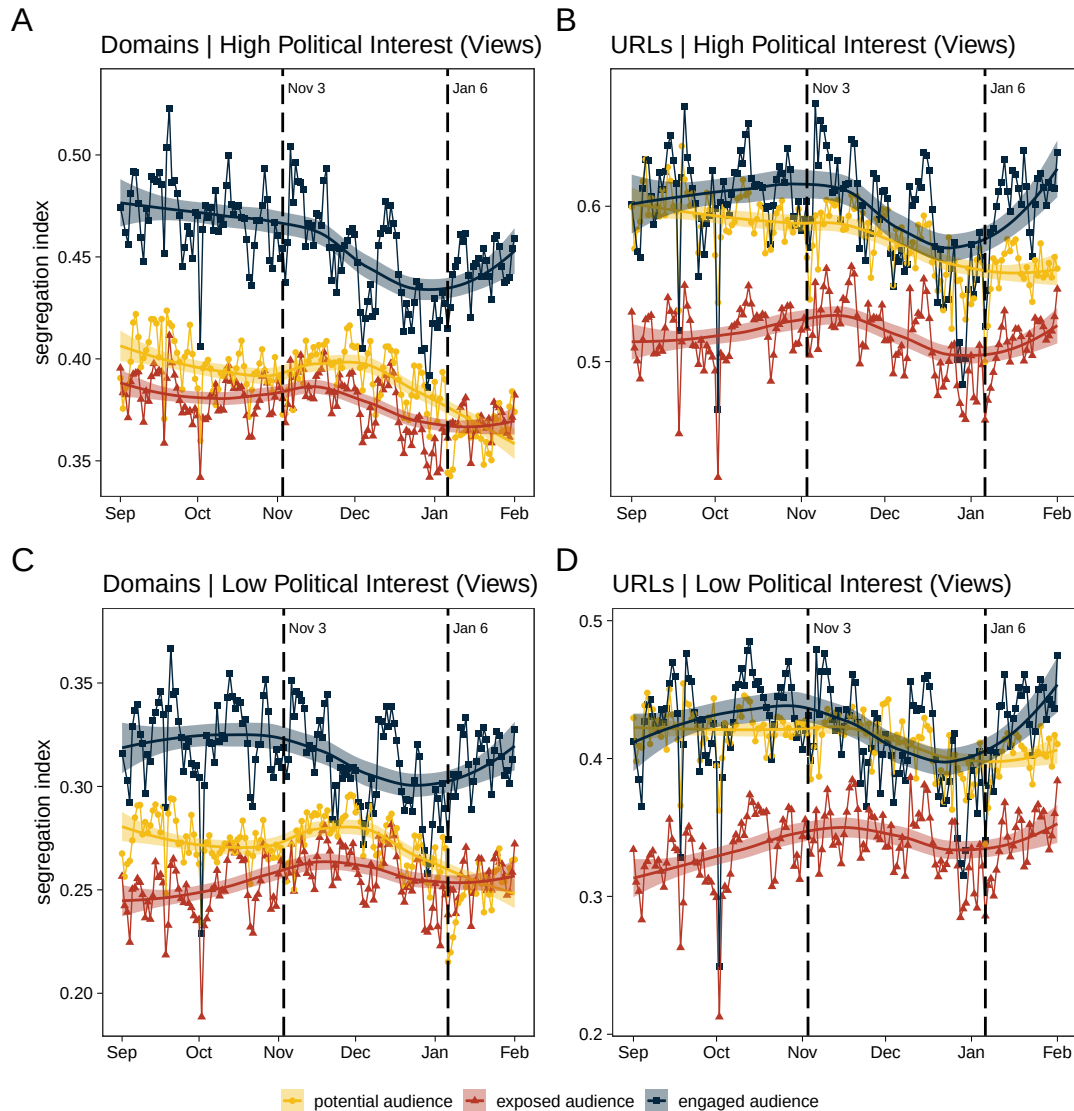


Figure S25: Segregation Scores for High (top 10%) and Low Political Interest Audiences (based on Views for Potential, Exposed, and Engaged Audiences). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. A vertical comparison of the plots confirms that high-political interest users are more strongly segregated, and that content is more polarized at the URL level than domain. However, under this operationalization of “political interest”, the difference between potential and exposed audiences disappears. For data aggregated at the URL-level, we see a reversal of trends in the potential and exposed audience scores (see text for discussion of possible mechanisms).

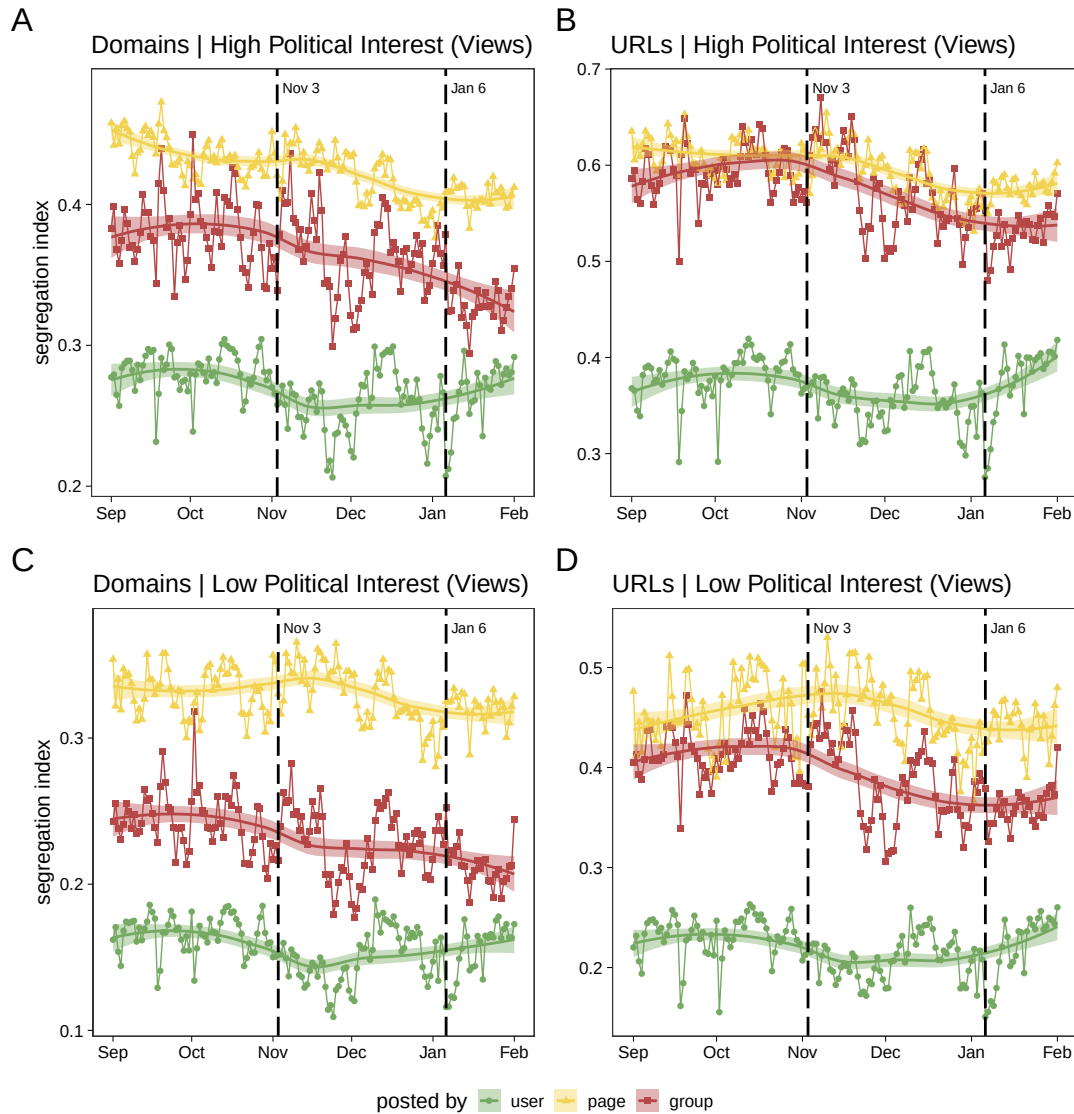


Figure S26: Segregation Scores for High (top 10%) and Low Political Interest Audiences (based on Views for Users, Pages, and Groups). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. Here the results are in line with what we showed using “engagement levels” to operationalize political interest: high-political interest users are more strongly segregated; and Pages and Groups are driving up segregation levels regardless of how the data are aggregated (domains/URLs).

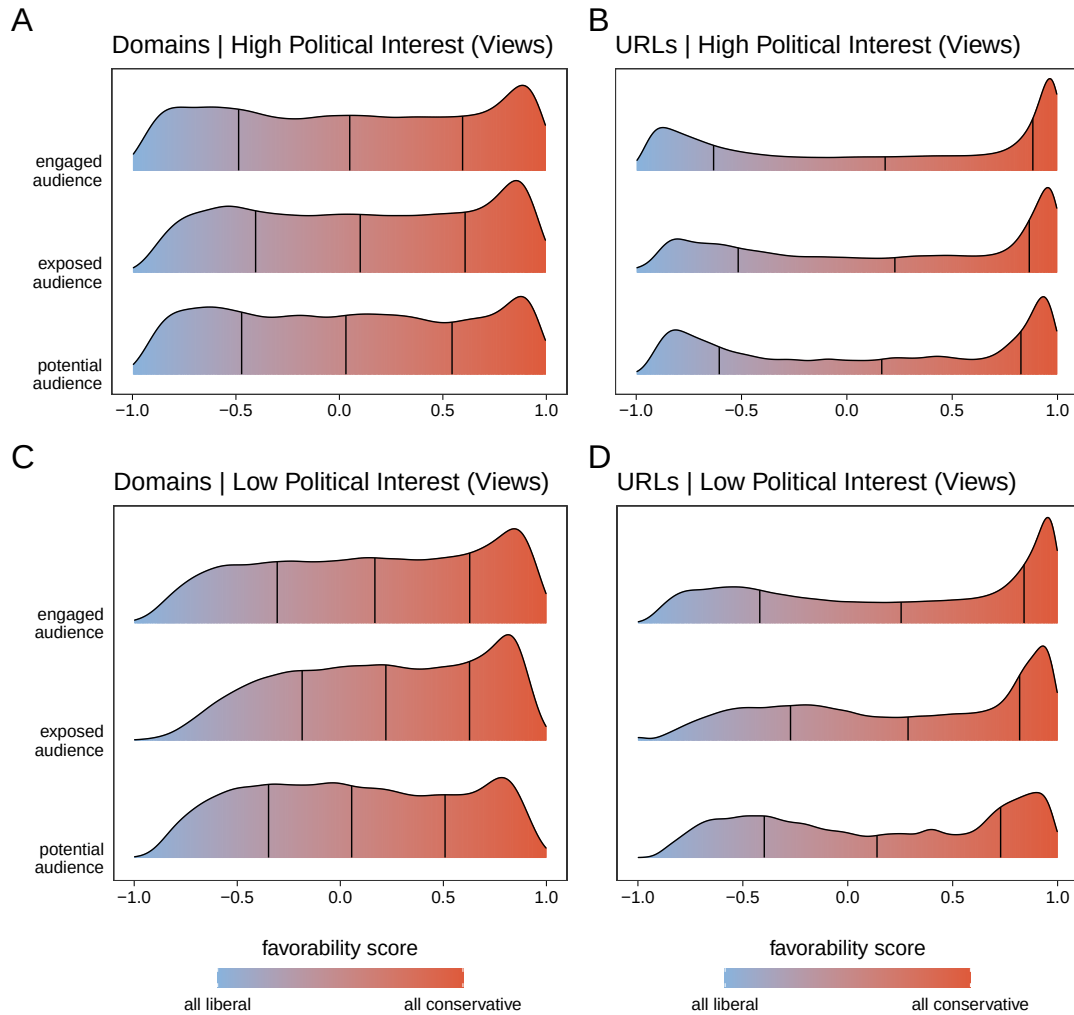


Figure S27: Favorability Scores for High (top 10%) and Low Political Interest Audiences (based on Views for Potential, Exposed, and Engaged Audiences). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. This figure confirms that high-political interest users are driving extreme polarization, a pattern that is, again, particularly visible at the URL-level.

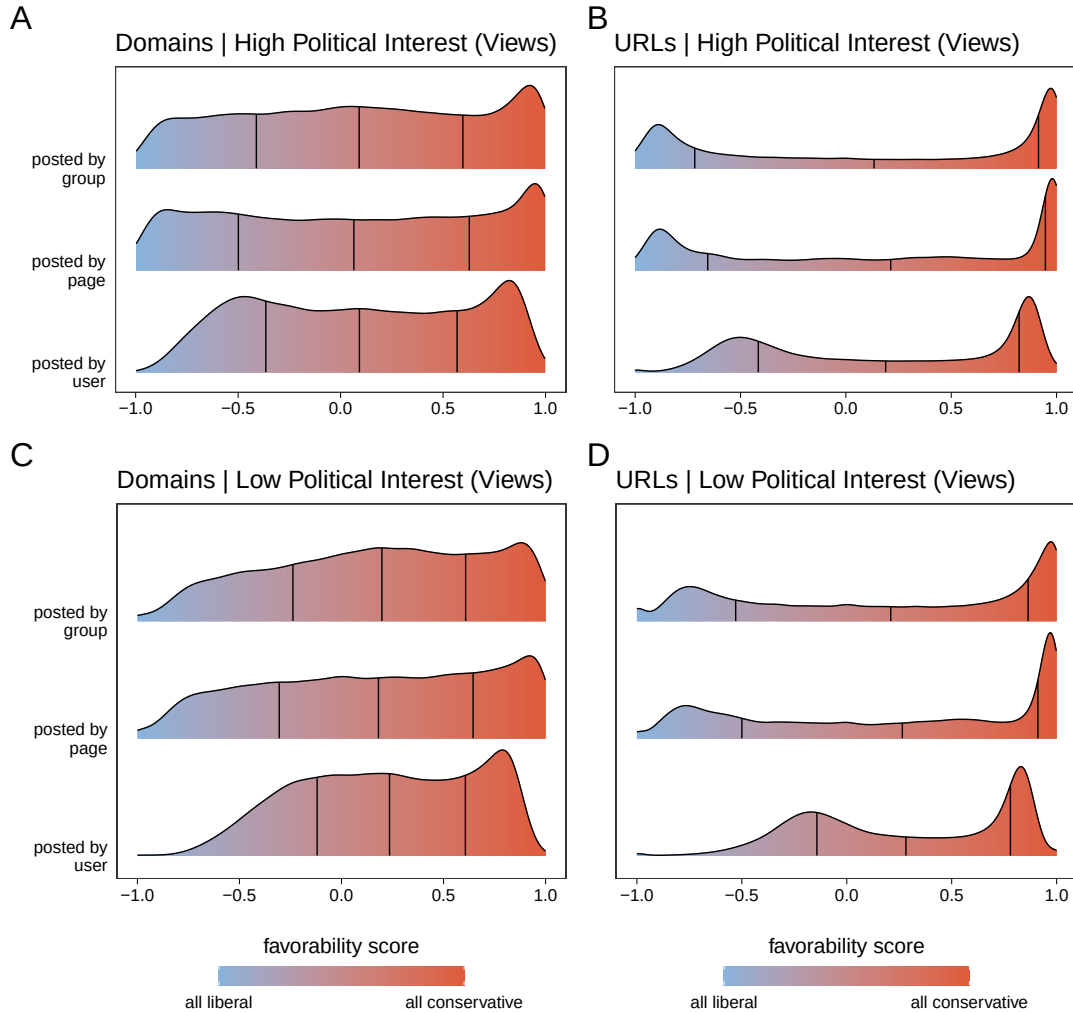


Figure S28: Favorability Scores for High (top 10%) and Low Political Interest Audiences (based on Views for Users, Pages, and Groups). The first column plots the scores aggregated at the domain level, the right column aggregates the scores at the URL-level. This figure confirms that high-political interest users are driving extreme polarization, a pattern that is particularly visible at the URL-level. Extreme right-leaning content is mostly posted by Pages and Groups.

S4.8 Subsets of Data by Cumulative Views

Table S15 shows the segregation index and average favorability scores at each audience level for all domains ($N \sim 35000$), domains with 99% and 90% of views ($N = 1235$ and $N = 8,387$, respectively), the top 100 domains, as well as an external list of 800 news domains that was compiled by aggregating lists used by previous published studies on similar topics as this

one (8, 12, 3, 16, 29). Figure S29 displays the entire favorability distributions for the same subsets of the data. For domains, we systematically find that segregation scores from exposed audiences are higher than those based on potential audiences, while being lower than those of engaged audiences. For URLs we find similar patterns for the most viewed. The right-leaning asymmetry is also visible regardless of which domains/URLs are included in the analyses.

	Segregation			Favorability		
	Potential	Exposed	Engaged	Potential	Exposed	Engaged
All political news URLs	0.47	0.47	0.55	0.12	0.16	0.13
URLs in political news domains with 99% of views	0.47	0.47	0.55	0.12	0.16	0.12
URLs in political news domains with 90% of views	0.46	0.47	0.55	0.10	0.14	0.10
URLs in top 100 political news domains	0.43	0.48	0.57	0.04	0.07	0.04
URLs in external list of 800 news domains	0.45	0.47	0.54	0.05	0.08	0.05
All political news domains	0.29	0.35	0.42	0.05	0.12	0.07
political news domains with 99% of views	0.29	0.35	0.42	0.07	0.14	0.11
political news domains with 90% of views	0.29	0.35	0.42	0.09	0.18	0.13
Top 100 political news domains	0.29	0.40	0.42	0.09	0.08	0.13
External list of 800 news domains	0.30	0.37	0.44	0.03	0.10	0.05

Table S15: **Segregation and favorability scores across subsets of domains.** This table displays segregation and favorability scores for political news URLs and domains for all three audience measures, and across different subsets of domains.

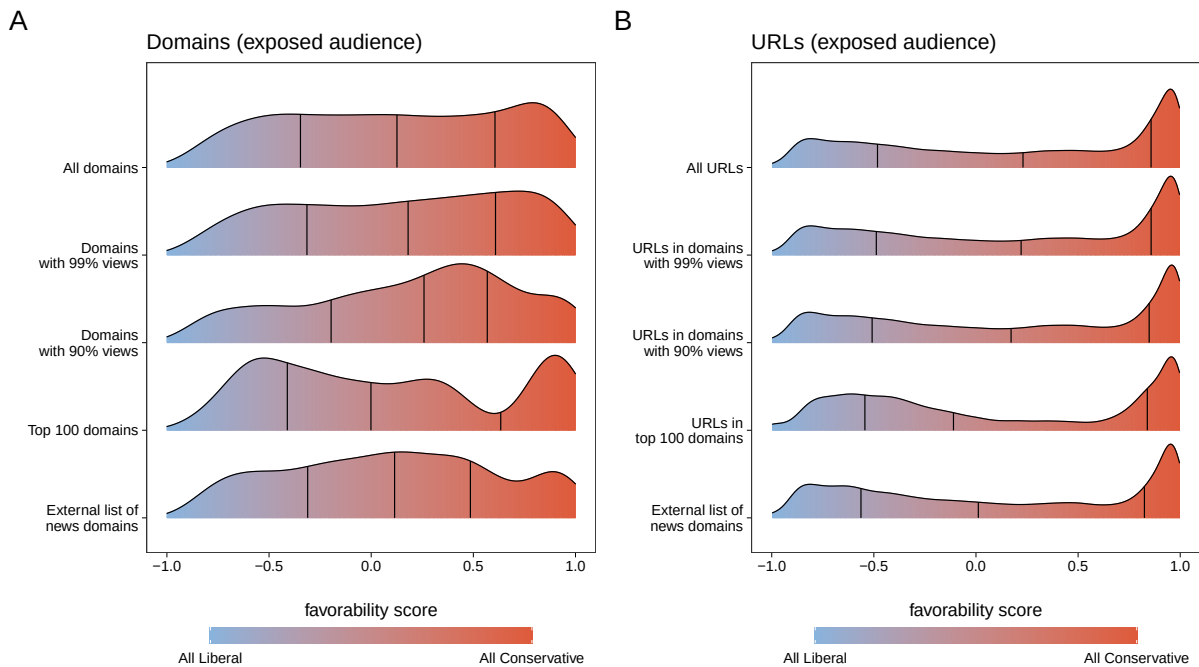


Figure S29: **Favorability Distributions for Different subsets of the Data.** The left column shows distributions for data aggregated at the domain-level; the right column shows data aggregated at the URL level. The right-leaning asymmetry is visible regardless of which domains/URLs are included, in terms of their percentile in the cumulative view distribution.

S4.9 Segregation Scores for Untrustworthy Content

Figure S30 shows results for the subset of domains classified as untrustworthy, and the subset of URLs flagged as misinformation. The results for domain-level data are consistent with what we find when analyzing all domains: segregation is higher for exposed audiences than for potential audiences (suggesting algorithmic amplification) but lower than for engaged audiences (suggesting social amplification). Pages and Groups are also generating higher segregation. The results for URL-level data are so noisy that no pattern is discernible.

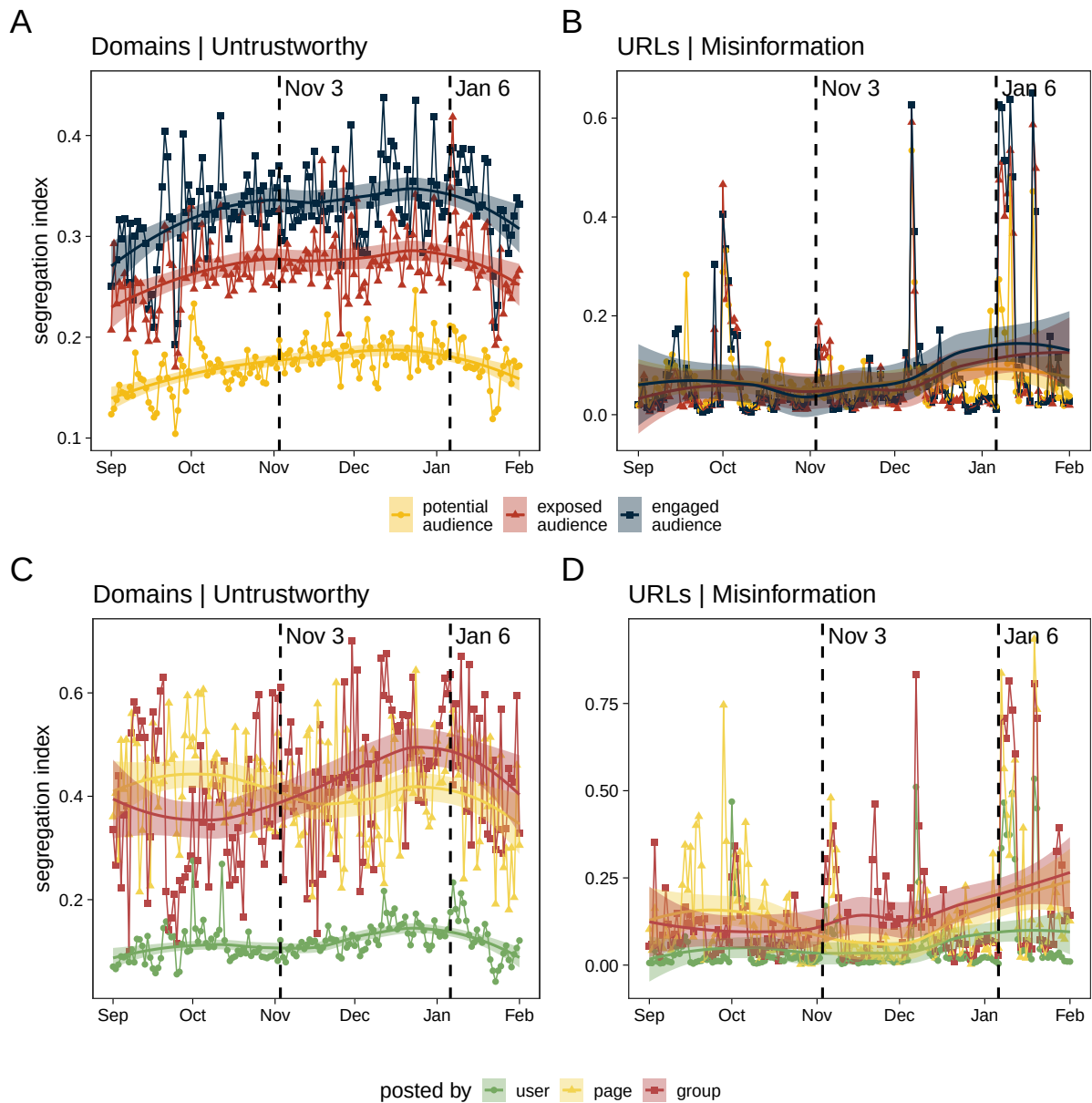


Figure S30: Segregation Scores for Untrustworthy Domains and Misinformation. This plot is an extension to Figure 3 in the main paper. The results for domain-level data are consistent with what we find when analyzing all domains: segregation is higher for exposed audiences than for potential audiences (suggesting algorithmic amplification) but lower than for engaged audiences (suggesting social amplification); segregation is also higher for untrustworthy content generated from Pages and Groups. The results for URL-level data are so noisy that no pattern is discernible.

S4.10 Different Ideology Thresholds

Figure S31 and Table S16 show the segregation and favorability scores for different versions of the ideology classifier. Our main results are based on a threshold of 0.35 or lower to classify users as ‘liberal’ and of 0.65 or higher to classify users as ‘conservative’. We rerun our analyses with two alternative thresholds to dichotomize users’ predicted ideology score: 0.5 and 0.10-0.90 (which only takes into account information from users predicted to be more extreme). The figure shows that, as expected, segregation scores became larger in magnitude the more stringent the ideology threshold is. Likewise, the extremity of the polarization we identify with the favorability scores is sensitive to the choice of a threshold, but not largely: the overall shape of the distributions (and the asymmetry they reveal) are consistent regardless of the threshold used.

Subset	Ideology threshold	Segregation			Favorability		
		Potential	Exposed	Engaged	Potential	Exposed	Engaged
Domains	(0.35, 0.65)	0.29	0.35	0.42	0.05	0.12	0.07
Domains	0.50	0.24	0.29	0.37	0.05	0.11	0.07
Domains	(0.10, 0.90)	0.41	0.49	0.52	0.14	0.16	0.11
URLs	(0.35, 0.65)	0.47	0.47	0.55	0.12	0.16	0.13
URLs	0.50	0.4	0.4	0.49	0.11	0.16	0.13
URLs	(0.10, 0.90)	0.63	0.62	0.67	0.2	0.2	0.16

Table S16: **Segregation and Favorability Scores under Different Ideology Thresholds.** To explore the robustness of our results to the use of different thresholds to dichotomize user’s ideology prediction, here we displays the segregation index and favorability scores for each audience level and aggregation (URLs or domains).

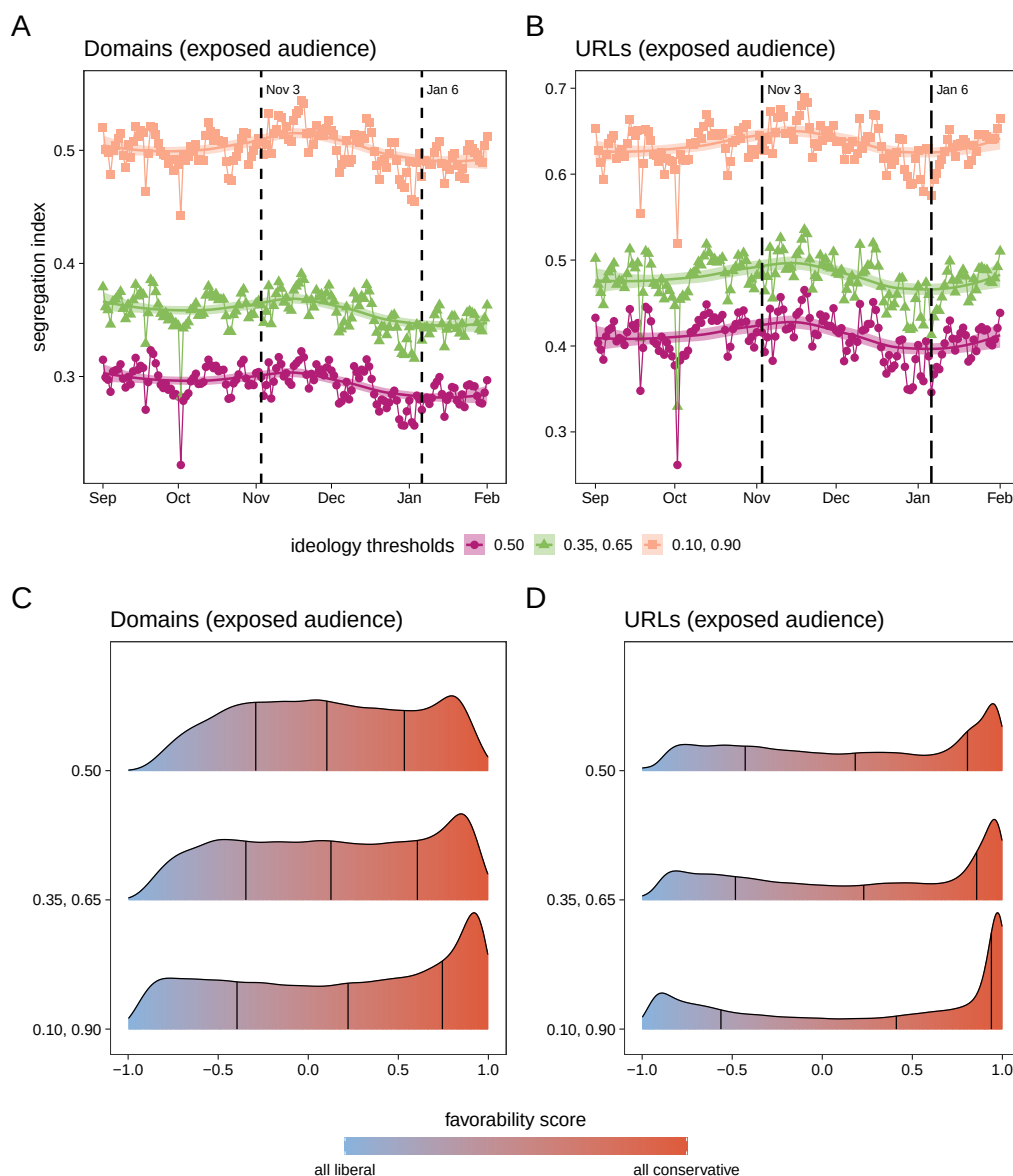


Figure S31: Segregation and Favorability Scores under Different Ideology Thresholds. We rerun our analyses with two alternative thresholds to dichotomize users' predicted ideology score: 0.5 and [0.10-0.90], which only takes into account information from users predicted to be more extreme. The figure shows that, as expected, segregation scores became larger in magnitude the more stringent the ideology threshold is. Likewise, the extremity of the audience polarization we identify with the favorability scores is sensitive to the choice of a threshold, but not largely: the overall shape of the distributions (and the asymmetry they reveal) is consistent regardless of the threshold used.

S4.11 Changes in Segregation Over Time

In our PAP (see SM section S7), we pre-registered a statistical test to examine whether segregation increased as election day approached. In particular, we proposed fitting a regression model with the segregation index as the dependent variable and the number of days since the beginning of the study period as explanatory variable. Tables S17 and S18 show the results of this analysis for URLs and domains, respectively. As shown in Figures 2 and 3 in the main text, the segregation index varies substantially from day to day. As a result, our estimates here can be interpreted as an approximation to whether segregation was on average increasing or decreasing during the campaign. In most cases, we don't find trends that are significantly different from zero; and when they are significant, the estimated coefficients are small in magnitude.

	Potential	Audience type	
		Exposed	Engaged
All political news domains/users	-0.0002 (0.0001)*	-0.0001 (0.0001)	-0.0005 (0.0002)*
Subset: High pol. int. users	-0.0002 (0.0001)*	-0.0001 (0.0001)	-0.0001 (0.0001)
Subset: Low pol. int. users	-0.0001 (0.0001)	0.0002 (0.0001)	0.0002 (0.0001)
Subset: Untrustworthy domains	0.0005 (0.0001)*	0.0008 (0.0002)*	0.0010 (0.0003)*

Table S17: **Estimated daily change in segregation index throughout election campaign, by audience type (Domains).** This table show the slope coefficients of an OLS regression model with daily segregation scores (for each of the subsets indicated in the row titles) as outcome variable and number of days since the beginning of the study period and until the election. Segregation is computed for the audience of political news URLs. Standard errors in parentheses. * $p < 0.05$

	Potential	Audience type	
		Exposed	Engaged
All political news URLs/users	-0.0001 (0.0001)	0.0002 (0.0002)	0.0001 (0.0003)
Subset: High pol. int. users	-0.0002 (0.0001)	0.0004 (0.0002)*	0.0003 (0.0002)
Subset: Low pol. int. users	0.0001 (0.0001)	0.0005 (0.0002)*	0.0005 (0.0002)*
Subset: COVID-related URLs	-0.0003 (0.0002)	-0.0002 (0.0002)	-0.0002 (0.0003)
Subset: Misinformation URLs	-0.0002 (0.0004)	-0.0001 (0.0005)	-0.0005 (0.0006)

Table S18: **Estimated daily change in segregation index throughout election campaign, by audience type (URLs).** This table show the slope coefficients of an OLS regression model with daily segregation scores (for each of the subsets indicated in the row titles) as outcome variable and number of days since the beginning of the study period and until the election. Segregation is computed for the audience of political news URLs. Standard errors in parentheses. * $p < 0.05$

S5 Web Tracking Data

As we explain in SM section S2.2, we have web tracking data for the subset of Facebook users who agreed to participate in the study and were part of the control group (for studies examining the effects of platform interventions). We measure their exposure to the list of domains that accumulate 90 and 99 percent of all views of political news within the platform. A total of 1,175 domains from the list in the top 90% of cumulative views appear on the web tracking data (so ~95% of the domains); and 5,042 domains from the list in the top 99% appear on the web tracking data (~60%). Part of this difference could be driven by the k -anonymization that was also applied to the web tracking data for privacy reasons and which involved excluding domains with visits by fewer than $k = 20$ unique panelists. With this data, we calculated segregation and favorability scores for both web and on-platform behavior, which allows us to run intra-person comparisons. In other words, with this data, we can answer the question: Is the information ecosystem created within Facebook more or less segregated than the information ecosystem created on the Web for the same subset of individuals?

As we report in figure S32 and table S19 (and as we mention in the main text) the segregation scores for the same set of individuals are at least three times larger on Facebook than on the Web. One relevant caveat here is that the subset of Facebook users who agreed to participate in the study, and who also agreed to share their web tracking data, are not representative of the broader Facebook user base: for starters, they are clearly more liberal (as panels B and D in figure S32 and the favorability scores in table S19 suggest). We try to address this concern by reporting weighted estimates in all the figures and tables below. As described in SM section S2.2, for the web tracking data we rely on our project-wide survey weights; for Facebook exposure, however, technical reasons limit our ability to adopt the same approach. Instead, we adopt a simpler approach that first groups users into nine buckets (defined by the intersection of their estimated ideology and political interest), then computes the relevant domain exposure counts within each of those buckets, and finally computes a new weighted count where the weights correspond to the relative size of each of those user buckets in the overall Facebook population. While imperfect, this simpler approach allows us to account for what we see as the two most important factors driving selection into our survey sample: being liberal and being interested in politics.

The segregation and favorability metrics in figure S32 are computed using the U.S. ideology classifier discussed in S3.4, which we use as source of ideology labels for the participants whose data we analyze. One potential concern with the use of this classifier is that it relies on different forms of engagement with content on Facebook as features for training, which could lead to circularity in our estimates of segregation and thus drive up its value. To explore whether this could affect our results, in figure S33 we take advantage of the fact that we can also use the survey data as a source of information about the ideology of participants who consented to share their passive tracking data, and thus compare how the estimates of segregation and favorability vary depending on whether we rely on the platform classifier or the survey self-reports to label the respondents' ideology.

The results here show that the segregation index is roughly the same regardless of whether we use classified or self-reported ideology. (Note that the y-axis here is expanded compared to figure S32.) If anything, it appears that segregation is slightly higher when computed using the self-reported survey data, which suggests that the use of the classifier does not inflate segregation. The distribution of favorability is more centered when computed using the classifier compared to the survey responses. This result is consistent with the performance metrics we report in SM section S3.4, which show that precision is higher for liberals than conservatives. However, the differences we observe here are small and thus unlikely to affect the ideological asymmetries we identify in our main analyses.

Another caveat is that the measures of web activity do not just take into account exposure to political news (as we do with on-platform data), but exposure to *any* URL within the domains identified as containing political news, regardless of whether the particular web page is about politics or not. To understand this difference, note that in our main analysis at the domain level we only include URLs within the domains that are predicted to be political news. For example, when we examine exposure to *nytimes.com*, we would only include exposure to political news URLs within that domain, and not URLs related to sports, entertainment, etc. Since our web tracking data is at the domain level, we cannot make the same distinction in this part of our analysis. This implies that some of the differences between the web tracking data and the Facebook exposure data for this set of users might be driven by differences in which URLs they choose to visit within each domain (e.g. if web browsing respondents gravitate towards non-political URLs within a domain such as *youtube.com*, but on Facebook they are only exposed to political *youtube.com* URLs.)

Despite these limitations, the data still allows us to run an intra-person and within-domain comparison that reveals substantially larger segregation levels when exposure to news happens on Facebook. In figure S34 we show the correlation of Facebook views and Web visits for the same set of domains. And in figure S35 we show the favorability scores on Facebook and the Web for the top 100 domains.

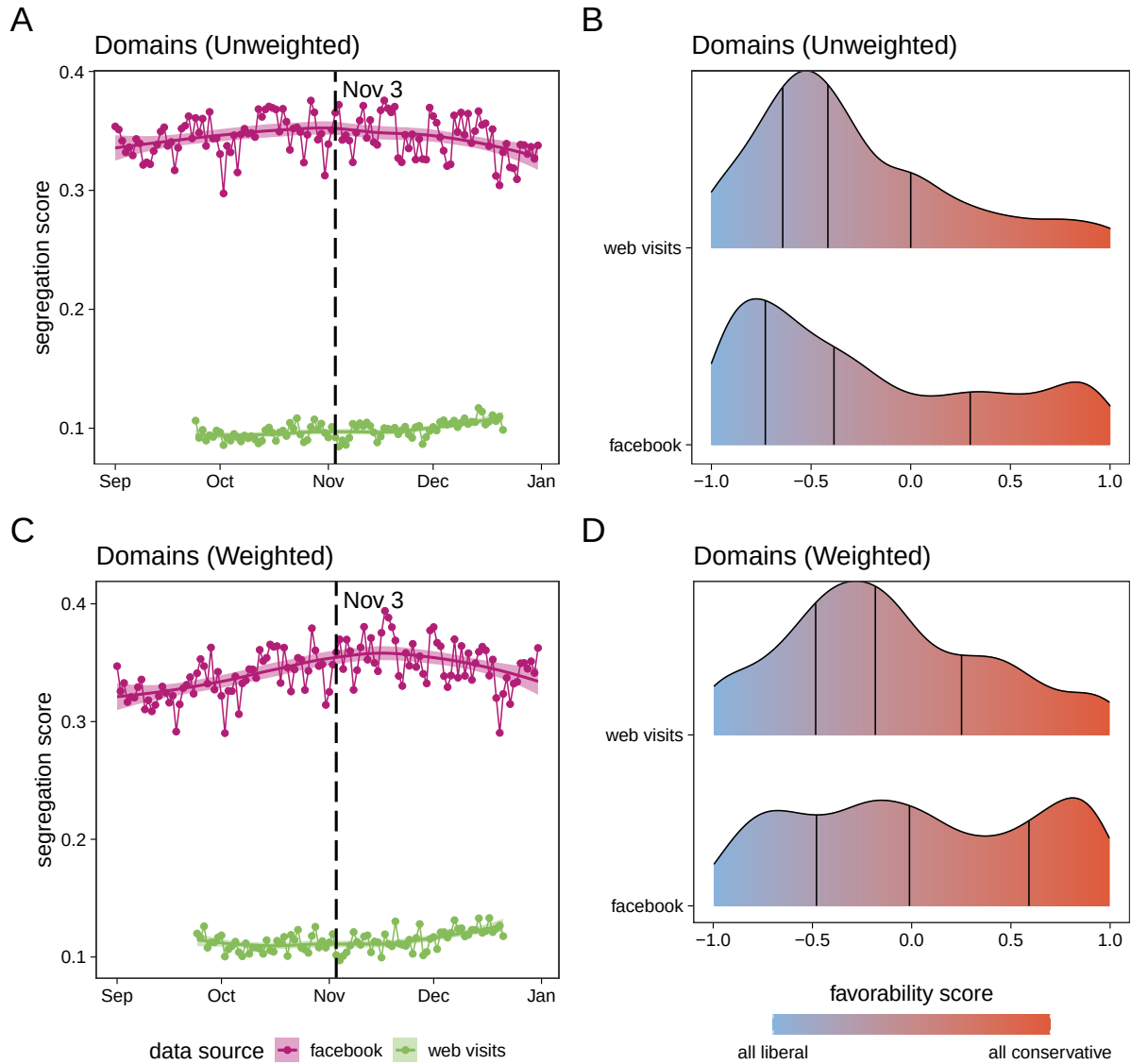


Figure S32: **Intra-person Comparison of Segregation and Favorability Scores.** The left column shows the segregation scores calculated for Facebook and Web news exposure. The right column shows the distribution of favorability scores.

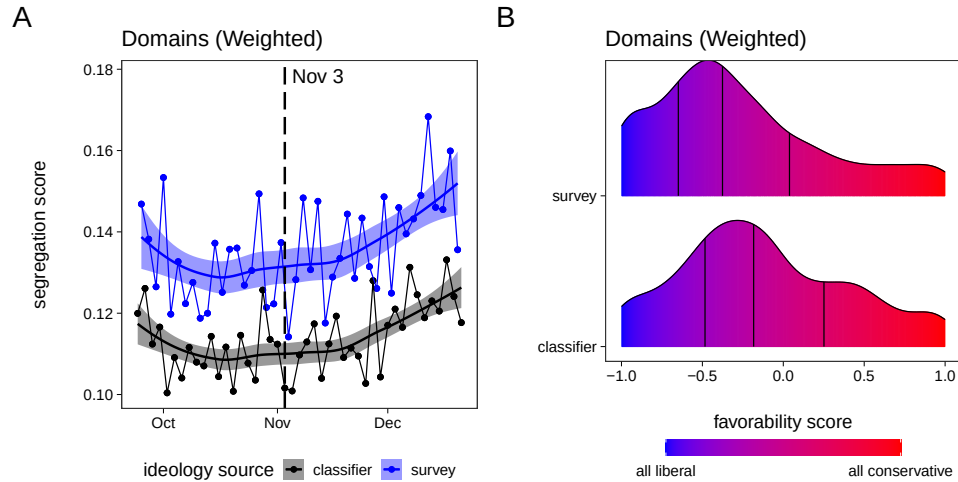


Figure S33: Intra-person Comparison of Segregation and Favorability Scores, by Source of Ideology Labels. The left column shows the segregation scores calculated for Web news exposure when using the ideology classifier or the self-reported ideology data from survey responses. The right column shows the distribution of favorability scores according to the same two sources of ideology labels.

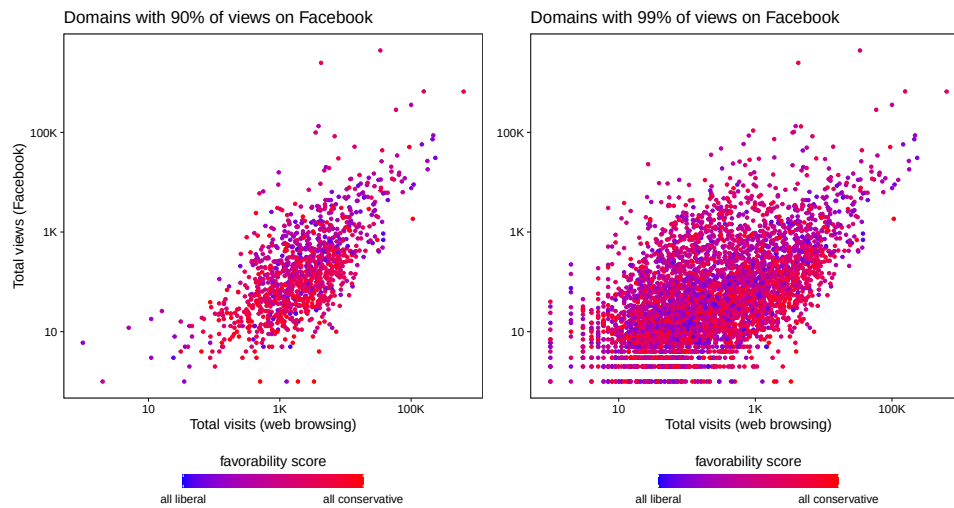


Figure S34: Correlation of Domain Views (Facebook) and Domain Visits (Web). These scatterplots show the association between on-platform views and off-platform web visits for the same set of domains: those accumulating 90% of views on platform (panel A) and those accumulating 99% of views (panel B).

Top 100 Domains by Views on Facebook

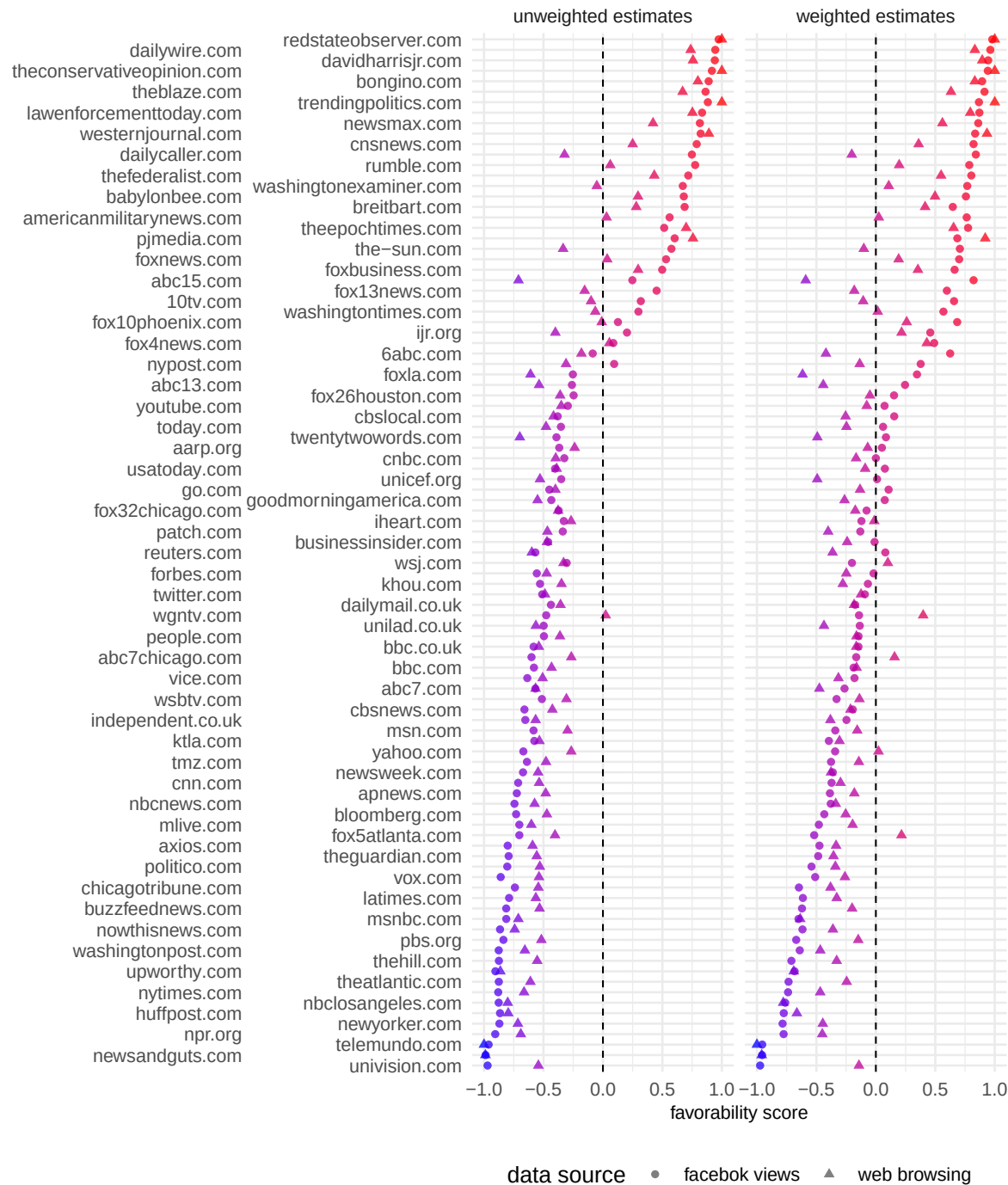


Figure S35: **Difference in Favorability Scores On- and Off-Platform.** This dot chart summarizes the difference in favorability scores for the top 100 domains by views on Facebook.

	Segregation		Favorability	
	Web visits	Facebook	Web visits	Facebook
Political news domains with 90% of views (unweighted)	0.04	0.30	-0.30	-0.22
Political news domains with 99% of views (unweighted)	0.06	0.30	-0.36	-0.23
Political news domains with 90% of views (weighted)	0.04	0.29	-0.12	0.03
Political news domains with 99% of views (weighted)	0.06	0.29	-0.20	-0.01

Table S19: **Segregation and favorability scores for respondent sample with web tracking data.** This table displays the same segregation and favorability metrics reported in Figures 2 and 3 of the main text, but computed based on the web tracking data and Facebook exposure for the sample of study participants who agreed to share their web tracking data. Estimates are presented without and with user-level weights.

S6 Co-Exposure Networks

In order to determine if Facebook users cluster around specific news sources, we built weekly co-exposure networks at the domain and URL levels. The nodes in these networks are unique domains/URLs, and the ties measure the number of unique users who were co-exposed to a given pair of sources (note that for this part of the analyses, we only take into account the exposed audience of domains/URLs). The purpose of analyzing these networks is to determine if we observe evidence of segregation even if we know nothing about the ideology of users. The presence of clusters in the network is one of the footprints of selective exposure and, as we show in the main paper (figure 4), we not only find that these clusters exist, but also that they map onto the two sides of the ideological scale. Here we provide more details about how the networks were built and a summary of their descriptive statistics.

S6.1 Backbone Extraction

Co-exposure networks are highly dense (i.e., there is a tie between two nodes even if only one user viewed both sources) so prior to analyzing the networks, we applied a two-step backbone extraction to only retain the most significant ties (i.e., the ties measuring an overlap unlikely to exist by random chance). To eliminate the weakest ties, we applied a well-established backbone extraction technique known as ‘disparity filter’ (21). This technique compares the observed tie weights (i.e., the count of users co-exposed to two news stories or domains) with randomized weights, normalized for each node such that each of its ties has equal weight. The technique then uses a parameter α to determine the statistical significance of the difference between observed and randomized weights. This parameter α acts as a conventional p -value: smaller p -values lead to a more aggressive filtering of ties. Instead of choosing an arbitrary α value, we opted for a data-driven, mathematical heuristic to choose the optimal threshold, i.e., an α that maximizes the number of edges removed while minimizing the number of nodes becoming isolates. This technique is based on the L-curve method from inverse problems (29), and the filtering resulted in the networks we analyze here. The following section offers basic descriptive statistics that add context to Figure 4 in the main text.

S6.2 Descriptive Statistics

We calculated the following statistics for the weekly networks: number of nodes; number of edges; number of nodes in the largest connected component (LCC); number of edges in the LCC; density; modularity; assortativity by ideology; degree assortativity; centralization; transitivity; minimum edge weight; and maximum edge weight (30, 31). Figure S36 plots these statistics for the networks formed by the domains that accumulate 99% of all views. Figure S37 shows the descriptive statistics for the domains that accumulate 90% of all views. And figure S38 shows the statistics for the networks formed by the top 100 domains ranked by views. Figure S39 offers the same descriptive statistics for the URLs that accumulate 90% of the views. In addition, we run a Quadratic Assignment Procedure (QAP) test with the domain networks

to determine whether the strength of audience overlap between two sources is significantly correlated with pairwise distance in the ideology scores (32, 33); figure S40 shows the results of this test (as expected, we found a negative correlation: i.e., the higher the ideology distance, the lower the audience overlap between two sources). This is a non-parametric test that relies on $N = 1000$ random permutations of the matrices (i.e., the network matrix and the matrix for pairwise ideology distance).^{S17}

^{S17}Due to computational constraints, we only report results based on 100 random permutations for the domain co-exposure network including domains that accumulate 99% of views; and we do not report results for the entire domain coexposure network.

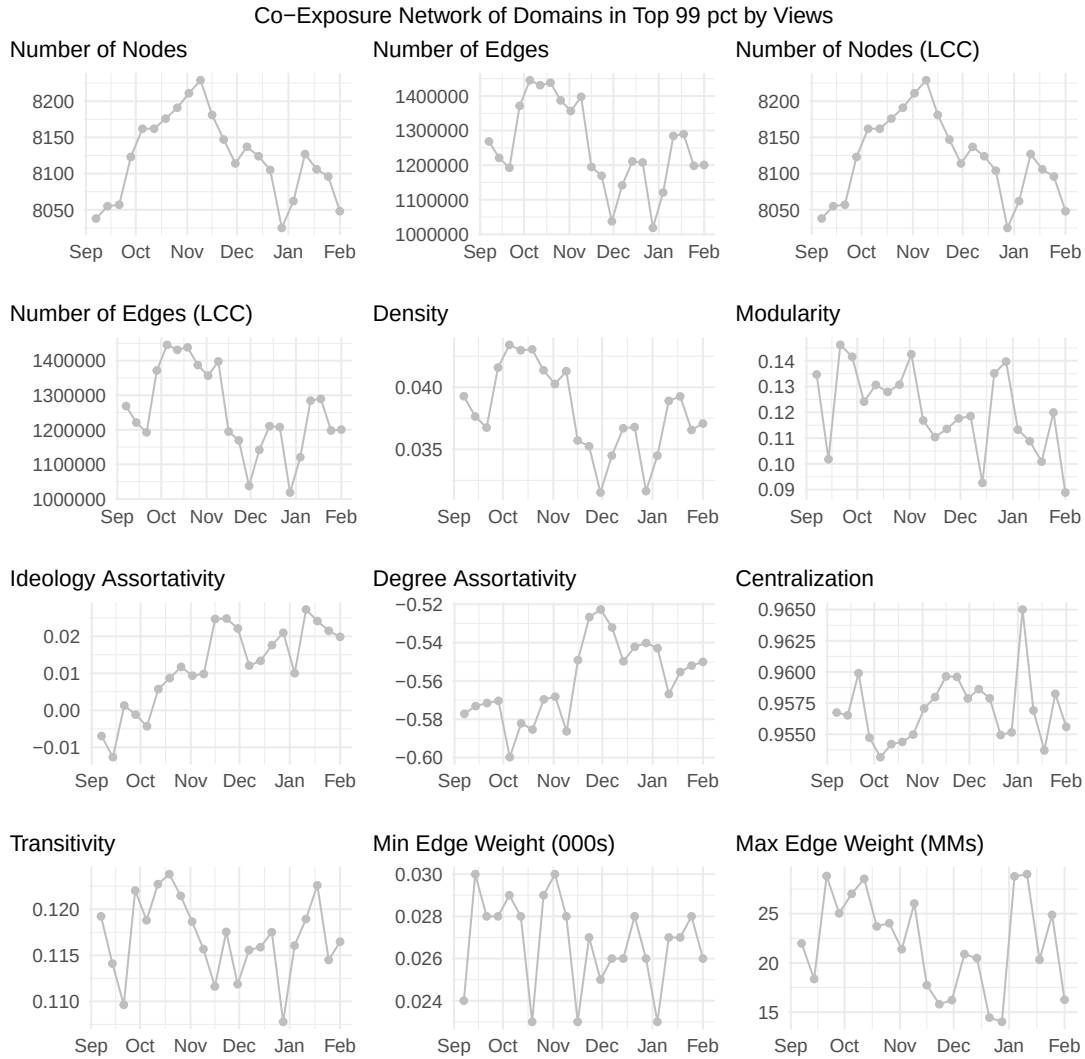


Figure S36: Descriptive Statistics for Domain Co-Exposure Network (99% of all Views).

The co-exposure networks, aggregated weekly and filtered according to the backbone extraction described in SM section S6.1, are highly centralized and tightly connected: all nodes are part of the largest connected component (LCC). The strength of audience overlap connecting pairs of domains varies substantially, from a few hundred users to millions. There is a measurable community structure (as assessed by the modularity score, and described in more detail on SM section S6.3), and co-exposure ties predominantly connect core domains (those with larger reach and higher centrality) with a much larger periphery of lower-reach outlets, as indicated by the negative degree assortativity. There is no strong evidence of strong assortative mixing (or homophily) along ideology, but see Figure S40 for an additional Quadratic Assignment Procedure (QAP) test. The size and composition of the networks changes from week to week, which limits direct comparison.

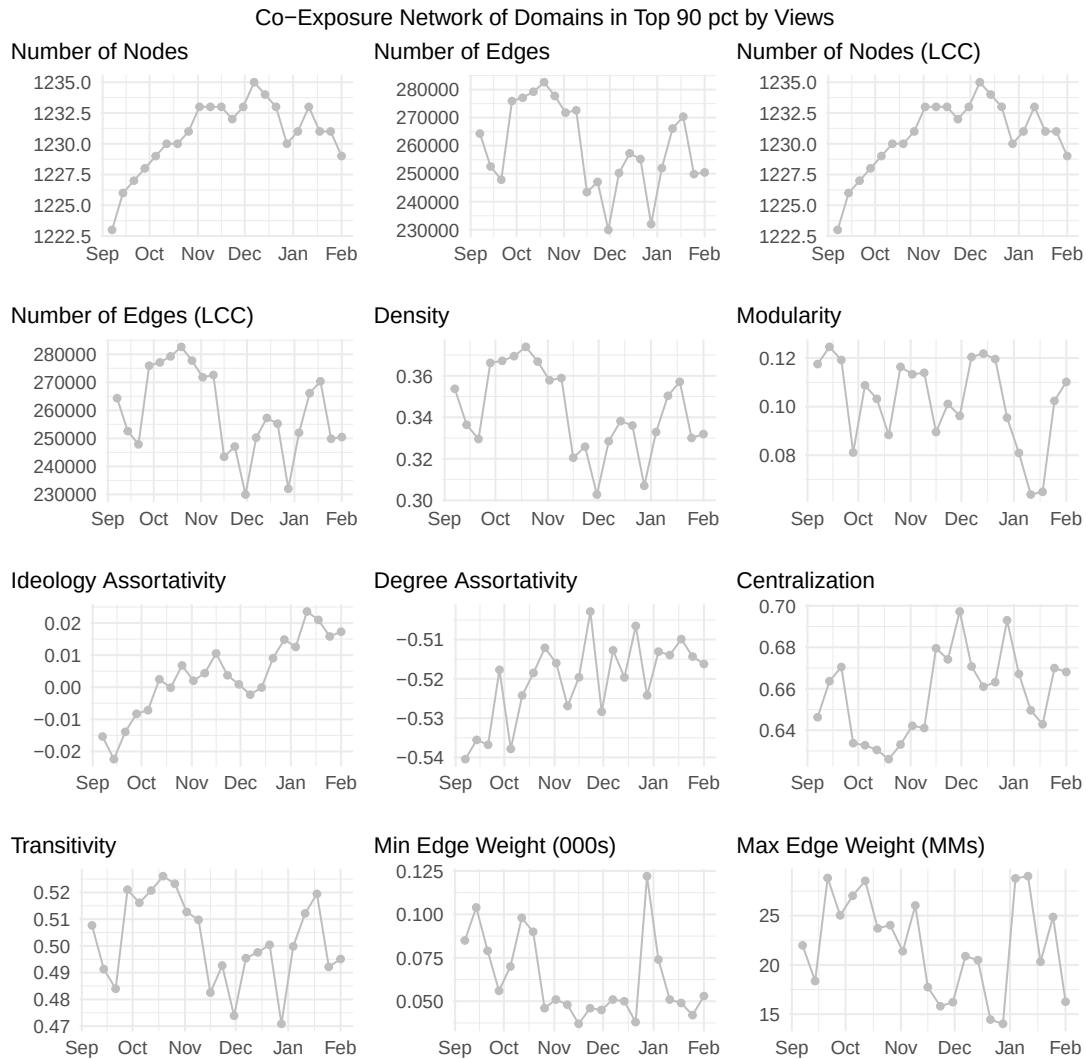


Figure S37: Descriptive Statistics for Domain Co-Exposure Network (90% of all Views). Descriptive statistics for the co-exposure networks that only retain the domains with 90% of all views offer a very similar picture to the descriptive statistics drawn from the larger network. As expected (since the networks are smaller), density is higher; centralization is also slightly lower.

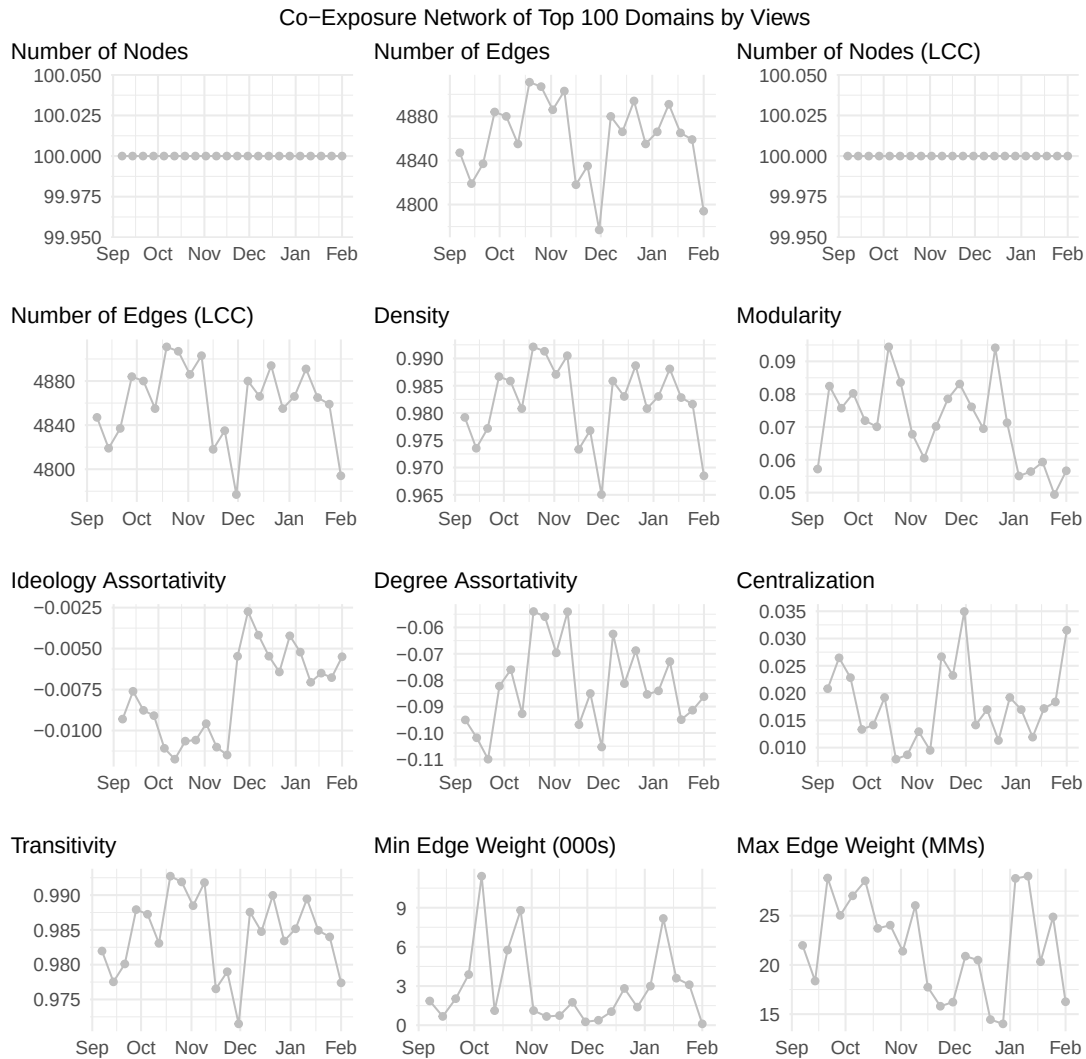


Figure S38: Descriptive Statistics for Domain Co-Exposure Network (Top 100 by Views). Descriptive statistics for the co-exposure networks formed by the top 100 by views: this sub-graph is extremely dense, which is to be expected: these are the most viewed domains, so there are many co-exposure ties across.

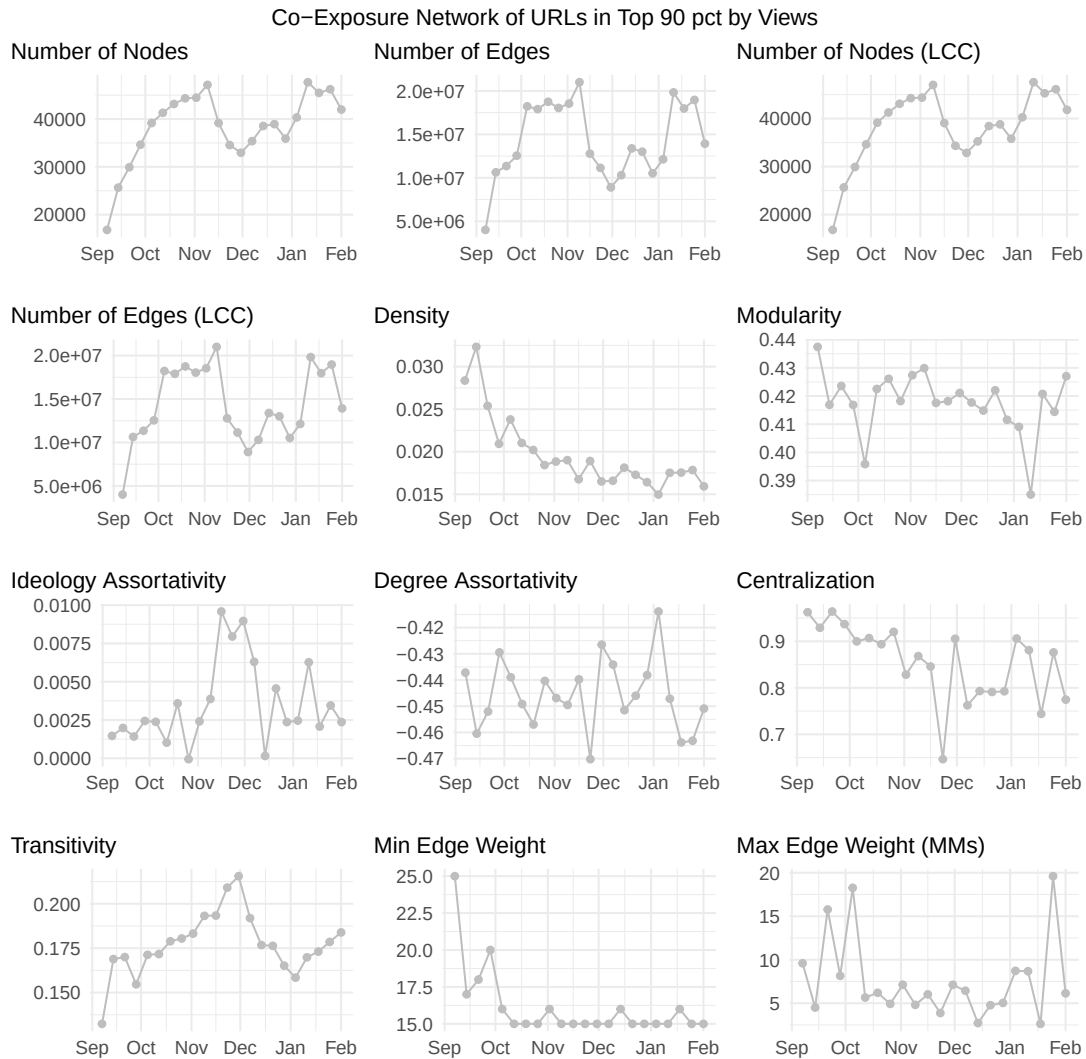


Figure S39: Descriptive Statistics for URL Co-Exposure Network (90% of all Views). The co-exposure networks for URLs (also aggregated weekly and filtered according to the backbone extraction described in SM section [S6.1](#)) are much larger than the domain networks but they are also tightly connected into a single large connected component (LCC). In this case, the network composition changes even more drastically from week to week, which limits direct comparison of the statistics as the networks change.

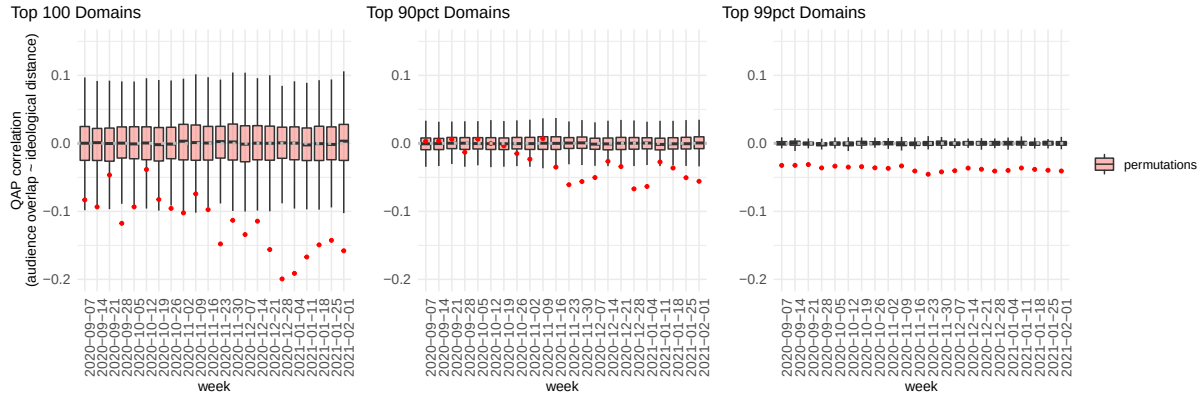


Figure S40: QAP Test with the Domain Networks. Results of a Quadratic Assignment Procedure (QAP) test to determine whether the strength of audience overlap is significantly correlated with pairwise distance in the ideology scores. As expected, the observed correlation (red dots) is negative, which means that domains that are further apart in the ideology distribution share less audience (i.e., less Facebook users are co-exposed). The significance of the correlation is assessed with a non-parametric test that relies on $N = 1000$ random permutations of the matrices (i.e., the network matrix and the matrix for pairwise ideology distance) for panels A and B; and $N = 100$ random permutations for panel C.

S6.3 Community Detection

As mentioned in the section above, we use a random walk algorithm to identify communities and compute the modularity score (20). This algorithm classifies domains (or URLs) into communities (or groups) that have better connectivity internally than externally – in the context of our data, this means that domains (or URLs) that are classified in the same community share a larger audience (i.e., are seen by the same Facebook users). The modularity score summarizes how well the division of nodes in communities captures the actual distribution of connections: the score computes the fraction of the edges that fall within the communities minus the expected fraction if the edges were randomly distributed (34). Figure S41 shows the modularity score, number of communities, network formed by those communities, and favorability scores for the top three communities in the weekly networks formed by the full set of domains. Figure S42 shows the same information for the subset of top 100 domains by views; figure S43 for the subset of domains that accumulate 90% of all views; and figure S44 for the subset of domains with 99% of all views. In figure S45 we show the same information for the URLs that accumulate 90% of all views. (Note: the first and last figure in this series inform Figure 4 in the main paper).

Overall, the figures show highly consistent patterns regardless of how the data is subset or aggregated. The modularity score is higher in the URLs network because of the nestedness of the data: URLs with the same parent domain are more likely to share an audience.

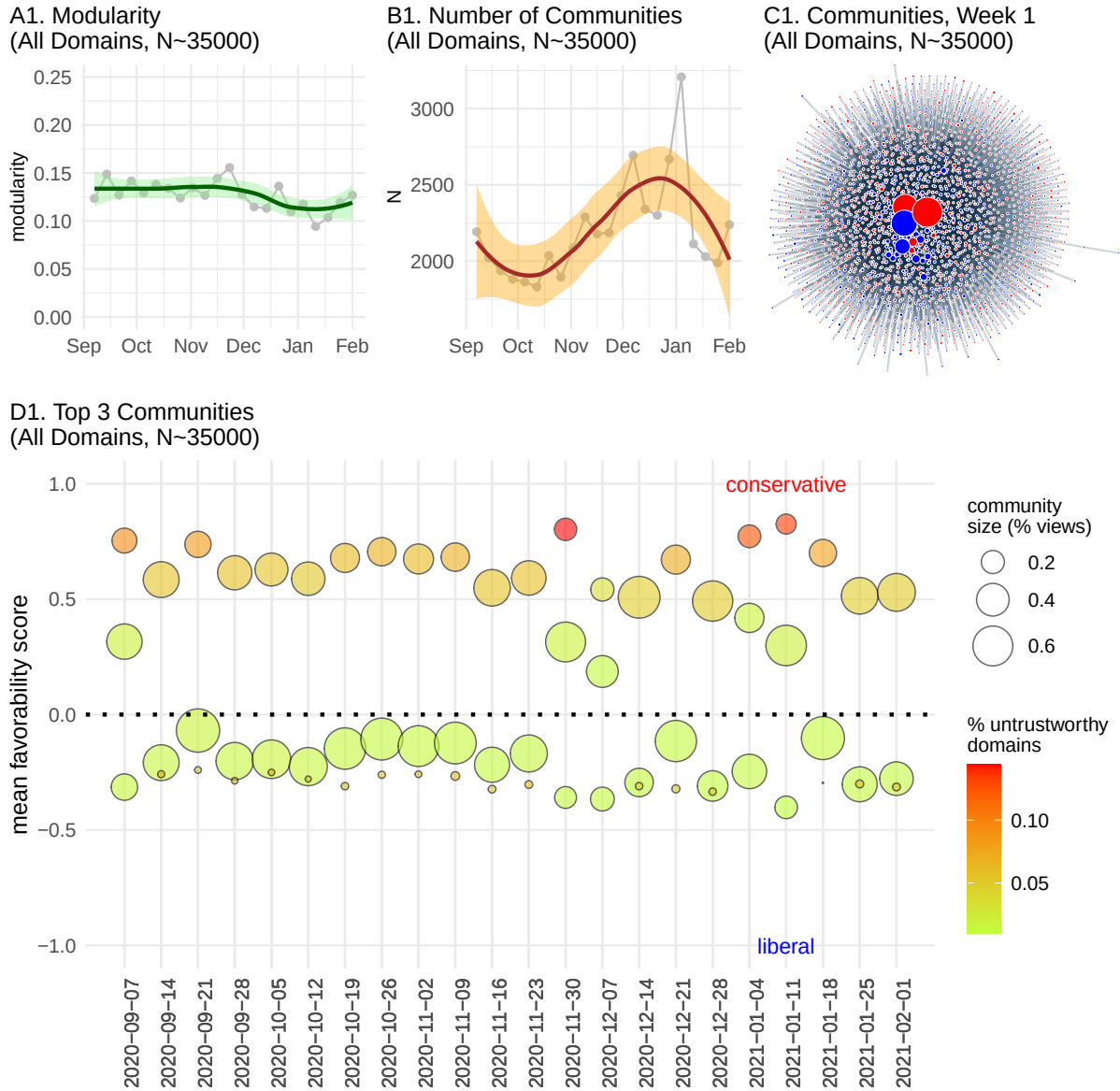


Figure S41: Modular Structure of the Weekly Networks (All Domains). (A1) The modularity score goes slightly down in the weekly networks. (B1) The number of communities identified with the random walk algorithm described in SM section S6.3. goes substantially up, but most of these communities are very small, with most domains classified in 2-3 communities that agglomerate most of the domains. (C1) Network for week 1 collapsed to the communities (i.e., every node is a group of domains), which shows that the structure of the network is dominated by two large communities. (D1) Top three communities in each weekly network as they fall on the ideology scale. Bubble size is proportional to community size and color encodes the fraction of domains classified in each community that is categorized as “untrustworthy”.

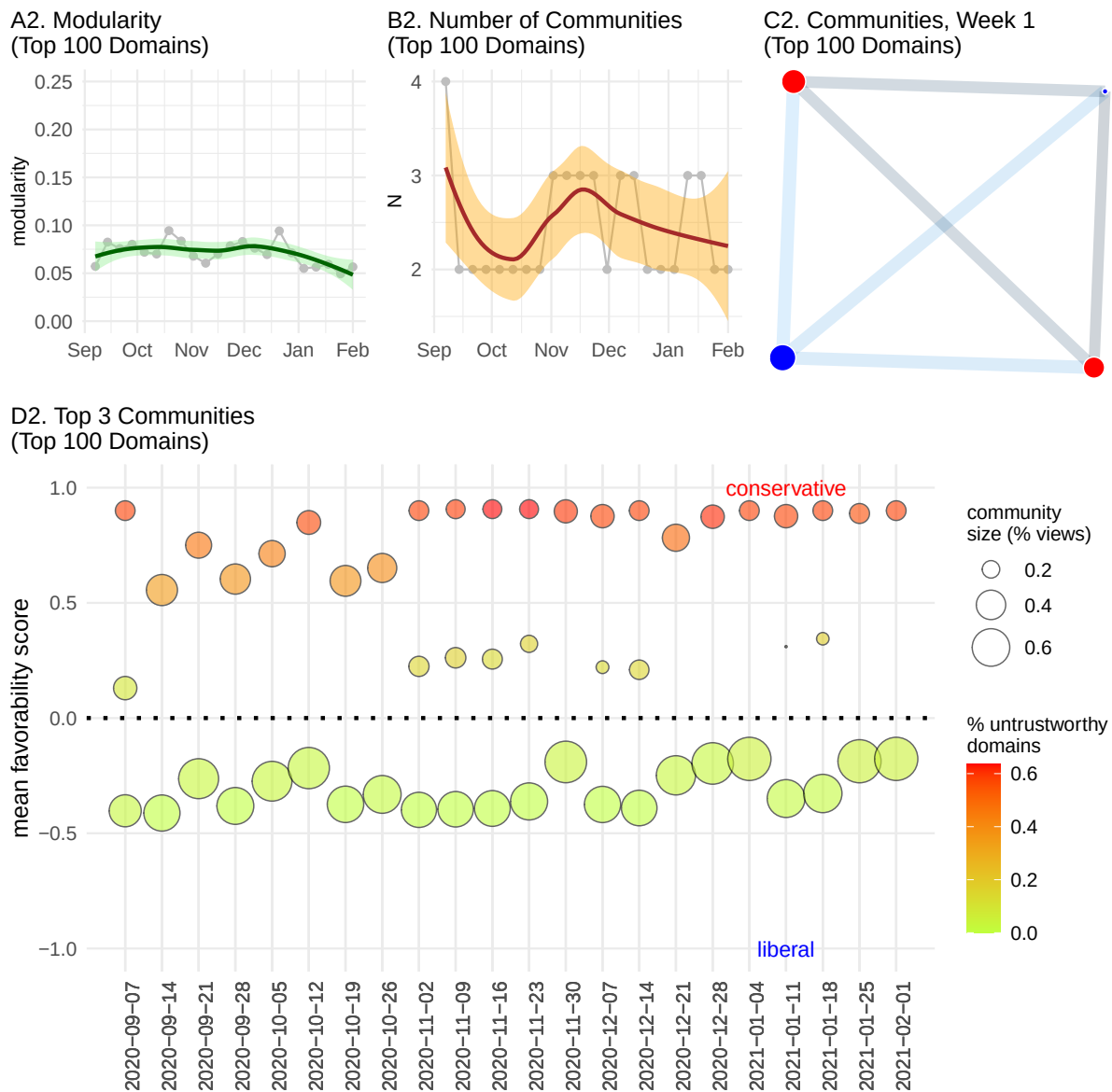


Figure S42: **Modular Structure of the Weekly Networks (Top 100 Domains by Views).** (A1) The modularity score goes slightly down in the weekly networks. (B1) The number of communities identified with the random walk algorithm goes substantially up, but most of these communities are very small, with most domains classified in 2-3 communities agglomerating most of the domains. (C1) Network for week 1, collapsed to the communities (i.e., every node is a group of domains), which shows that the structure of the network is dominated by two large groups. (D1) Top three communities in each weekly network as they fall on the ideology scale. Bubble size is proportional to community size and color encodes the fraction of domains classified in each community that is categorized as “untrustworthy”

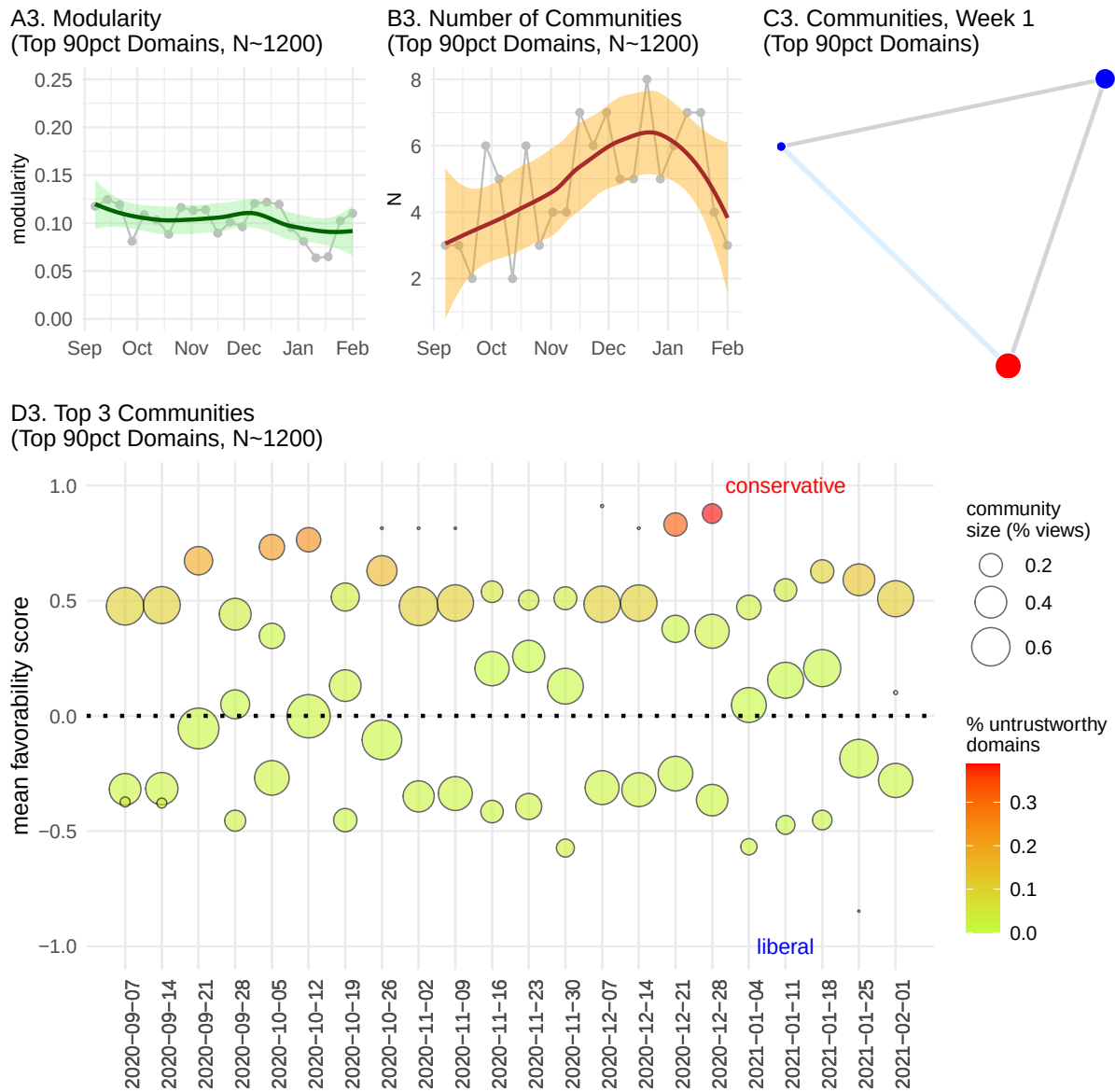


Figure S43: **Modular Structure of the Weekly Networks (Domains with 90% of all Views).** Patterns are very similar to those identified for the network of all domains.

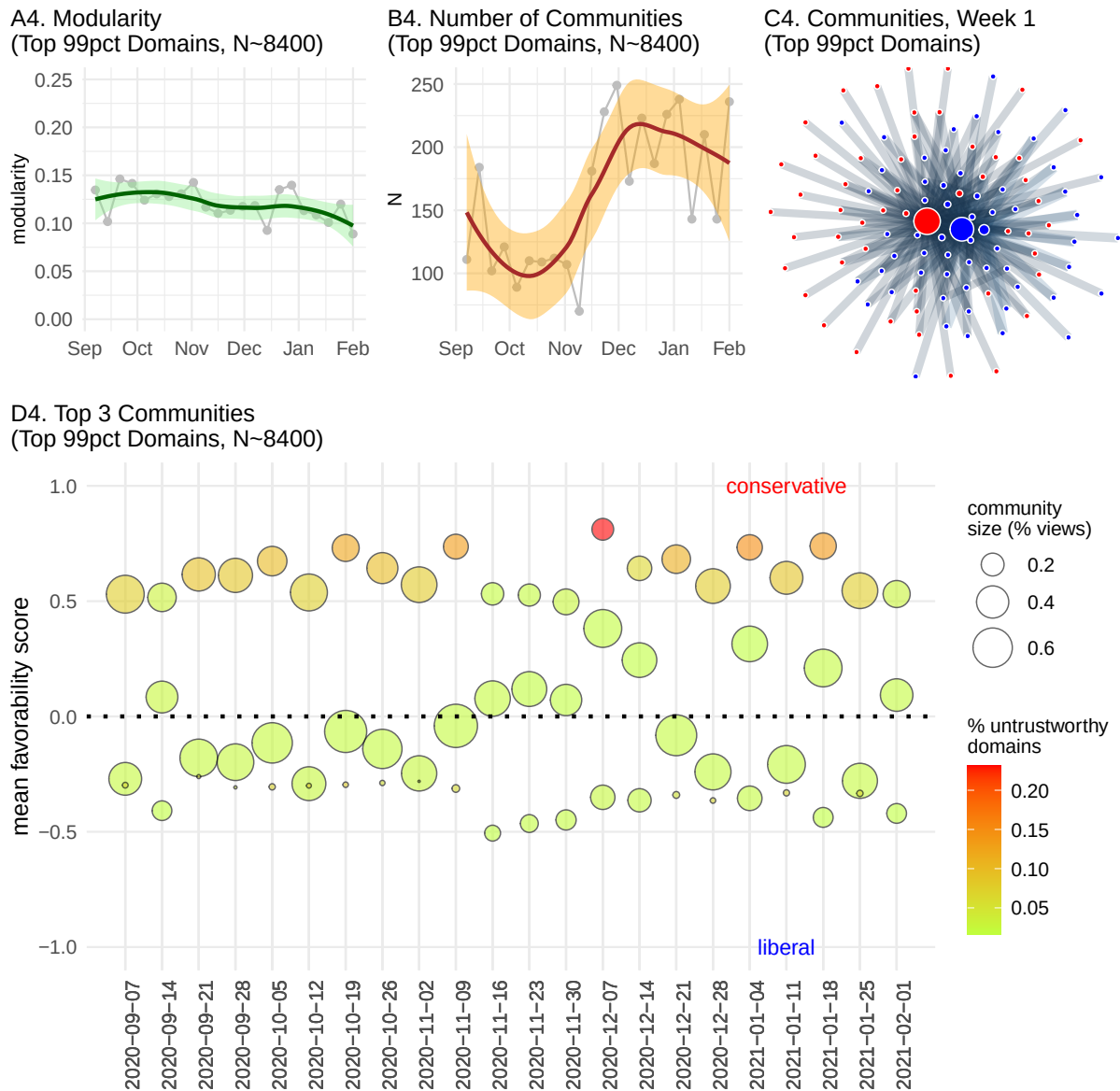


Figure S44: Modular Structure of the Weekly Networks (Domains with 99% of all Views). Patterns are very similar to those identified for the network of all domains.

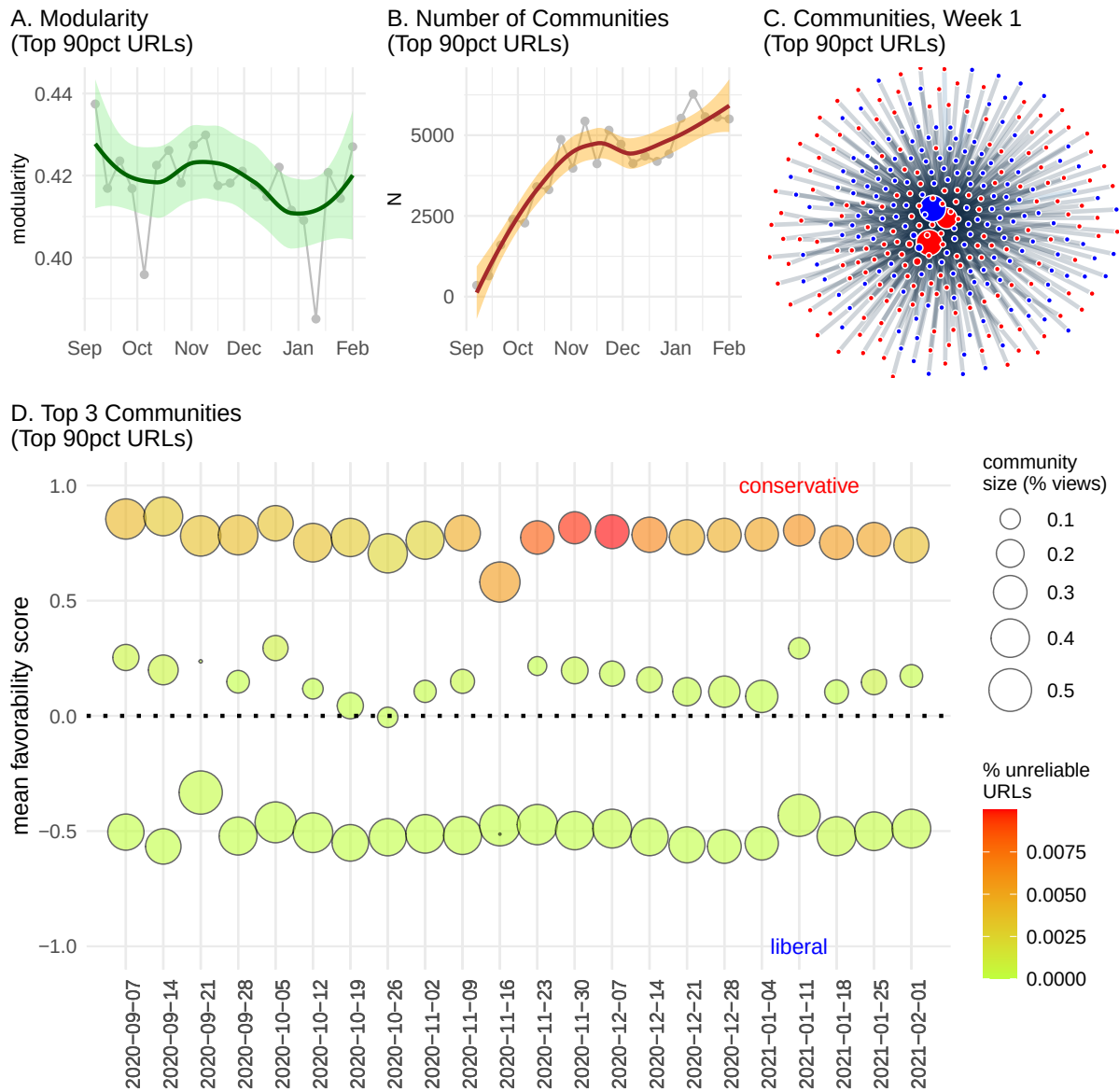


Figure S45: **Modular Structure of the Weekly Networks (URLs with 90% of all Views).** The URL network is (unsurprisingly) more modular than the domains network, but overall patterns are very similar (as explained in the main text).

S7 Pre-Analysis Plan

The PAP included in the next section was registered with the OSF in 2020, before the start of data collection (see SM section [S1.2](#) for more details).

Ideological Segregation in Exposure to News

0. Control Rights and Auditing

The lead authors of this study are Sandra González-Bailón (Penn) and David Lazer (Northeastern). They retain final discretion over what is reported in the paper and how it is reported. In the event of disagreement within the research team about the appropriate presentation or interpretation of the results, the final decision will rest with them.

Facebook researchers Pablo Barbera and Winter Mason are Facebook co-authors of the study. Talia Stroud and Joshua A. Tucker are lead researchers of the FB 2020 Election Research Study, of which this PAP is a part, and will serve as co-last authors on the paper.

1. Introduction

1.1. Summary

The main goal of this paper is to determine how Facebook shaped access to political news in the months leading up to the 2020 Election. This research will quantify segregation levels in exposure to political news, and compare those segregation levels with equivalent measures off-platform (web and Twitter). We will use behavioral indicators of exposure to news content measured at the level of news stories (URLs), news sources (web domains), and the media landscape (co-exposure networks). Further, we will also characterize the potential, active and engaged audience of news stories/news sources to determine how the funnel of engagement unfolds on the platform. This paper will complement the feed experiment demoting like-minded exposure by (a) offering aggregated measures characterizing Facebook's news ecosystem and (b) building indicators of selective exposure that can be compared across channels (Facebook, Twitter, the web).

1.2. Motivation

Digital technologies have increased the choices users have on which news sources to consume. Many theories suggest that more choices translate into more fragmented audiences (with small groups of users distributed in the long tail of media options) and higher segregation levels (with groups forming along ideological lines). Digital technologies, according to these accounts, encourage selective exposure, which risks eroding the common ground on which democracies rely. Here we aim to test those claims by measuring segregation on Facebook and contextualizing that measure with what we observe elsewhere off-platform (on the web, other social media). This paper will analyze exposure to political content on Facebook; determine if ideology is the main driver of exposure patterns (especially as it interacts with political interest); and compare the observed segregation levels with those observed elsewhere in the media environment.

1.3. Prior Work

The metrics and analytical approach we will follow in this paper are based on these prior studies [1-4].

2. Research Design

2.1. Study Overview

We want to analyze segregation in exposure to political news content on Facebook to determine if (a) ideology drives news consumption through selective exposure; (b) segregation levels vary for users with high and low political interest; (c) segregation levels change significantly over time (before/after election); and (d) segregation is higher on Facebook than on other platforms (Twitter, the web). This study will provide behavioral indicators of exposure to news on Facebook; offer a benchmark to situate Facebook activity within the larger media landscape; and compare segregation along ideological lines with segregation around political interest (arguably a more relevant divide in how audiences gain exposure to news: a large fraction of the online population do not proactively consume any news, they choose entertainment instead).

The analyses will focus on exposure to news stories (URLs to civic news domains, as defined by Facebook's internal classifier -- see definitions below). One set of analyses will use the URLs as the unit of observation (to calculate segregation metrics at the news story level). A second set of analyses will aggregate the data to the top-level domain (to calculate segregation metrics at the news source level). A third set of analyses will focus on the structure of the news ecosystem (and determine if there is significant clustering in the co-exposure to news sources). Figure 1 shows a schematic representation of the study. Our analyses will offer context to the patterns identified with the subset of users participating in the like-minded demotion experiment. We will also offer measures that can be compared with observed segregation in Twitter and the web as determined through the data generated by users who agree to participate in the survey. This comparison can also be achieved using other data sources tracking news exposure on the web and Twitter -- although in this case, the data will derive from different user panels. Ultimately, the goal is to analyze statistics characterizing (a) the audience reach of news stories/sources (*actual audience*); (b) the wider range of users that could have been potentially exposed to such news stories/sources (*potential audience*); and the narrower range of news stories/sources with which users interacted (*engaged audience*).

unit of analysis	constructs	aggregated metrics	platform data	survey subset	experiments subset	external data
news stories (URLs)	exposure	potential audience	✓		✓	
		actual audience	✓		✓	✓
		engaged audience	✓		✓	✓
	co-exposure	actual audience overlap	✓		✓	✓
news sources (web domains)	exposure	potential audience	✓	✓	✓	
		actual audience	✓	✓	✓	✓
		engaged audience	✓	✓	✓	✓
	co-exposure	actual audience overlap	✓	✓	✓	✓

Figure 1. Summary of the study. The core of the analyses reported in this paper will rely on platform metrics aggregated at the URL and web domain levels. Blue columns indicate the metrics that can also be calculated using other sets of data (pending their availability as part of the project) so that measures of segregation in exposure can be (1) connected to attitudes and opinions and (2) compared with other online platforms. Our analyses will rely on three definitions of “audience”. We define *actual audience* as the number of unique users who view the story/domain; we define *potential audience* as the number of users that could have been exposed to such news stories/sources; and we define engaged audience as narrower range of users that interacted with the news stories/sources (liked or commented). More detailed definitions can be found in section 2.3.

2.2. Research Questions

- How large (or small) is the level of segregation of exposure to news?
- How does segregation for the active audience compare to segregation for the potential and engaged audiences?
- Do segregation levels vary over the course of the election campaign?
- Do segregation levels vary between users with high and low political interest, as defined by their overall levels of engagement with civic content?
- Are segregation levels higher in original or in reshared posts?
- Does segregation happen at the news source or at the news story level? (How much selection do we observe within news outlets?)
- Does segregation vary across reliable and unreliable news domains?
- How do segregation levels on Facebook compare with off-platform (web, Twitter) segregation levels?
- Is segregation along ideological lines more relevant (in terms of magnitude and prevalence) than segregation along political interest?

2.3. Data

The key variable in this project is “views”, which we use to count impressions and the *actual audience* (unique users) exposed to URLs/domains. We will also calculate metrics of *potential audience* (i.e., users who had those URLs/domains in their inventory but did not view the content) to contextualize observed levels of segregation in news consumption. And we will also calculate engagement statistics (for the *engaged audience*) to add a layer of “meaningful exposure” to news (“views” offer a less strict definition of news exposure than

“engagement”, which presumes a deeper examination of the content by the user; engagement includes clicks, reactions, likes, shares and comments). We will use the inventory data as a benchmark to determine if segregation is higher/lower in the offer of content (*potential audience*) or in the selection of content (*actual audience*). We will use the engagement data to determine if segregation is higher/lower when users react to the content (*engaged audience*).

The key type of content are URLs from civic news domains, which we will then categorize into topics (politics, sports, business, arts...). We will also pay specific attention to covid-related content, given the relevance of information related to the pandemic for the 2020 elections and the dynamics of segregation and misinformation that exist around covid-related content. Section 2.5 gives more information about the content classification methods we will use.

Following the schematic representation in Figure 1, we will analyze data aggregated in the following Data Frames (full definitions of variables below). Data will be collected from September 1 to February 1 with preliminary analyses conducted half-way through the observation window to make sure data processing produces the right measurements. Only URLs shared more than 100 times during this period will be included in the data.

The following data frames describe the relevant variables on which our analyses will be based, using different aggregation levels. These Data Frames summarize the data on which this study will be based; they are not output tables but a description of the variables that we will use to produce the results outlined in sections 3 and 4.

- Data Frame 1: actual audience of URLs/news stories.
 - Variables: <URL>; <URL_id>; <content_type_misinformation>; <content_origin>; <untrustworthy>; <%_original>; <news_topic>; <covid_related>; <domain_id>; <views>; <unique_users>; <%_active_users>; <%_high_political_interest>; <%_low_political_interest>; <Liberals>; <Conservatives>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 1A: this version has the same variables as Data Frame 1 but it is aggregated for users classified as having high political interest based on engagement with civic content.
- Data Frame 1B: this version has the same variables as Data Frame 1 but it is aggregated for users classified as having low political interest.
- Data Frame 2: potential audience of URLs/news stories.
 - Variables: <URL>; <URL_id>; <content_type_misinformation>; <content_origin>; <untrustworthy>; <%_original>; <news_topic>; <covid_related>; <domain_id>; <unique_users>; <%_active_users>; <%_high_political_interest>; <%_low_political_interest>; <Liberals>; <Conservatives>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 2A: this version has the same variables as Data Frame 2 but it is aggregated for users classified as having high political interest.

- Data Frame 2B: this version has the same variables as Data Frame 2 but it is aggregated for users classified as having low political interest.
- Data Frame 3: engaged audience of URLs/news stories.
 - Variables: <URL>; <URL_id>; <content_type_misinformation>; <content_origin>; <%_original>; <news_topic>; <covid_related>; <domain_id>; <clicks>; <reaction>; <likes>; <shares>; <comments>; <unique_users>; <%_active_users>; <%_high_political_interest>; <%_low_political_interest>; <Liberals>; <Conservatives>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 3A: this version has the same variables as Data Frame 3 but it is aggregated for users classified as having high political interest.
- Data Frame 3B: this version has the same variables as Data Frame 3 but it is aggregated for users classified as having low political interest.
- Data Frame 3C: this version has the same variables as Data Frame 3 but it only includes clicks to measure the proportions (e.g. <%_active_users> refers to the fraction of active users that clicked on posts with the URL)
- Data Frame 3D: this version has the same variables as Data Frame 3 but it refers only to engagement excluding clicks to measure the proportions (e.g. <%_active_users> refers to the fraction of active users that liked, reacted, commented on, and reshared posts with the URL)
- Data Frame 4: actual audience of domains/news sources.
 - Variables: <domain>; <domain_id>; <views>; <unique_users>; <untrustworthy>; <%_active_users>; <%_high_political_interest>; <%_low_political_interest>; <Liberals>; <Conservatives>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 4A: this version has the same variables as Data Frame 4 but it is aggregated for users classified as having high political interest.
- Data Frame 4B: this version has the same variables as Data Frame 4 but it is aggregated for users classified as having low political interest.
- Data Frame 5: potential audience of domains/news sources.
 - Variables: <domain>; <domain_id>; <unique_users>; <untrustworthy>; <%_active_users>; <%_high_political_interest>; <Liberals>; <Conservatives>; <%_low_political_interest>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 5A: this version has the same variables as Data Frame 5 but it is aggregated for users classified as having high political interest.
- Data Frame 5B: this version has the same variables as Data Frame 5 but it is aggregated for users classified as having low political interest.

- Data Frame 6: engaged audience with domains/news sources.
 - Variables: <domain>; <domain_id>; <untrustworthy>; <clicks>; <reaction>; <likes>; <shares>; <comments>; <unique_users>; <%_active_users>; <%_high_political_interest>; <%_low_political_interest>; <Liberals>; <Conservatives>; <%_Liberals>; <%_Conservatives>; <time_stamp>.
- Data Frame 6A: this version has the same variables as Data Frame 6 but it is aggregated for users classified as having high political interest.
- Data Frame 6B: this version has the same variables as Data Frame 6 but it is aggregated for users classified as having low political interest.
- Data Frame 7: co-exposure to URLs (weighted edgelist).
 - Variables: <URL_id_1>; <URL_id_2>; <co-exposed_users_count>; <Liberals>; <Conservatives>; <%_co-exposed_Liberals>; <%_co-exposed_Conservatives>; <week>time_stamp.
- Data Frame 7A: this version has the same variables as Data Frame 7 but it is aggregated for users classified as having high political interest.
- Data Frame 7B: this version has the same variables as Data Frame 7 but it is aggregated for users classified as having low political interest.
- Data Frame 8: co-exposure to news sources (weighted edgelist).
 - Variables: <domain_id_1>; <domain_id_2>; <co-exposed_users_count>; <Liberals>; <Conservatives>; <%_co-exposed_Liberals>; <%_co-exposed_Conservatives>; <date>.
- Data Frame 8A: this version has the same variables as Data Frame 8 but it is aggregated for users classified as having high political interest.
- Data Frame 8B: this version has the same variables as Data Frame 8 but it is aggregated for users classified as having low political interest.
- Data Frame 9: population size (used only for calculations, and not reported as part of the study)
 - Variables: <time_stamp>; <total_active_users>; <news_active_users>; <total_active_Liberals>; <total_active_Conservatives>.

Definition of variables (sorted alphabetically):

- <%_active_users>: fraction of all active users (US adults) exposed to the URL/domain (or with the URL in their inventory, or engaging with it).
- <%_co-exposed_Liberals>: fraction of users co-exposed to a pair of news sources that are classified as liberal.

- <%_co-exposed_Conservatives>: fraction of users co-exposed to a pair of news sources that are classified as conservative.
- <%_Conservatives>: fraction of users exposed that are classified as Conservative using Facebook's internal ideology classifier. (See description above)
- <%_high_political_interest>: fraction of users exposed classified as having a high level of political interest based on their on-platform civic engagement (more details on classifier in section 2.5).
- <%_Liberals>: fraction of users exposed that are classified as liberal using Facebook's internal ideology classifier. Users are categorized based on a supervised machine learning model that predicts ideology at the user level based on demographic characteristics, and engagement with civic and non-civic content
- <%_low_political_interest>: fraction of users exposed classified as having a low level of political interest (more details on classifier in section 2.5).
- <%_original>: the proportion of shares of the URL as part of original posts (and not just reshares of someone else's post).
- <clicks>: number of clicks on the URL.
- <co-exposed_users_count>: number of unique users co-exposed to a pair of news sources (edge weight).
- <comments>: number of comments.
- <content_type_misinformation>: binary, content rated "false" by the Third Party Fact-Checking Program (3PFC).
- <content_origin>: is the URL posted by a friend, a page, a group?
- <Conservatives>: number of unique users exposed that are classified as conservative (or with the news URL/source in their inventory, or engaging with it).
- <covid_related>: yes/no.
- <date>: date during which the co-exposure Data Frame was compiled.
- <domain_id>: numeric key.
- <domain>: top-level domain of news stories (URLs). For domains that may serve many different types of content (e.g. twitter.com, youtube.com, etc) we will make our best effort to use a meaningful subdomain.
- <Liberals>: number of unique users exposed that are classified as Liberal (or with the news URL/source in their inventory, or engaging with it).
- <likes>: number of likes.
- <news_active_Democrats>: total number of users (US adults) exposed to URL news content on Facebook classified as Democrat.
- <news_active_Republicans>: total number of users (US adults) exposed to URL news content on Facebook classified as Republican.
- <news_active_users>: total number of users (US adults) exposed to URL news content on Facebook.
- <news_topic>: as defined by Facebook's topic classification; e.g. sports, business, politics, art...
- <reaction>: emotions expressed with emoticon.
- <shares>: number of shares.
- <time_stamp>: data aggregation period (daily, by default).
- <total_active_Liberals>: total number of active users (US adults) on Facebook classified as Democrat.
- <total_active_Conservatives>: total number of active users (US adults) on Facebook classified as Republican.

- <total_active_users>: total number of active users (US adults) on Facebook.
- <unique_users>: number of unique users who view the story/domain (or have it in their inventory, or engage with it).
- <untrustworthy>: binary, whether news domain is untrustworthy, as defined by the experiment that seeks to demote such content (domains with 2+ lifetime strikes for misinformation)
- <URL_id_1>: head of co-exposure tie (URLs that were shared more than 100 times).
- <URL_id_2>: tail of co-exposure tie (URLs that were shared more than 100 times).
- <URL_id>: numeric key.
- <URL>: canonical web address of news content posted on Facebook. This can be traced in Twitter data and linked with web browsing behavior (if data availability allows).
- <views>: number of impressions.
- <week>: week of the year during which co-exposure Data Frame was compiled

2.4. Measures

We will calculate and analyze three measures used in prior research that complement each other in their assessment of selective exposure.

- Segregation index:

$$S_t = \sum_{n \in N} \left(\frac{C_n}{C} \times \frac{C_n}{v_n} \right) - \sum_{n \in N} \left(\frac{L_n}{L} \times \frac{C_n}{v_n} \right)$$

where n refers to news URLs (Data Frames 1, 2, 3) or news domains (Data Frames 4, 5, 6), C_n and L_n are the number of Conservative and Liberals exposed to n , C is the total number of Conservatives and L the total number of Liberals on Facebook, and v_n is the total number of users exposed to n ($v_n = C_n + L_n$). This score approximates 1 when there is strong selective exposure, and it is 0 when Republicans and Democrats visit the same outlets.

- Favorability scores:

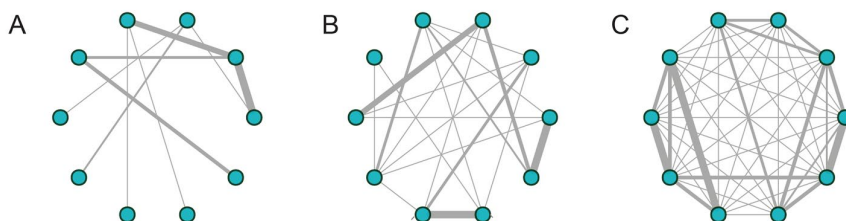
$$f_{av}(n_t) = \frac{(C - L)}{(C + L)}$$

Assigned to news site n in time t . The score equals 1 when n is visited exclusively by Conservatives and -1 when it is visited exclusively by Liberals (0 means that Conservatives and Liberals are equally likely to visit the site n).

- Co-exposure networks: we will measure these structures using weekly statistics (size, density, weight distribution, centralization, transitivity, modularity). Figure 2

illustrates our levels of analysis and the types of co-exposure structures these metrics will help differentiate.

1. Co-exposure to News Sources



2. Co-exposure to News Stories



Figure 2. Schematic representation of co-exposure networks. Our analyses will characterize co-exposure to news at the source level (web domains) and the story level (URLs). Nodes are news sources/stories, and edges map audience overlap. The analysis of these networks will allow us to determine whether there is significant clustering in exposure to sources and, within sources, in exposure to stories. The networks in the figure are all random examples used for illustrative purposes. Panels A-C, for instance, do not show strong evidence of clustering or selective exposure, but the networks in panels D-E do.

2.5. Classifiers

Some of the variables included in Data Frames 1-6 and the ideological segregation measures require applying classifiers to raw platform data. What follows is a description of how these classifiers operate (a full description of these classifiers can be found in the global document “List of Classifiers”):

- Content type: whether URL is from a domain identified as civic news by Facebook’s internal classifier. Civic content is defined as content that relates to politics or social issues. News content is defined as content that reports on current events and follows journalistic standards (features such as presence of a byline, original content, etc.).
- Covid-content: whether URL is predicted to be related to the Coronavirus Disease 2019 (content that explicitly mentions COVID-19, health concepts related to the virus, policy or public health guidance or the economic impact of the virus, as well as people’s reactions to the situation).
- High political interest: users in the top 20% of exposure to/engagement with civic content on Facebook. We will use two operationalizations for robustness test

purposes: (1) a measure based on exposure to civic content on the platform; and (2) a measure based on engagement with civic content (comments, likes, shares). Definition (2) will give us a more stringent measure of 'political interest'. We will also consider a measure that increases this threshold to the top 10%.

- Untrustworthy content: URLs from domains with 2+ lifetime strikes for misinformation. A strike is defined as a post rated "False" by a third-party fact-checker or a post that matches another post rated "False" by a third-party fact-checker.
- Misinformation: URLs rated "False" by a third-party fact-checker.
- Reliable content: we will use two operationalizations for robustness checks. (1) content rated "true" by the Third Party Fact-Checking Program (3PFC); and (2) content classified as "reliable" if it is published by a source included in publicly available databases of reliable sources (e.g., NewsGuard). We will also analyze unlabeled content (missing data) and compare to the reliable/unreliable categories.
- URL topic: whether the URL points to content classified as any of the following topics identified by Facebook's internal classifiers, if available: (1) health; (2) science and tech; (3) religion & spirituality; (4) business; (5) education & learning.
- User ideology: the labels "liberal" and "conservative" are assigned to users based on Facebook's internal ideology classifier. For additional details, see the global "List of Classifiers" document.

3. Descriptive Statistics

Figure shells with descriptive statistics.

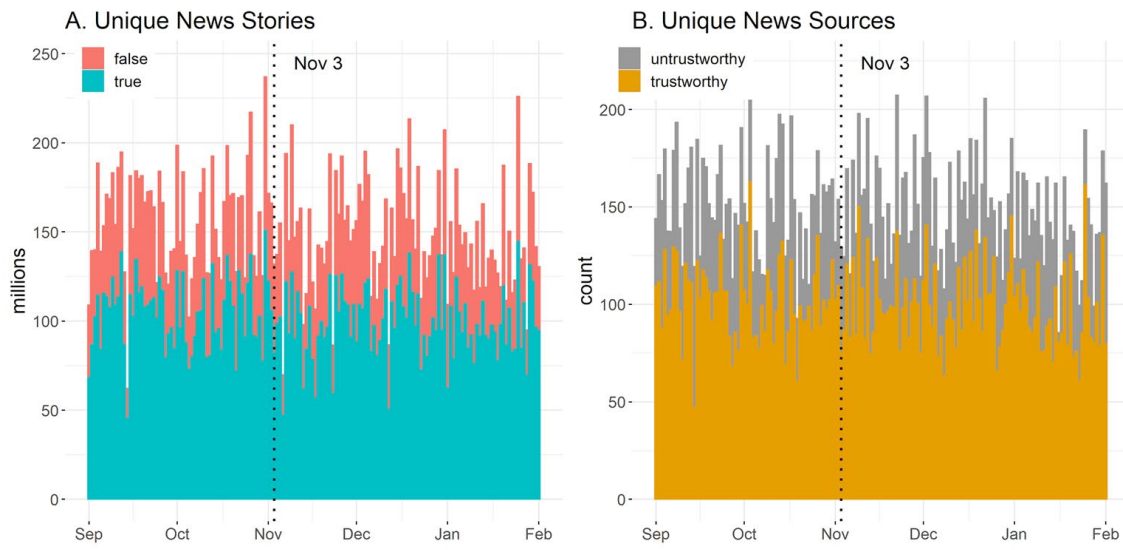


Figure 3. Number of news stories and sources analyzed. Random data. This figure illustrates volume trends in our two main units of analysis (news URLs and domains).

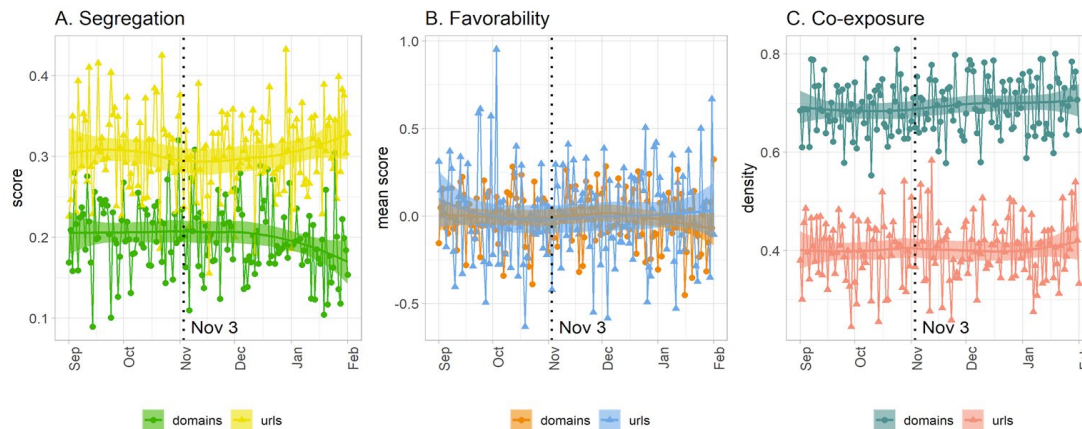


Figure 4. Changes in segregation scores and co-exposure density. Random data. This figure will plot temporal trends for our three measures of interest, based on metrics for active audience: (A) segregation scores at the URL and domain level (to determine if segregation rises as election day approximates); (B) mean favorability scores at the URL and domain level (to determine if there is an ideological slant in active audiences); and (C) density scores for the co-exposure networks built with URLs and domains (denser networks mean higher co-exposure to diverse sources/stories).

4. Empirical Strategy

The analyses will focus on three levels of data aggregation: news stories, news sources, and the media landscape. To the extent that sample size is not an issue, and pending its availability as part of the project, we will also compute these measures for Twitter and Web exposure at the domain level and for the experimental and control groups in the experimental studies.

4.1. Segregation in Exposure to News Stories (URLs)

- T1: compare segregation and favorability scores for three base audiences: potential audience (users who could potentially see the URL); actual audience (users who actually see the URL); and engaged audience (users who click, react, like, reshare or comment on the post). Null hypothesis: segregation levels around news stories do not change (i.e., we cannot identify algorithmic ‘funneling’ effects that nudge users to ideologically-aligned content). Use non-parametric test with bootstrapped CI.
 - T1a: same as T1 but for users classified as having high political interest.
 - T1b: same as T1 but for users classified as having low political interest.
 - T1c: same as T1 but for covid-related URLs only.
 - T1d: same as T1 but for URLs classified as misinformation only.
- T2: compare segregation and favorability scores for URLs grouped in three categories: posted by individual user, posted by group, or posted by page. Null hypothesis: segregation levels around news stories do not change, regardless of who posted the content. Use non-parametric test with bootstrapped CI.
 - T2a: same as T2 but for users classified as having high political interest.
 - T2b: same as T2 but for users classified as having low political interest.
 - T2c: same as T2 but for covid-related URLs only.
 - T2d: same as T2 but for URLs classified as misinformation only.
- T3: test if segregation (calculated using potential audience, actual audience, and engaged audience) increases over time. Null hypothesis: segregation does not increase as election day approaches. Fit regression model with segregation score as DV and t as explanatory variable.
 - T3a: same as T3 but for users classified as having high political interest.
 - T3b: same as T3 but for users classified as having low political interest.
 - T3c: same as T3 but for covid-related URLs only.
 - T3d: same as T3 but for URLs classified as ‘misinformation’ only.

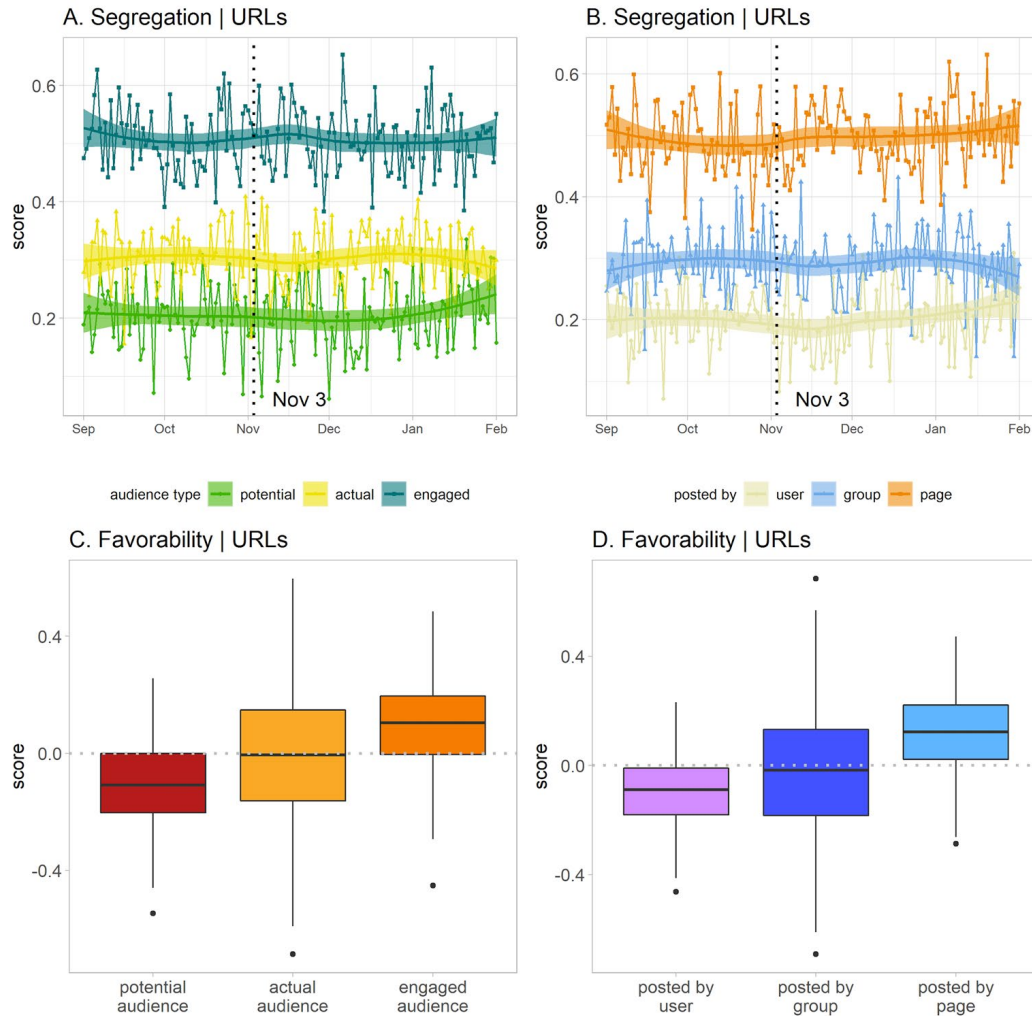


Figure 5. Segregation and favorability scores for grouped URLs. Random data. Panels A and B will plot changes in the segregation and favorability scores of URLs according to their potential audience, actual audience, and engaged audience. Panels B and D will plot changes in the segregation and favorability scores of URLs according to the type of account that posted the content: users, groups, or pages. The time series in panels A and B will be modeled to determine if there are any significant trends before and after election day. The boxplots in panels C and D will be aggregated for the full time period and also for the period before/after election day. We will also generate these figures for the subset of users classified as having high/low political interest, covid-related URLs, and URLs classified as misinformation.

4.2. Segregation in Exposure to News Sources (Domains)

- T4: compare segregation and favorability scores for three base audiences: potential audience (users who could potentially see the domain); actual audience (users who actually see the domain); and engaged audience (users who click, react, like, reshare or comment on the domain-related post). Null hypothesis: segregation levels around news domains do not change (i.e., we cannot identify algorithmic ‘funneling’ effects that nudge users to ideologically-aligned content). Use non-parametric test with bootstrapped CI.
 - T4a: same as T4 but for users classified as having high political interest.
 - T4b: same as T4 but for users classified as having low political interest.
 - T4c: same as T4 but for domains classified as ‘untrustworthy’ only.
- T5: compare segregation and favorability scores for domains grouped in three categories: posted by individual user, posted by group, or posted by page. Null hypothesis: segregation levels around news domains do not change, regardless of who posted the content. Use non-parametric test with bootstrapped CI.
 - T5a: same as T5 but for users classified as having high political interest.
 - T5b: same as T5 but for users classified as having low political interest.
 - T5c: same as T5 but for domains classified as ‘untrustworthy’ only.
- T6: test if segregation (calculated using potential audience, actual audience, and engaged audience) increases over time. Null hypothesis: segregation does not increase as election day approaches. Fit regression model with segregation score as DV and t as explanatory variable.
 - T6a: same as T6 but for users classified as having high political interest.
 - T6b: same as T6 but for users classified as having low political interest.
 - T6c: same as T6 but for domains classified as ‘untrustworthy’ only.

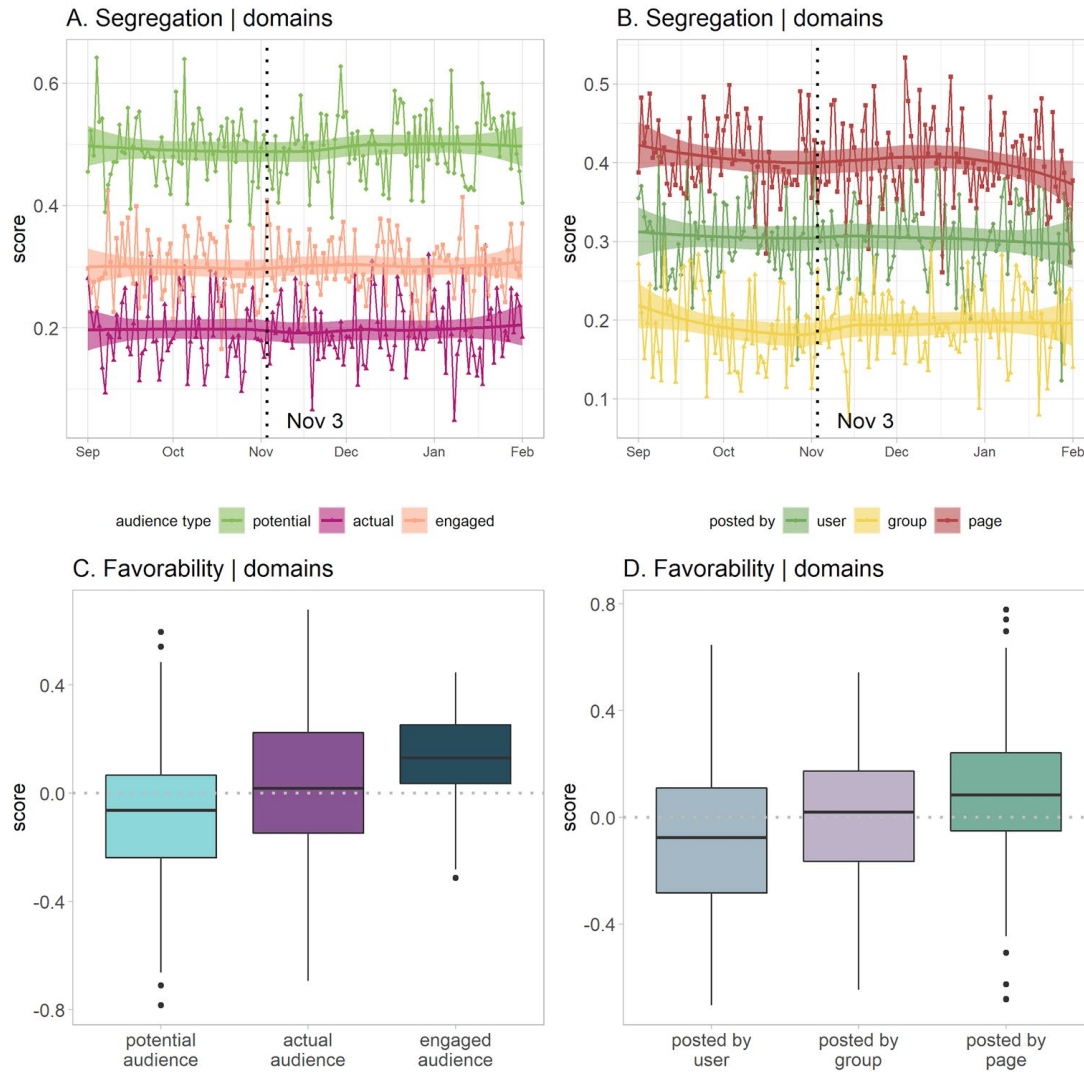


Figure 6. Segregation and favorability scores for grouped domains. Random data. Panels A and B will plot changes in the segregation and favorability scores of domains according to their potential audience, actual audience, and engaged audience. Panels B and D will plot changes in the segregation and favorability scores of domains according to the type of account that posted the content: users, groups, or pages. The time series in panels A and B will be modeled to determine if there are any significant trends before and after election day. The boxplots in panels C and D will be aggregated for the full time period and also for the period before/after election day. We will also generate these figures for the subset of users classified as having high/low political interest, and for domains classified as misinformation.

4.3. Segregation in Co-Exposure (Media Landscape)

- T7: determine if co-exposure to URLs reveals significant clusters. Calculate weekly measures of modularity, assortativity (with ‘untrustworthy’ attribute), and correlation between pairwise distance in favorability scores and pairwise audience overlap. For robustness, run analyses under different thresholding rules (to eliminate weakest overlapping ties, e.g., edges with low weight).
 - T7a: same as T7 but for users classified as having high political interest.
 - T7b: same as T7 but for users classified as having low political interest.
- T8: determine if co-exposure to domains reveals significant clusters. Calculate weekly measures of modularity, assortativity (with ‘untrustworthy’ attribute), and correlation between pairwise distance in favorability scores and pairwise audience overlap. For robustness, run analyses under different thresholding rules (to eliminate weakest overlapping ties, e.g., edges with low weight).
 - T8a: same as T8 but for users classified as having high political interest.
 - T8b: same as T8 but for users classified as having low political interest.

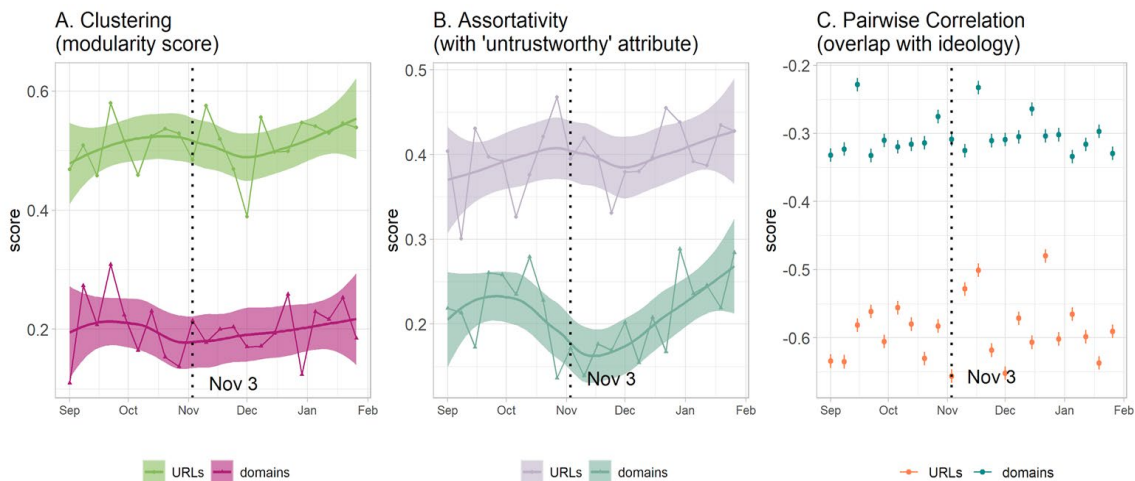


Figure 7. Clustering in co-exposure networks. Random data. Panel A will plot changes in the modularity score of the URL and domain co-exposure networks. A higher modularity score means the networks reveal stronger clustering. Panel B will plot changes in the assortativity score with the “untrustworthy” attribute at the URL and domain levels. If audience overlap is stronger within the group of sources classified as “trustworthy”/“untrustworthy”, the score will be higher. Panel C will plot pairwise correlations between audience overlap and distance in ideology (based on audience composition). We will also generate these figures for the subset of users classified as having high and low political interest.

5. References

1. Gentzkow, M. and J.M. Shapiro, "Ideological Segregation Online and Offline". *The Quarterly Journal of Economics*, 2011. 126: p. 1799-1839.
2. Tyler, M., J. Grimmer, and S. Iyengar, "Partisan Enclaves and Information Bazaars: Mapping Selective Exposure to Online News". *Working Paper*, 2019.
3. Grinberg, N., et al., "Fake news on Twitter during the 2016 U.S. presidential election". *Science*, 2019. 363(6425): p. 374-378.
4. Yang, T., et al., "Exposure to News Grows less Fragmented with Increase in Mobile Access", *PNAS*, in press.

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S8 Deviations and Clarifications

We disclose the following deviations and clarifications from the registered PAP, along with the justifications for these alterations.

- Deviation #1. Page 3 (figure 1 and text): we wrote “How do segregation levels on Facebook compare with off-platform (web, Twitter) data?”. We only analyze web tracking data due to data availability: we do not have exposure data for Twitter (the PAP stated that these analyses were contingent on availability of the data as part of the project, e.g., caption to figure 1, and page 12, first paragraph).
- Deviation #2. Page 12 (T1-T2) and page 14 (T4-T5): we wrote “Use non-parametric test with bootstrapped CI.” This was not necessary in general because we have population data. The only bootstrapped confidence intervals we calculated are part of the QAP tests in the Appendix (figure 7C in the PAP).
- Deviation #3. Page 16 (section 4.3): In section 4.3 and in figure 7 we wrote: “We will also generate these figures for the subset of users classified as having high and low political interest.” Due to computational constraints, we only analyzed co-exposure networks for the full aggregated exposed audience (domains and URLs).
- Deviation #4. Page 16 (section 4.3): The PAP made no statements about analyzing subsets of the co-exposure network based on cumulative views of domains/URLs. In the final analyses reported in this paper, we run the same analyses for different versions of the co-exposure network (all domains, domains accumulating 90% of all views; domains accumulating 99% of all views; and top 100 domains by views). This allowed us to gain insight into the core of the network (when we registered the PAP, we did not expect to have to work with a network formed by ~29000 unique domains).
- Deviation #5. Page 16 (section 4.3): We wrote we would “calculate weekly measures of modularity, assortativity (with ‘untrustworthy’ attribute), and correlation between pairwise distance in favorability scores and pairwise audience overlap.” We run these analyses, but also added additional descriptive statistics for the different subsets of networks (as displayed in the SM). The reason is that this allowed us to provide more descriptive statistics to assess change in the co-exposure networks during our observation window.
- Clarification #1. Pages 9-10: the variable ‘high political interest’ was defined as “top 20% of exposure to/engagement with political content on Facebook”. We use a more granular definition based on different percentiles (10% and 1%) and based on a different operationalization (activity) drawing data from the three months prior to the period of analysis (June 1 to August 31 2020). These modifications allowed us to capture aggregate behavior of users in the tail (high-end) of the distribution and to run robustness tests with different thresholds of what counts as “high political interest”.

- Clarification #2. Page 10: ideology was ambiguously defined as “assigned to users based on Facebook’s internal ideology classifier”. To ensure the robustness of our findings, the analyses we report in this paper apply three different versions of the ideology classification:
 - v1 – [0, 0.35) liberal, [0.35, 0.65] moderate, (0.65, 1] conservative.
 - v2 – [0, 0.5) liberal, (0.5, 1) conservative.
 - v3 – [0, 0.1) liberal, (0.9, 1] conservative.
- Clarification #3. Pages 4-6: exposure was ambiguously defined. The findings we report in this paper measure exposure to content on Feed (which represents 90% of all exposures; other surfaces where exposures happen are user profiles, Groups, Pages). The PAP lacked clarity on how this variable would be made operational.
- Clarification #4. Page 10: the PAP stated that URLs/domains would be labeled as “misinformation” and “untrustworthy content”. We now clarify that only URLs (not domains) can be classified as misinformation (when rated ‘false’ by 3PFC). The variable “untrustworthy” is defined at the domain level (2+ URLs rated false by an independent third-party fact-checker as of 2021-02-01). The PAP lacked clarity on how these variables would be made operational.
- Clarification #5. Page 15: the thresholding of co-exposure networks was ambiguously defined [we wrote “For robustness, run analyses under different thresholding rules (to eliminate weakest overlapping ties, e.g., edges with low weight)”]. As explained in the Appendix, the co-exposure networks we analyze in this paper result from a two-step thresholding: first, we apply the disparity filter for backbone extraction and then we use an L-curve method to choose the alpha threshold. The PAP lacked specific details on how to implement thresholding rules. The two relevant papers not cited in the PAP are:
 - Serrano, M. Á., Boguñá, M., and Vespignani, A. (2009). Extracting the multiscale backbone of complex weighted networks. *Proceedings of the National Academy of Sciences*, 106(16), 6483-6488.
 - Mukerjee, S., Yang, T., Stadler, G., and González-Bailón, S. (2021) What Counts as a Weak Tie? A Comparison of Filtering Techniques to Analyze Co-Exposure Networks. *Social Networks*, 68, 386-393.
- Clarification #6. Pages 11-16: figures 3 to 7 were designed as shells for final figures. We modified the figures as originally envisioned (i.e., we use density plots instead of boxplots) and added additional figures to facilitate the comparisons proposed in the PAP but not visualized. The PAP did not contain the full list of figures the proposed analyses required, and it did not differentiate main figures (i.e., those in the main text) from supplementary figures (those in the appendix).

References and Notes

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2. D. Lazer, Studying human attention on the Internet. *Proc. Natl. Acad. Sci. U.S.A.* **117**, 21–22 (2020). [doi:10.1073/pnas.1919348117](https://doi.org/10.1073/pnas.1919348117) [Medline](#)
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